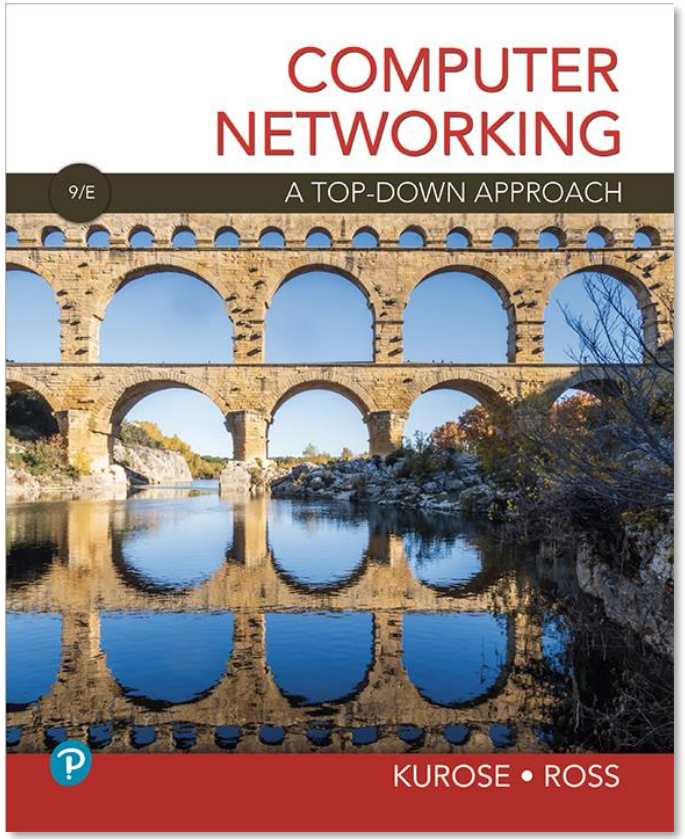
Architecture of Computer Networks



中文版



【第0章学前必看】0.0.0 课程概述_哔哩哔哩_bilibili

湖科大教书匠的个人空间-湖科大教书匠个人主页-哔哩哔哩视频 (bilibili.com)

Computer Networking: A Top-Down Approach

9th edition
Jim Kurose, Keith Ross
Pearson, 2025

Course Content

- Chapter 1 – Introduction
- Chapter 2 – Application Layer
- Chapter 3 – Transport Layer
- Chapter 4 – Network Layer: Data Plane
- Chapter 5 – Network Layer: Control Plane
- Chapter 6 – Link Layer and LANs
- Chapter 7 – Wireless & Mobile Net



Chapter 1: introduction

Unit goal:

- Get "feel," "big picture," introduction to terminology
  - more depth, detail *later* in course



Overview/roadmap:

- What *is* the Internet? What *is* a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, Internet structure
- Performance: loss, delay, throughput
- Protocol layers, service models
- Security
- History

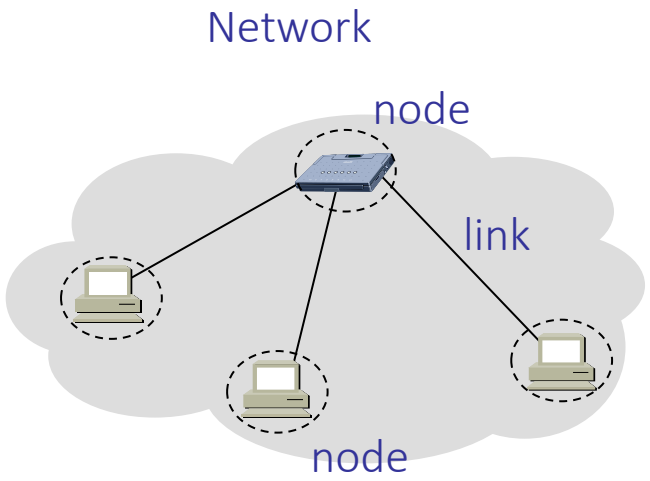
Chapter 1: roadmap

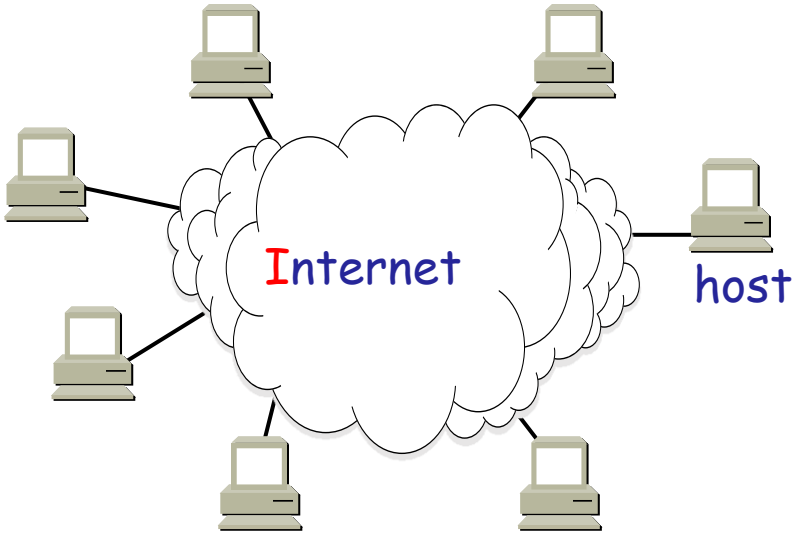
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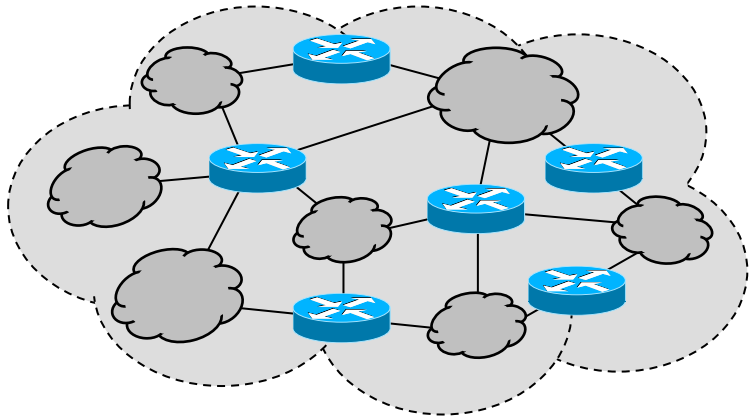
What is a network?

Q: What does a cloud represent?





Interconnection network
(Network of networks)



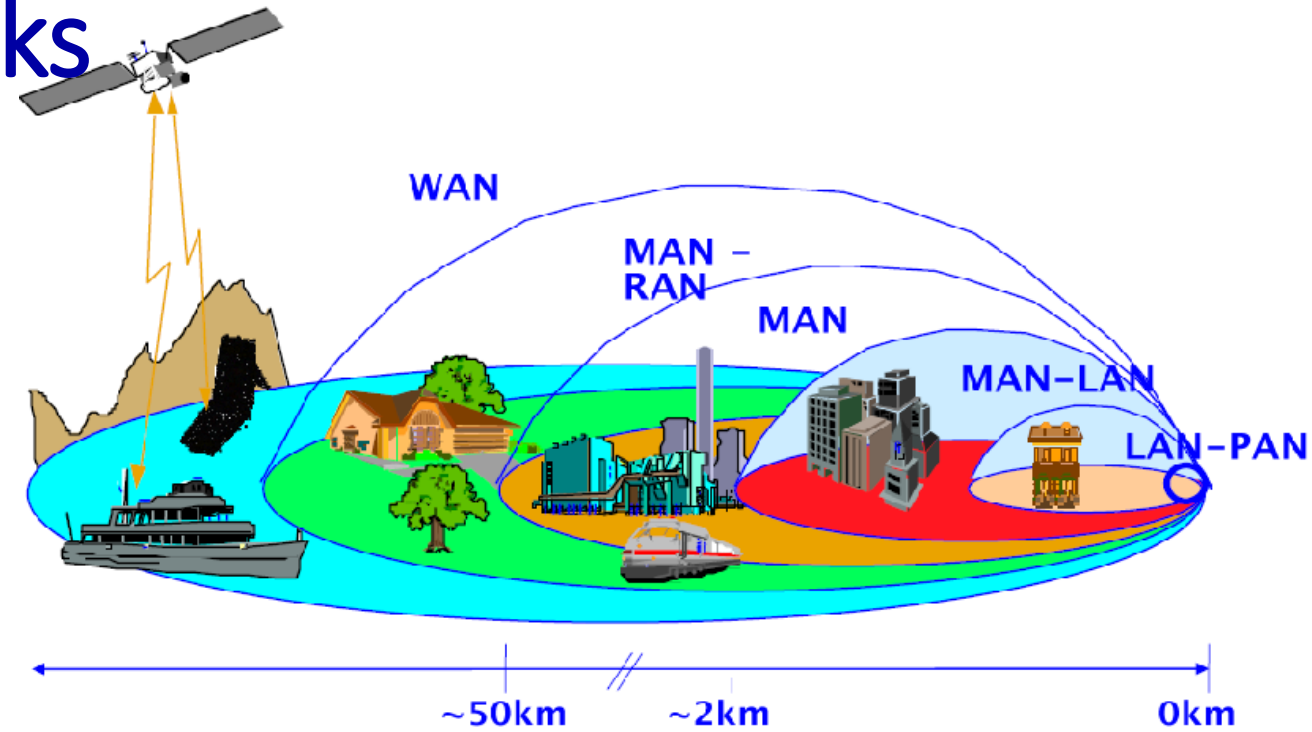
- Networks connect many nodes.
- The Internet connects many networks.

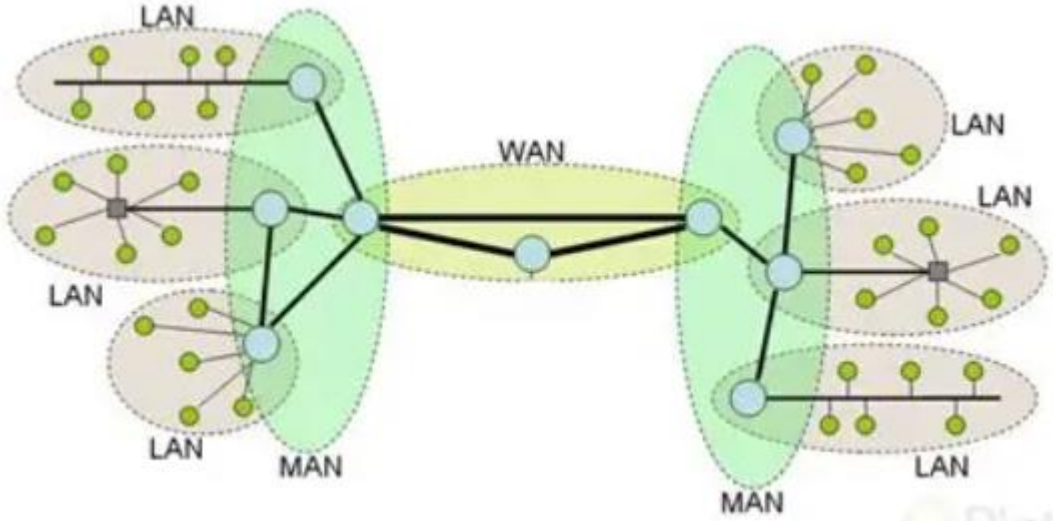
Q: Except the Internet, any other kinds of networks?

Classification of Networks

Classification by **Distance**

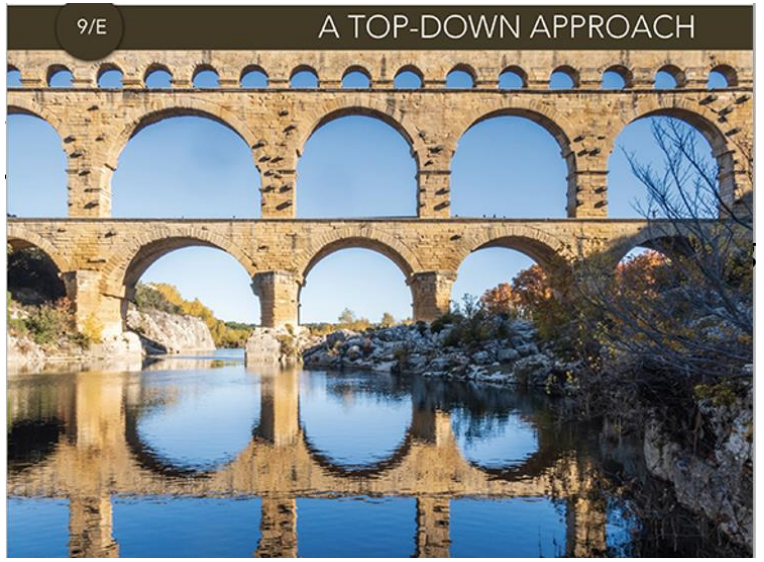
1 m	Personal Area Network (PAN)
10 m Room	
100 m Building	Local Area Network (LAN)
1 km Campus	
10 km Town	Metropolitan Area Network (MAN)
100 km Country	
1000 km Continent	Wide Area Network (WAN)
10000 km Planet	Internet





Chapter 1: roadmap

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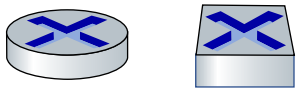


The Internet: a "nuts and bolts" view



Billions of connected computing *devices*:

- *hosts* = *end systems*
- running *network apps* at Internet's "edge"



Packet switches: forward packets (chunks of data)

- *routers*, *switches*

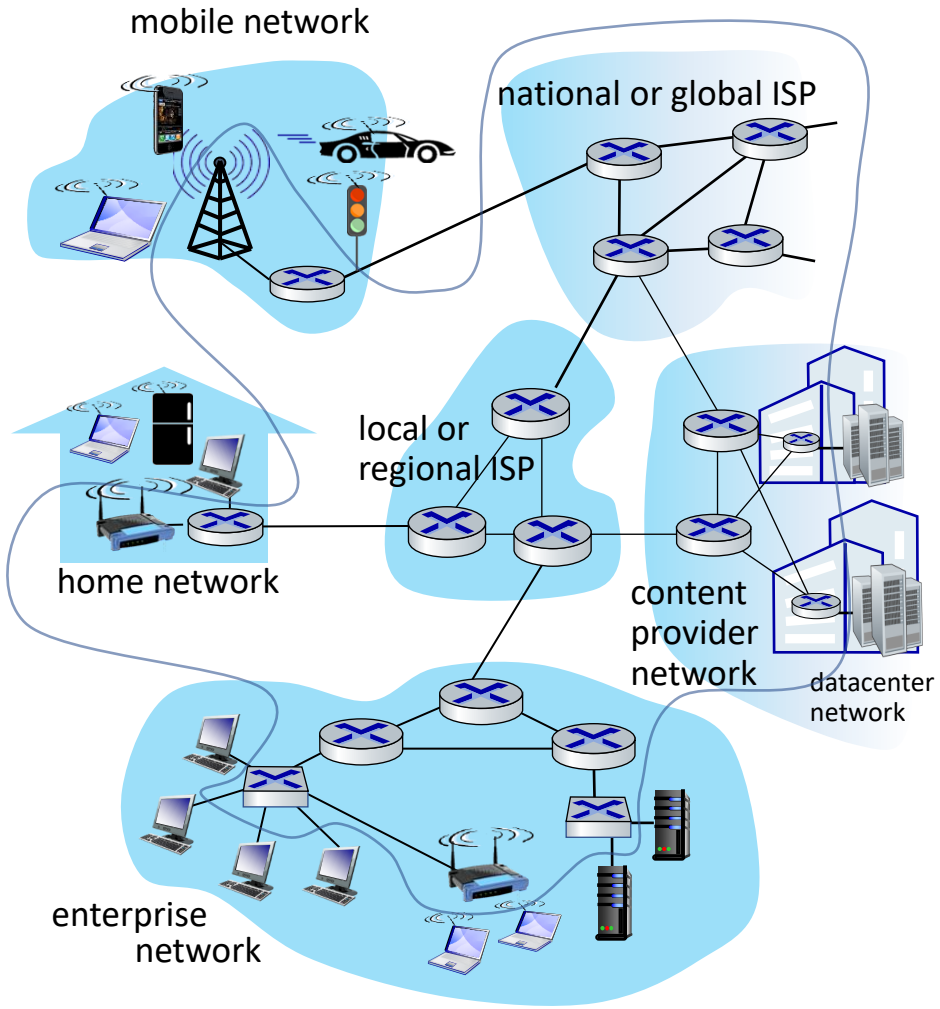


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks

- collection of devices, routers, links: managed by an organization



“Fun” Internet-connected devices

Business Models and IoT



Amazon Echo



Internet refrigerator



IP picture frame



Pacemaker & Monitor



Tweet-a-watt: monitor energy use

bikes





cars



scooters



Security Camera



Slingbox: remote control cable TV



Web-enabled toaster + weather forecaster



AR devices



Internet phones



Gaming devices



sensorized, bed mattress



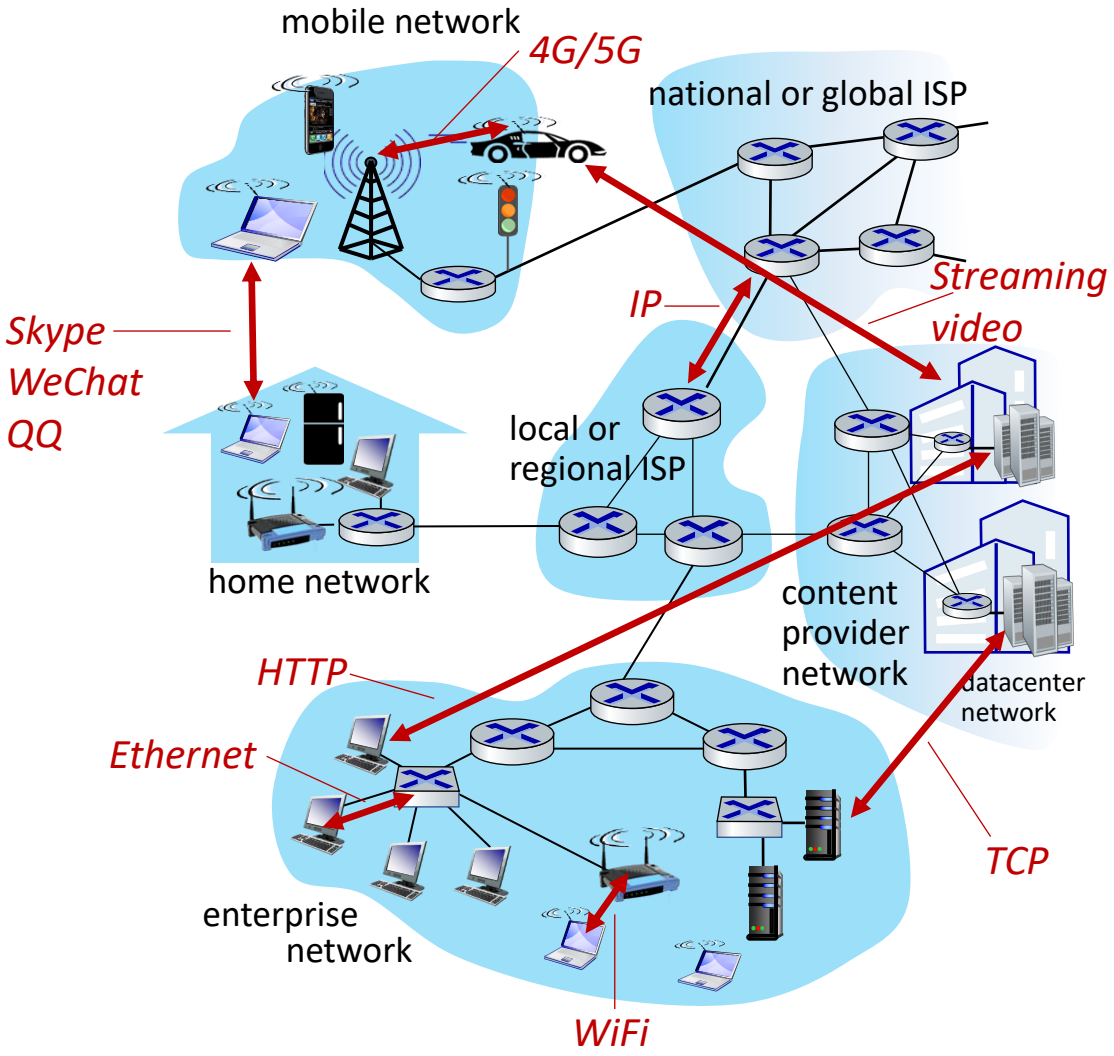
Fitbit

Others?

IoT

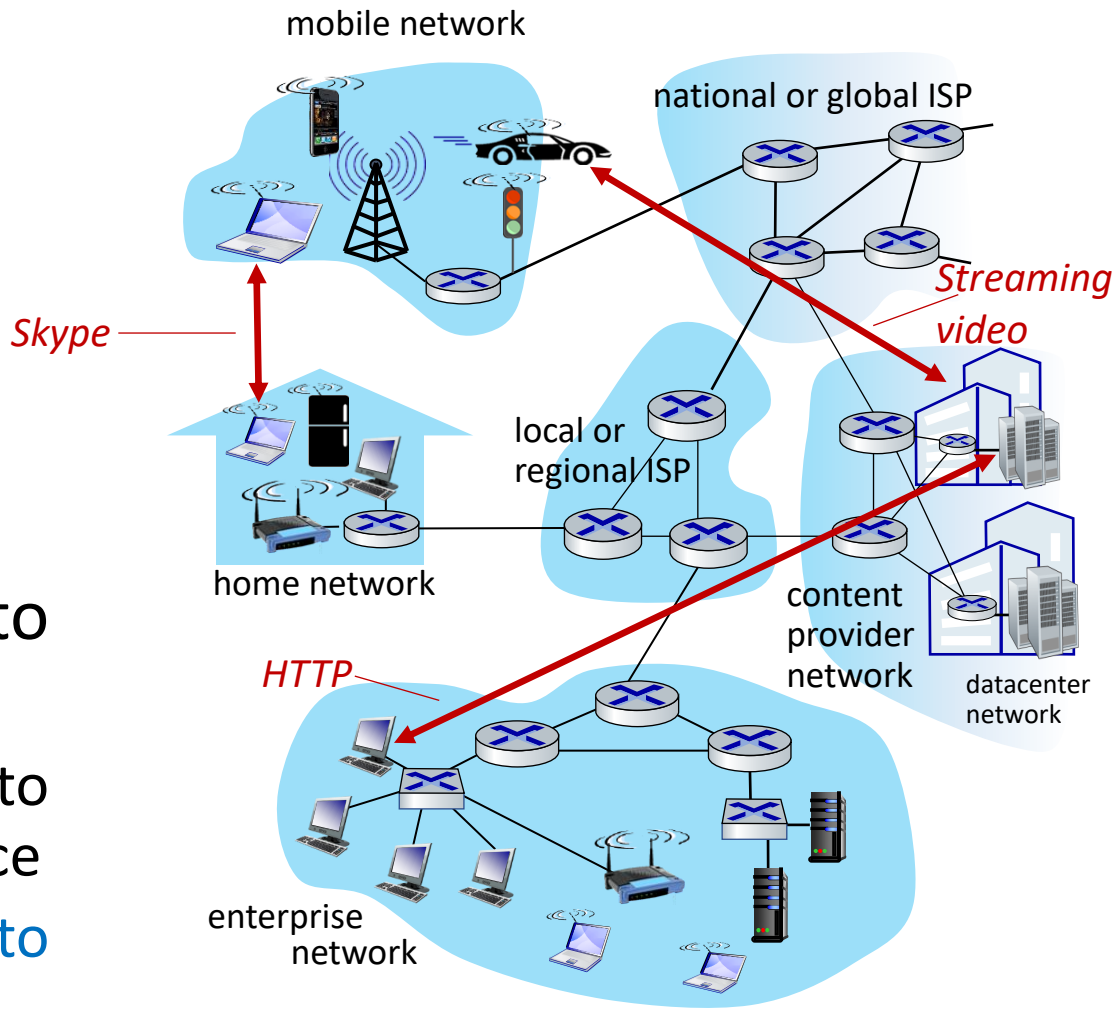
The Internet: a "nuts and bolts" view

- *Internet:* "network of networks"
  - Interconnected ISPs
  - loosely hierarchical
- *protocols* are *everywhere*
  - control sending, receiving of messages
  - e.g., HTTP (Web), streaming video, Zoom, TCP, IP, WiFi, 4G/5G, Ethernet
- *Internet standards*
  - RFC: Request for Comments
  - IETF: Internet Engineering Task Force



The Internet: a "services" view

- *Infrastructure* that provides services to applications:
  - **Traditional applications:** email, web surfing
  - **Distributed applications:** mobile smartphone and tablet apps, like Internet messaging, real-time road-traffic info, streaming video, video conferencing, multi-person games, online social media, location-based recommendation systems, …
- provides *programming interface* to distributed applications:
  - "hooks" allowing sending/receiving apps to "connect" to, use Internet transport service
  - provides service options, analogous to postal service



What's a protocol?

Human protocols:

- "what's the time?"
- "I have a question"
- introductions

Rules for:

… specific messages sent

… specific actions taken when message received, or other events

Network protocols:

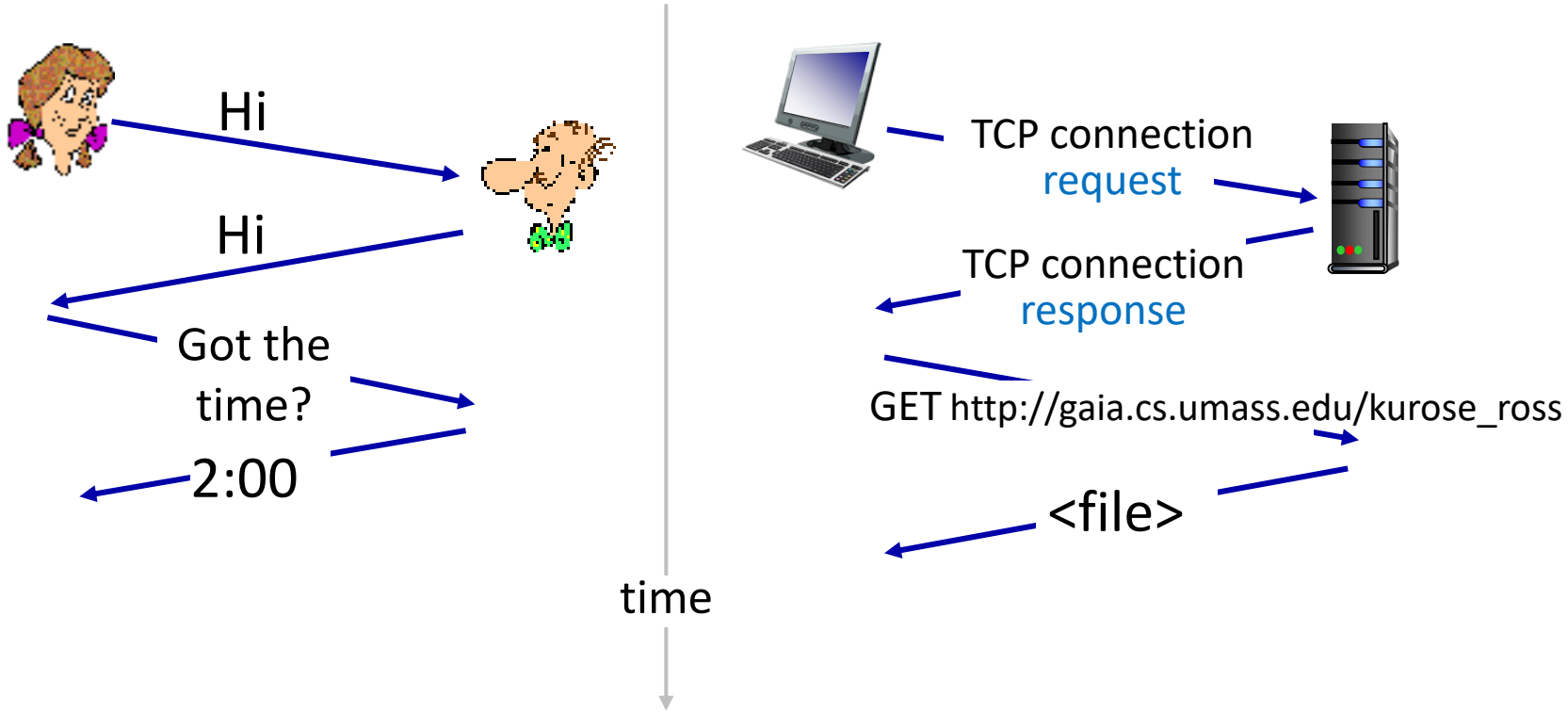
- computers (devices) rather than humans
- all communication activity in Internet governed by protocols



*Protocols define the **format**, **order** of messages sent and received among network entities, and **actions** taken on message transmission, receipt*

What's a protocol?

A human protocol and a computer network protocol:



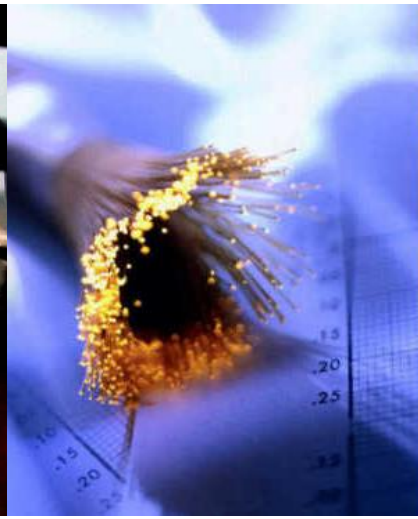
Q: other human protocols?

Examples: Optical Communications

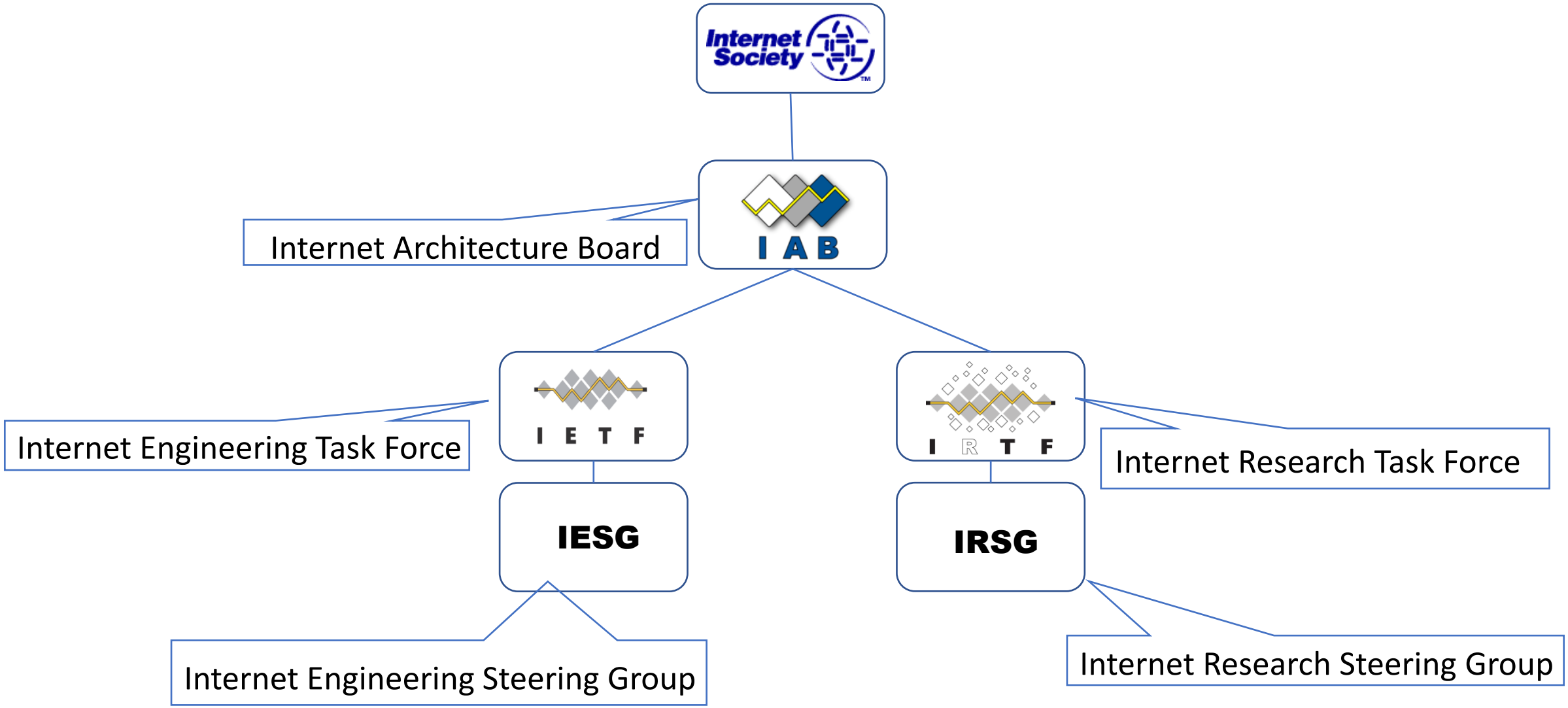
- Smoke signals
- Traffic lights
- Naval signals (ship flags)
- Landing lights for aircrafts
- …



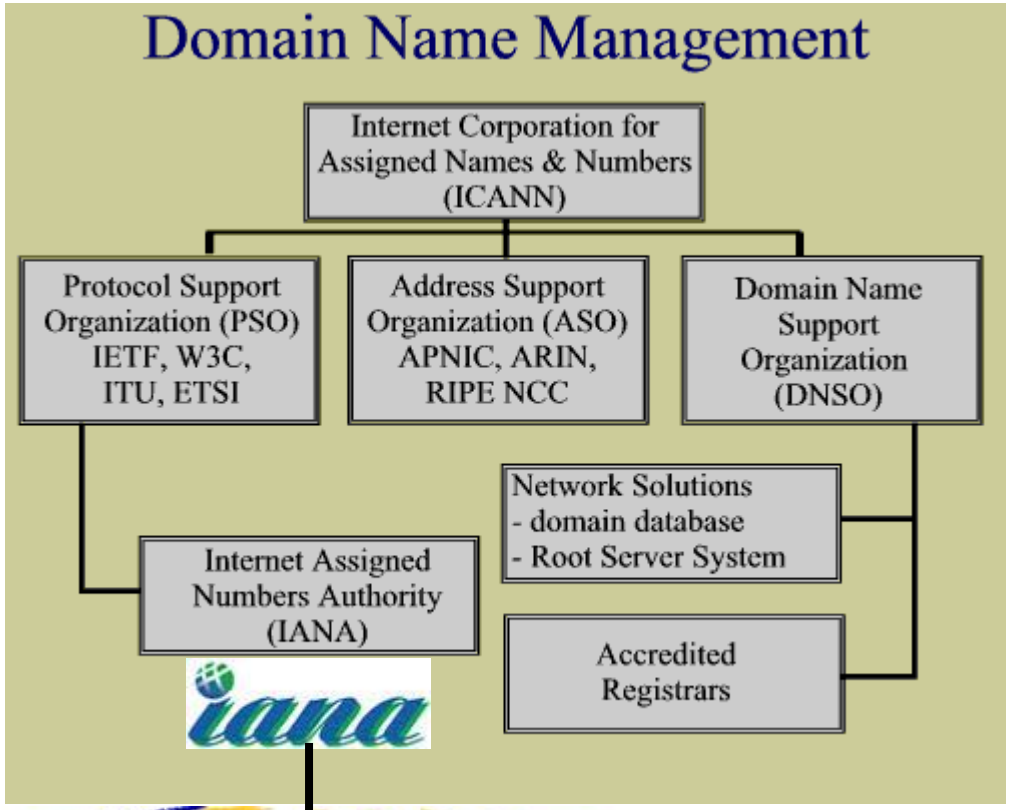




Internet Architectural Management



Domain Name Management

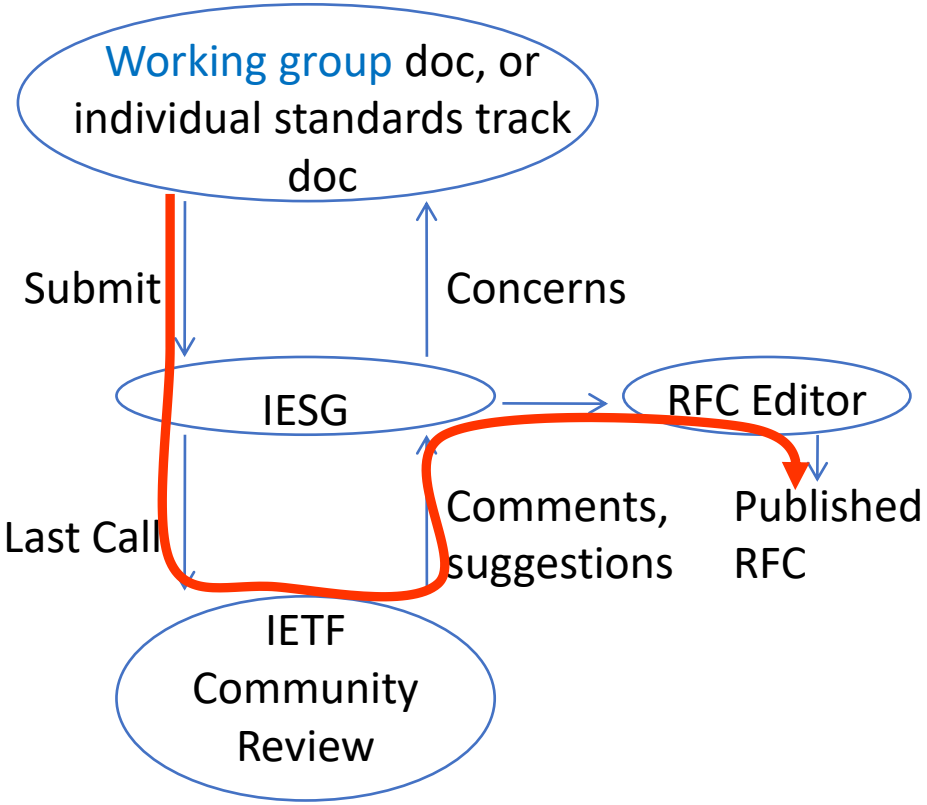




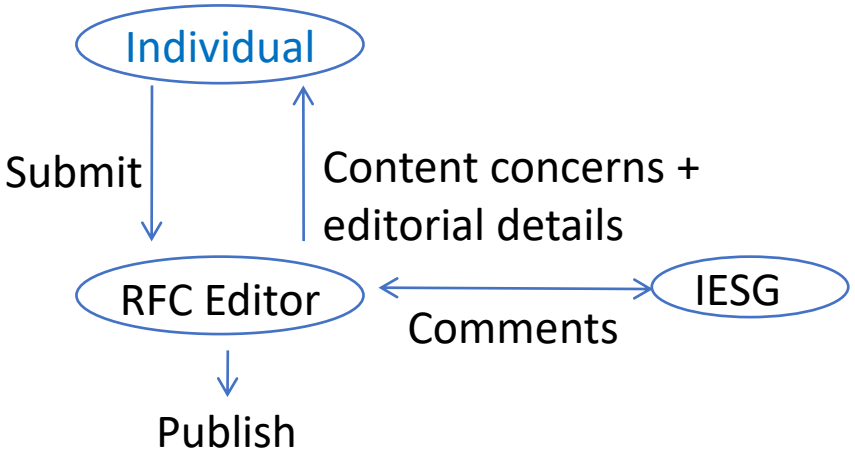
- ICANN generally sets up the system but delegates all of the work to IANA
- IANA
  - operates at the highest international level of organization
  - but divides the work up further among the five Regional Internet Registry's (RIRs).
- These five organizations give out blocks of numbers to ISP's.

IETF and non-IETF Submission





IETF submission



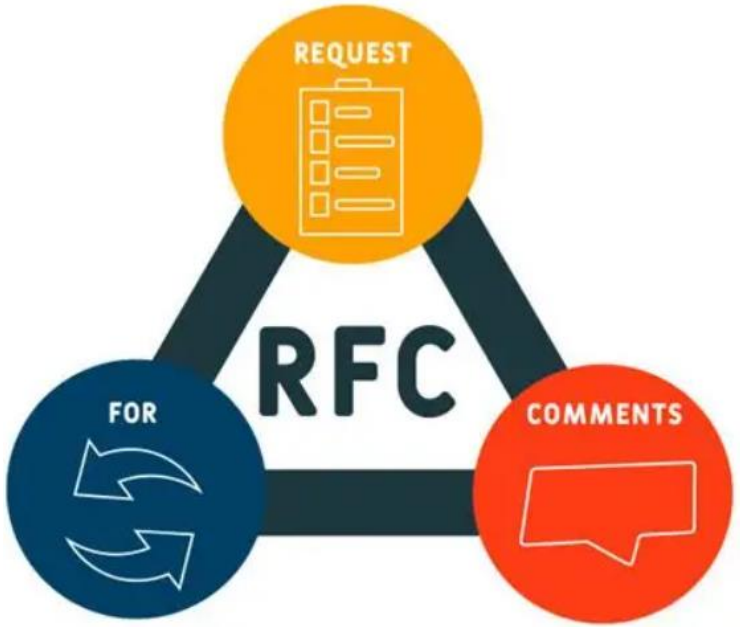
Non-IETF submission

IETF Documents – Two Types



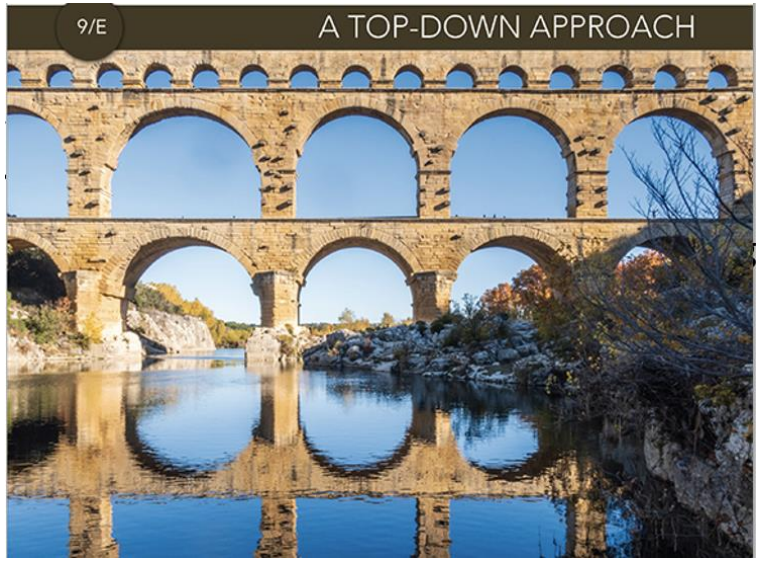
- All IETF documents are **open**, i.e., anyone can download and copy

Internet Draft (ID)	Request For Comment (RFC)
• Active working documents • Not finalized! Not stable! • Anyone can submit e.g., draft-yourname-... • Only some IDs are WG documents! e.g., draft-ietf-wgname-...	• Archival publications Never change once published • Not all RFCs are standards! **Standards track:** Proposed Standard Draft Standard Internet Standard Other types: Informational Experimental Historical Best-Current-Practice (BCP)



Chapter 1: roadmap

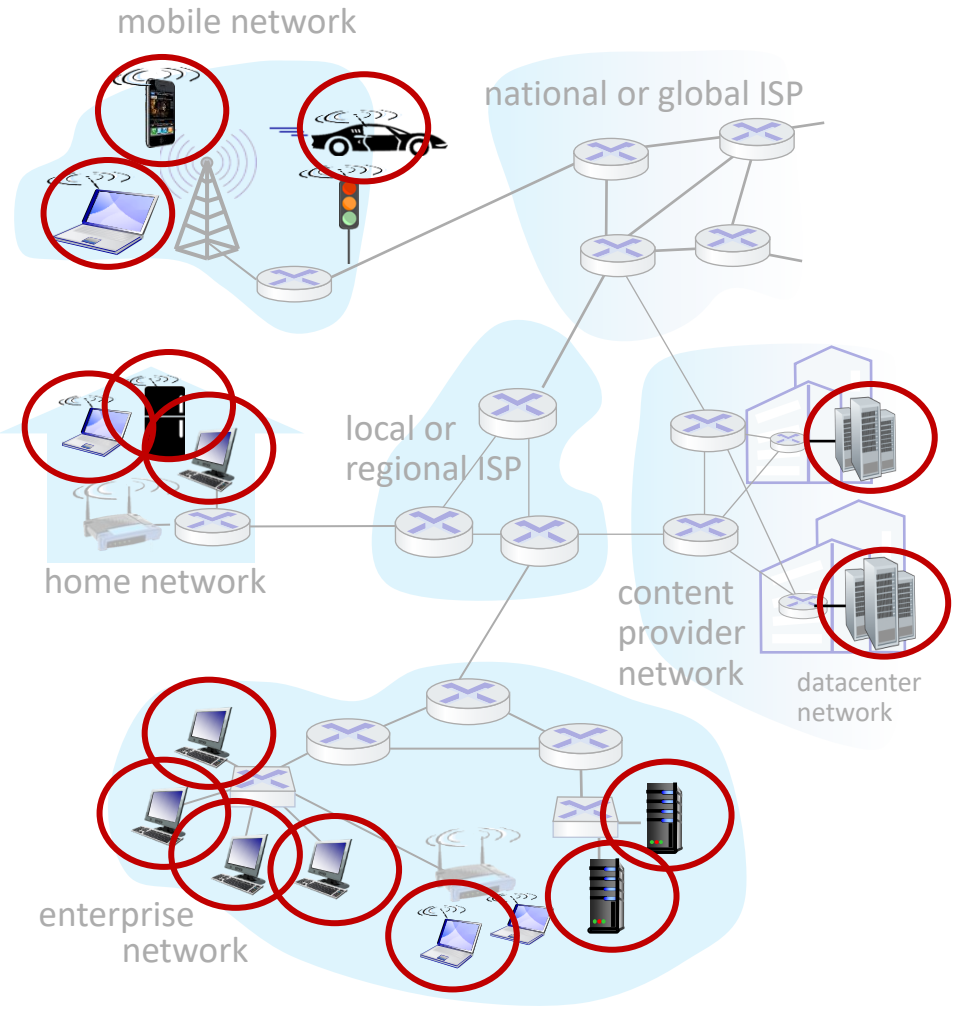
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A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers



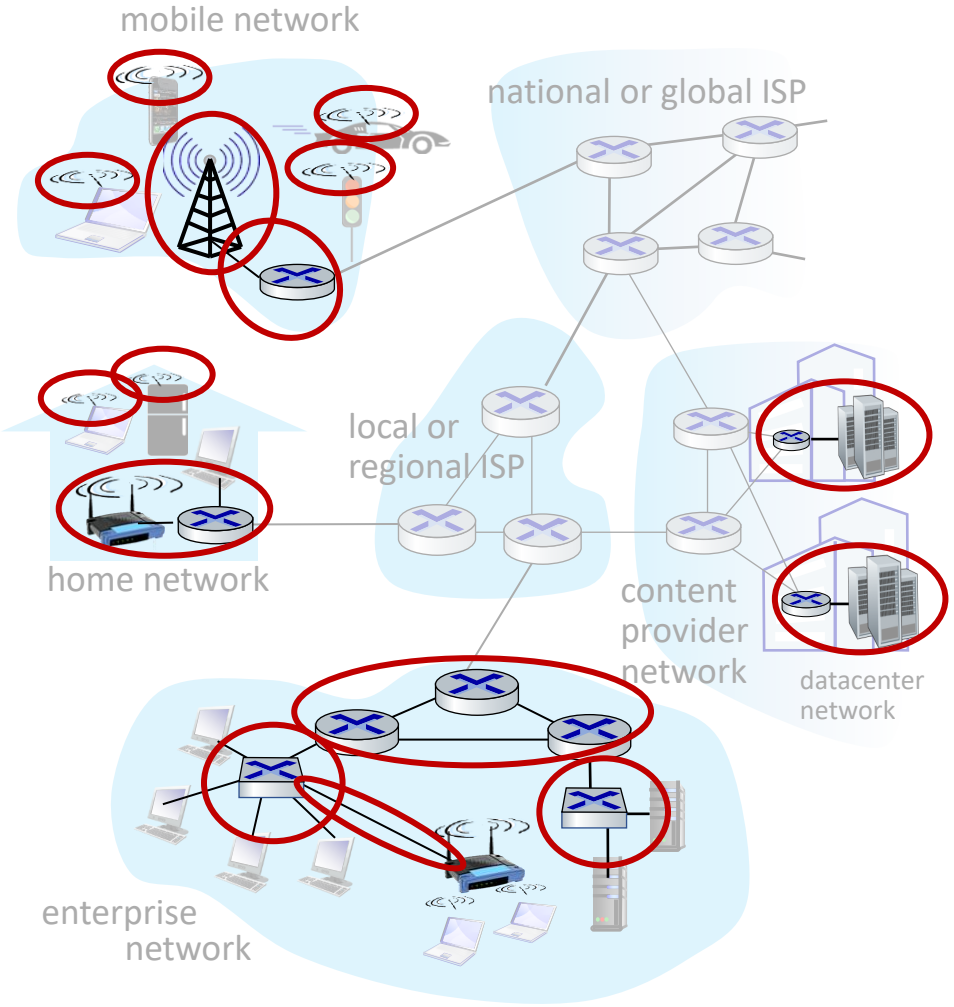
A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links



A closer look at Internet structure

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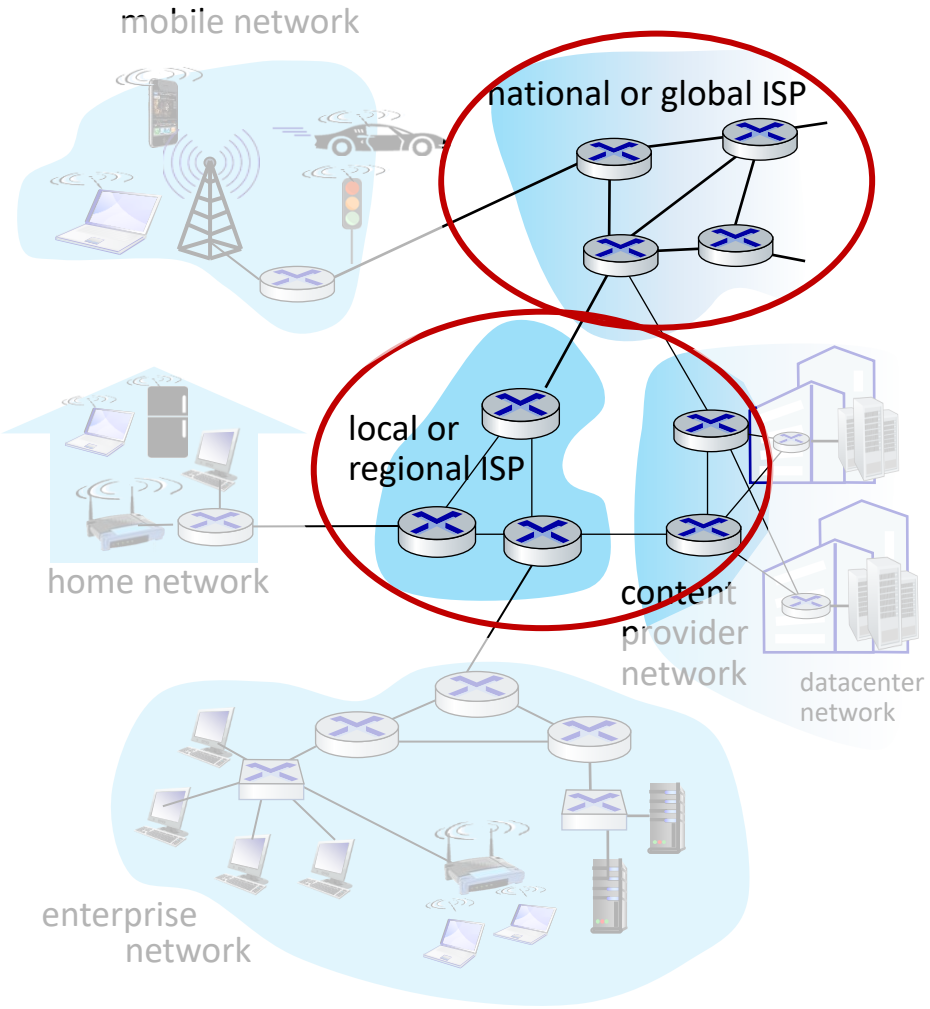
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Access networks, physical media:

- wired, wireless communication links

Network core:

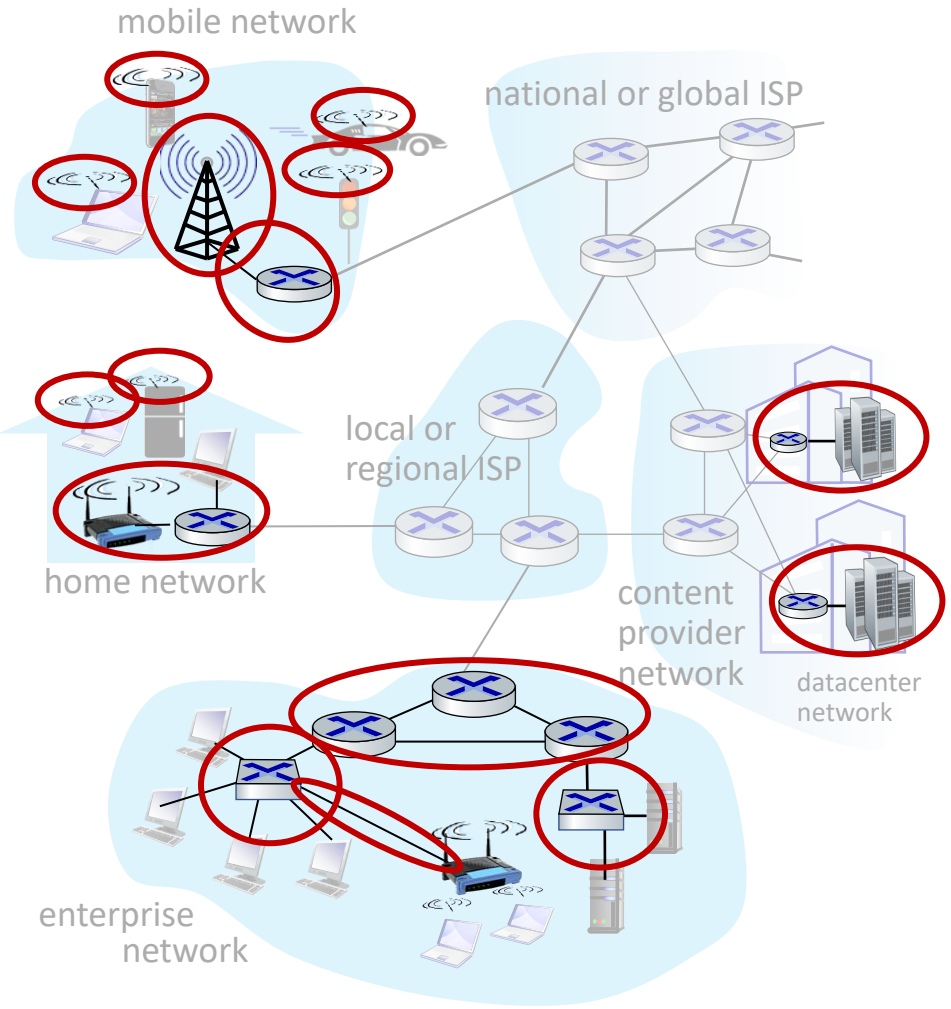
- interconnected routers
- network of networks



Access networks and physical media

Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)

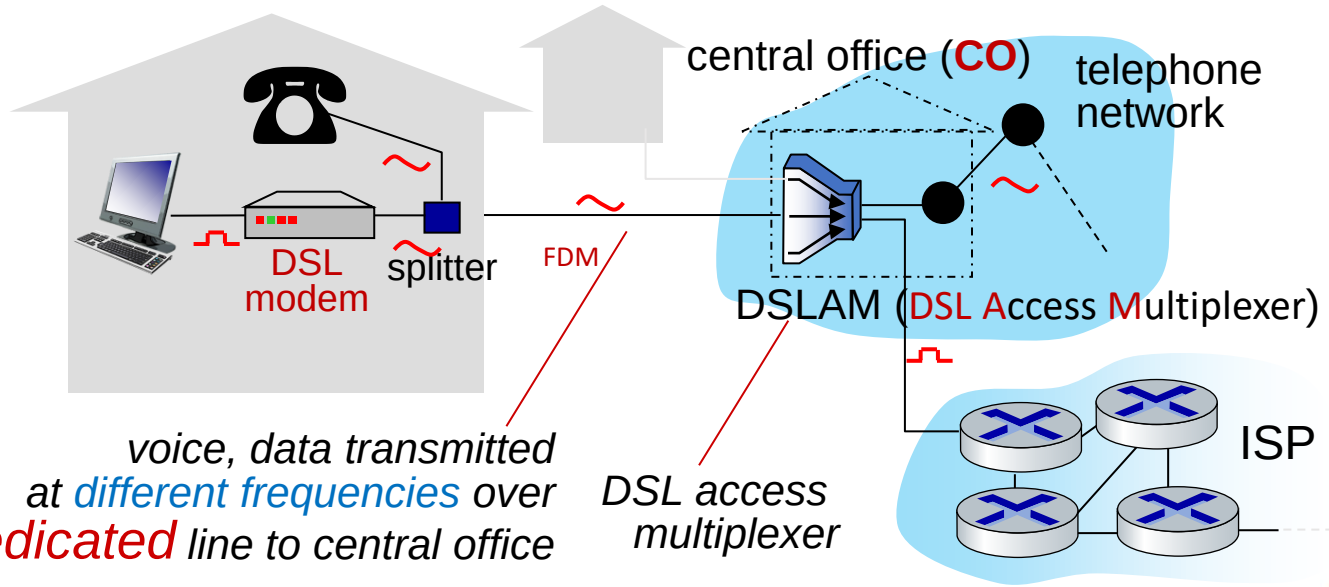


DEFINITION

access network

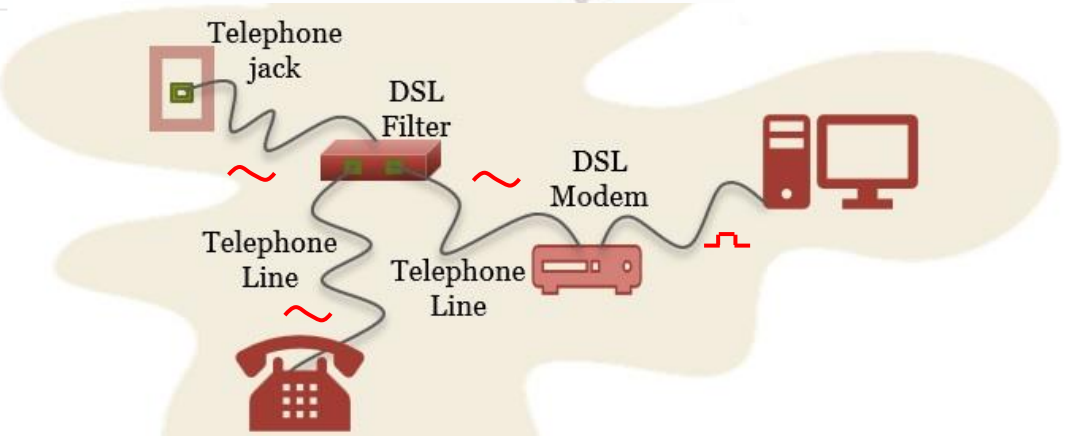
the network that physically connects an end system to the first router (a.k.a. the "edge router") on a path from the end system to any other distant end system.

Access Approach: digital subscriber line (DSL)



Plain Old Telephone Service (POTS)

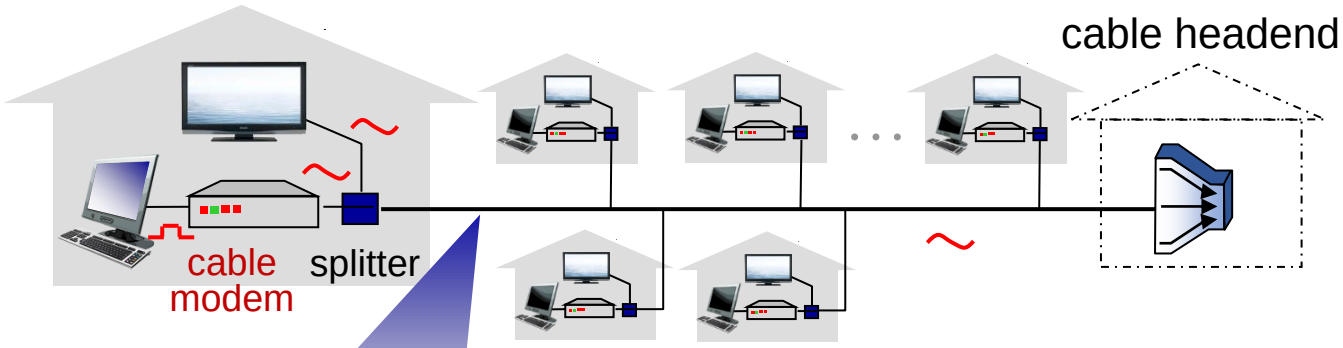


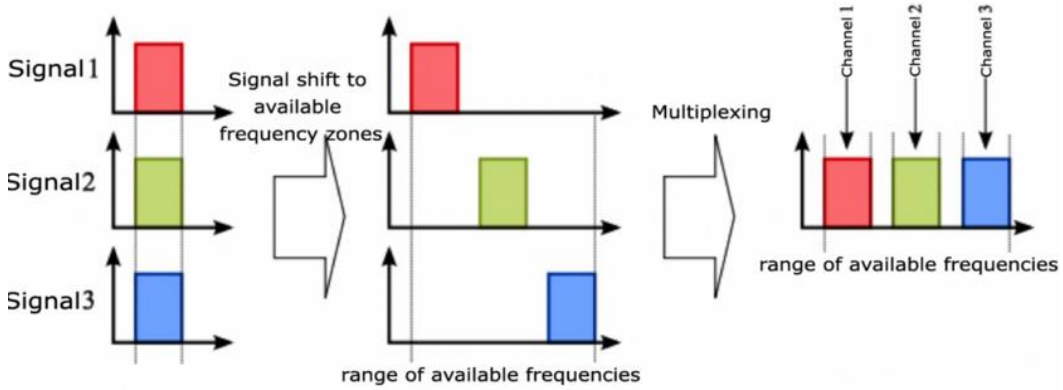


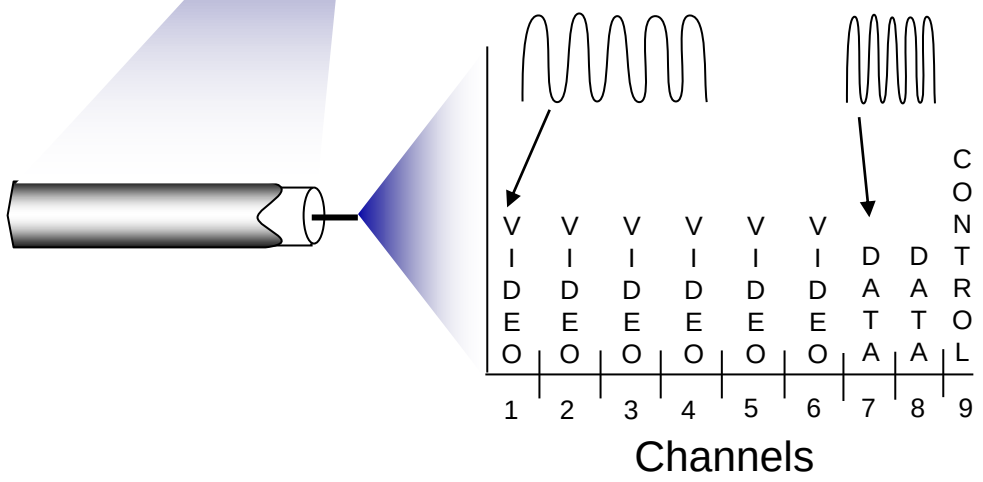
The method of transmission over the copper telephone wire is analog.

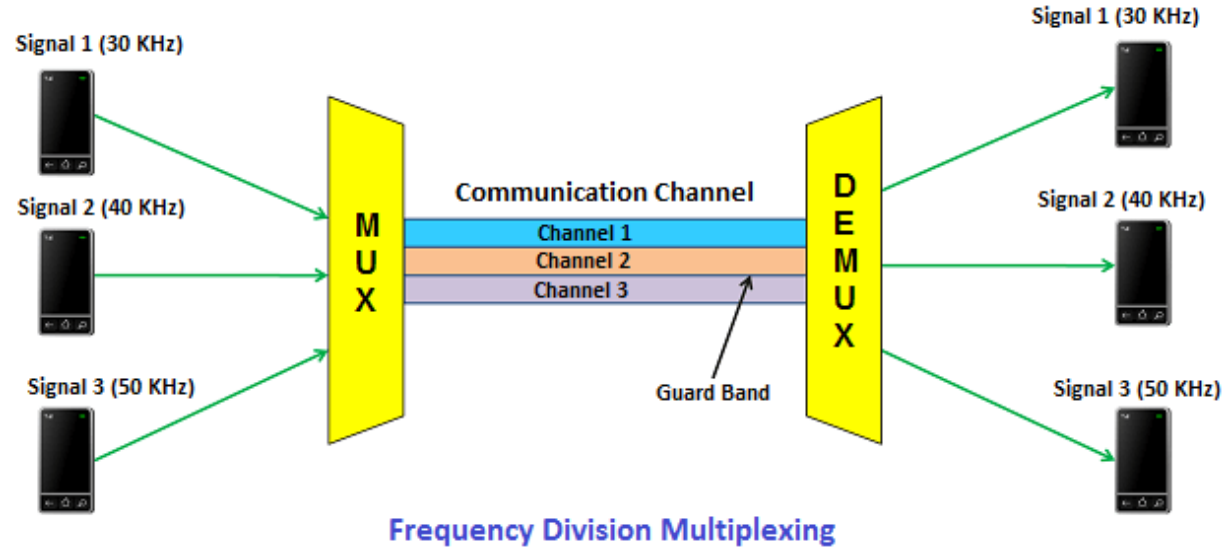
- use *existing* telephone line to central office DSLAM
  - data over DSL phone line goes to Internet
  - voice over DSL phone line goes to telephone net
- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate

Access via cable modem



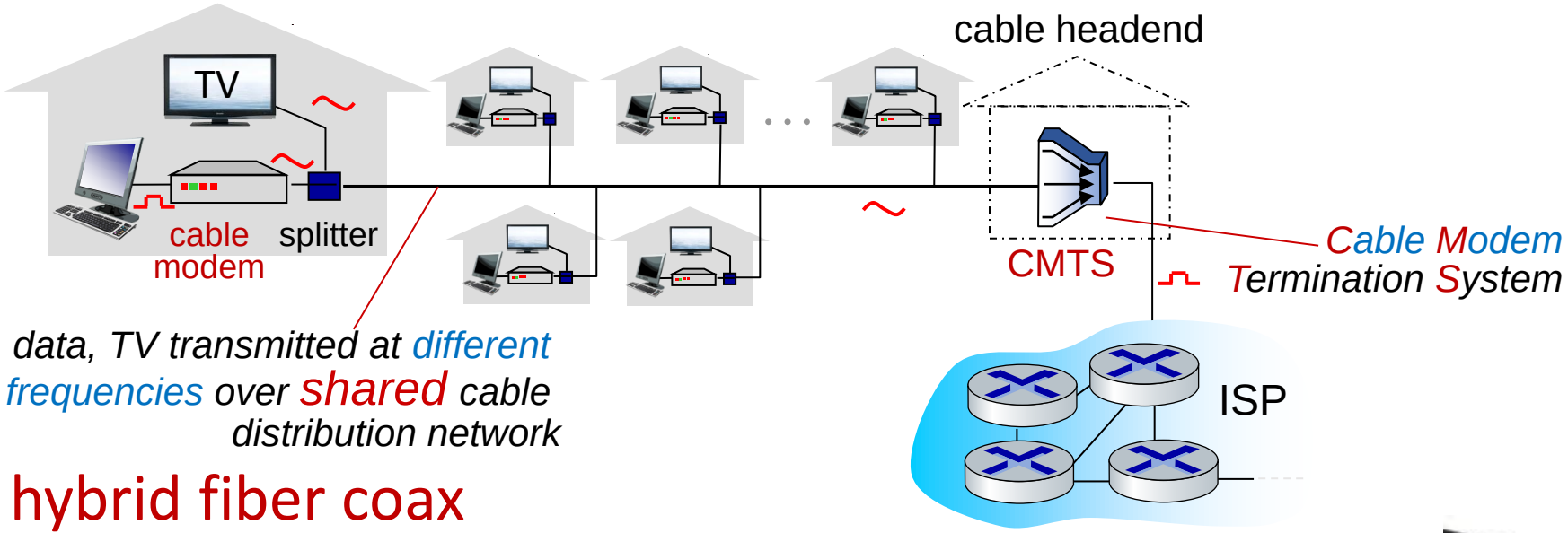






frequency division multiplexing (FDM):
different channels transmitted in different frequency bands

Access networks: cable-based access

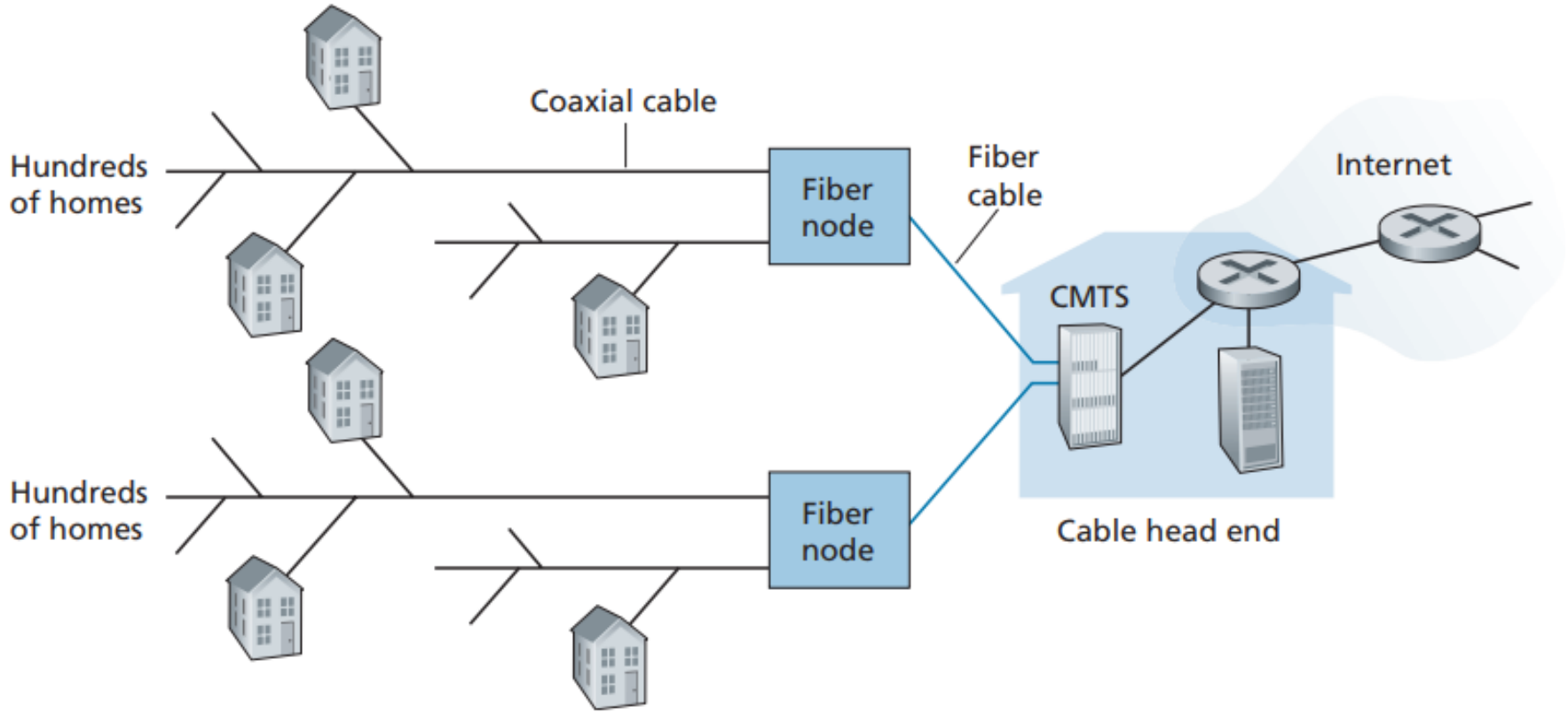


- **HFC**: hybrid fiber coax
  - Asymmetric downstream and upstream transmission rates
  - Downstream: up to 40 Mbps – 1.2 Gbps
  - Upstream : 30-100 Mbps



- network of cable, fiber attaches homes to ISP router
  - homes ***share*** *access network* to cable headend
  - Each neighborhood junction typically supports 500 to 5,000 homes.

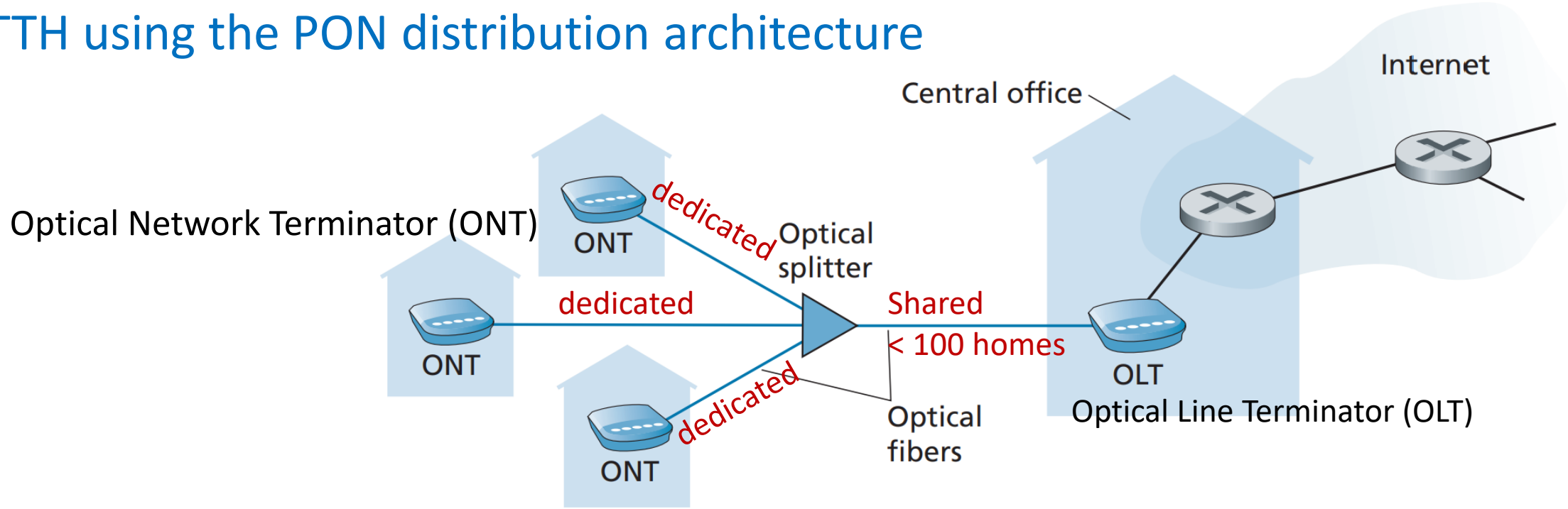
A hybrid fiber-coaxial (HFC) access network



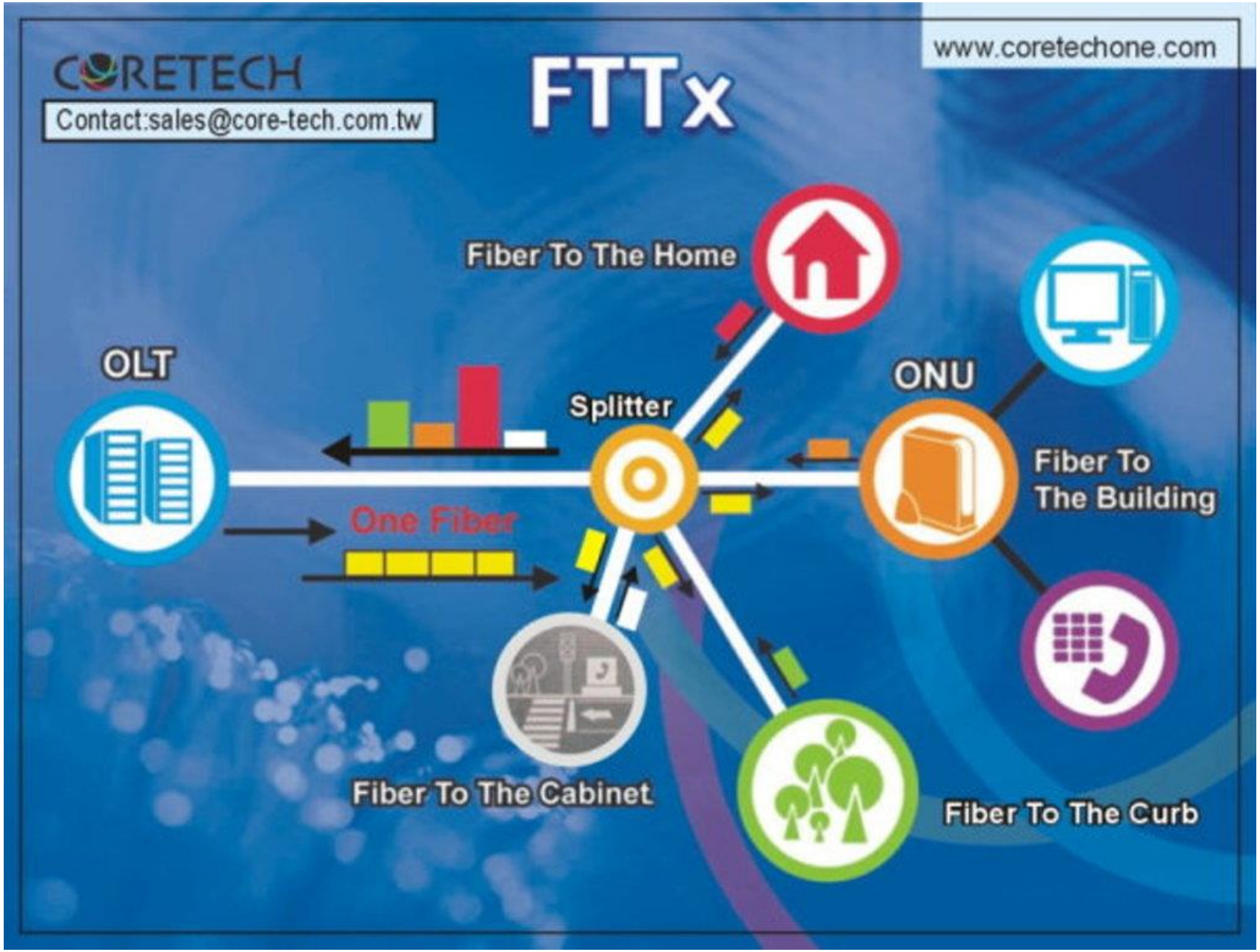
- Each neighborhood junction typically supports 500 to 5,000 homes.
- Because both fiber and coaxial cable are employed in this system, it is often referred to as hybrid fiber coax (HFC).

Access Approach: FTTH Internet

FTTH using the PON distribution architecture



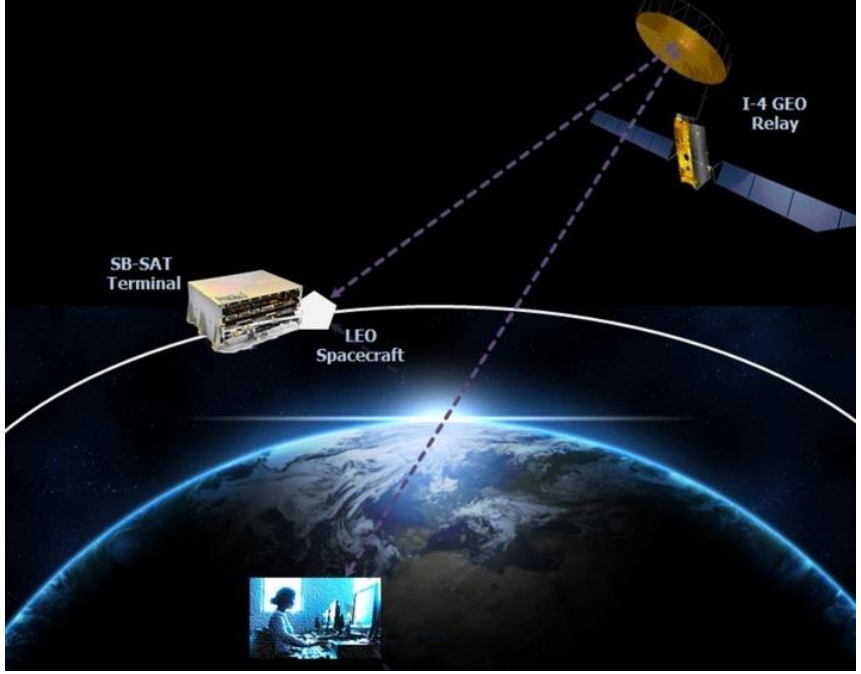
- Optical links from central office to the home
- Two competing optical technologies:
  - Passive Optical network (PON)
  - Active Optical Network (AON)
- Much higher Internet rates; fiber also carries television and phone services

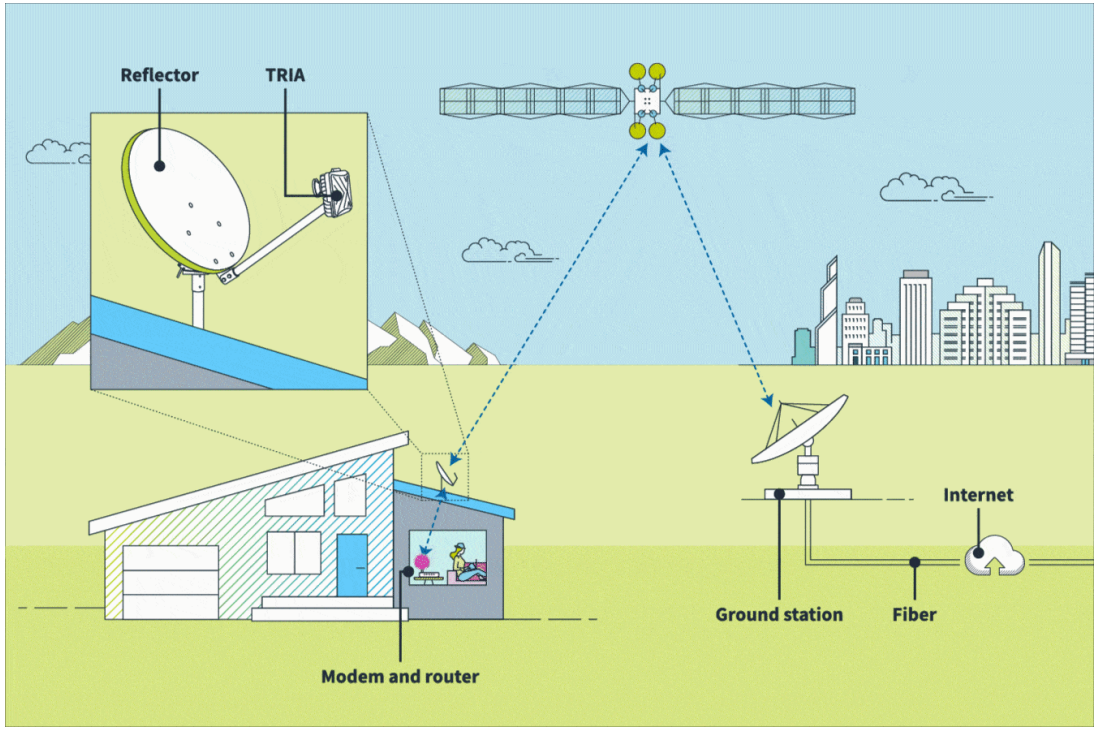


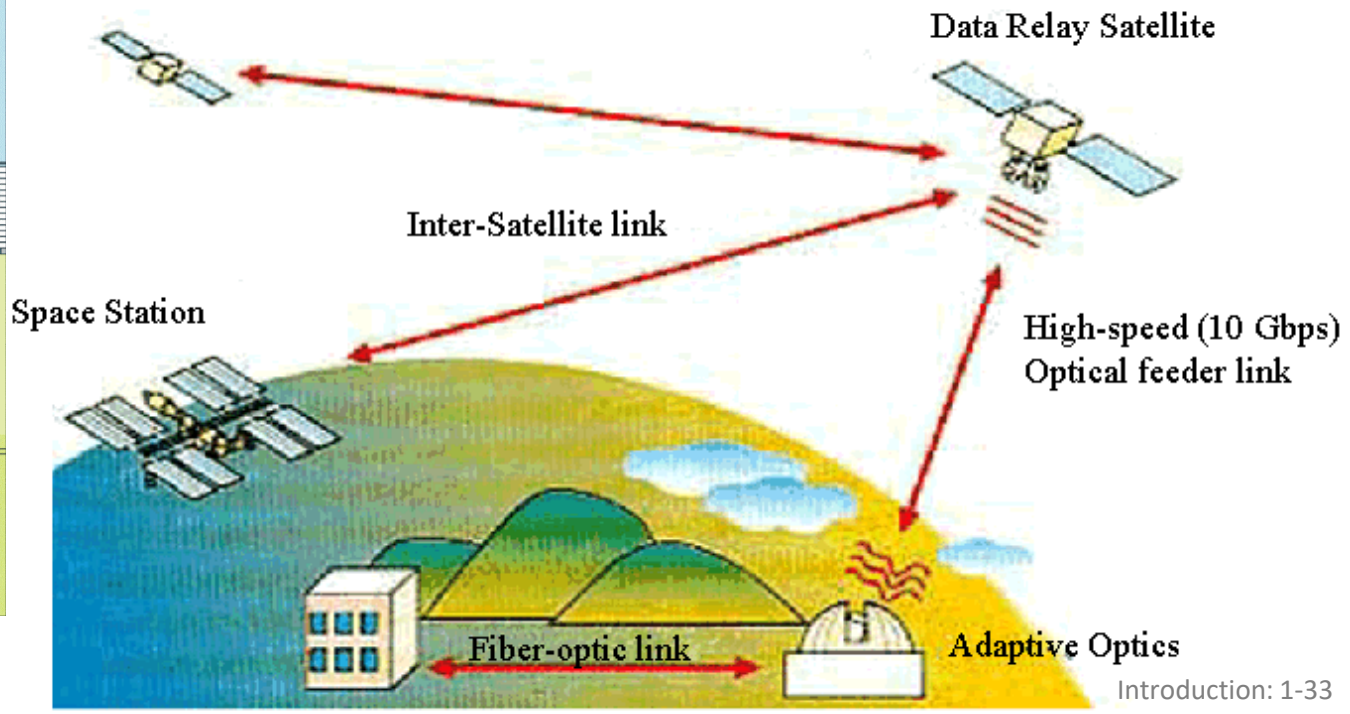
光纤宽带与ADSL的区别

Access Approach: Satellite

- A satellite link can be used to connect a residence to the Internet at speeds of more than 15Mbps (downstream ~ 100Mbps).



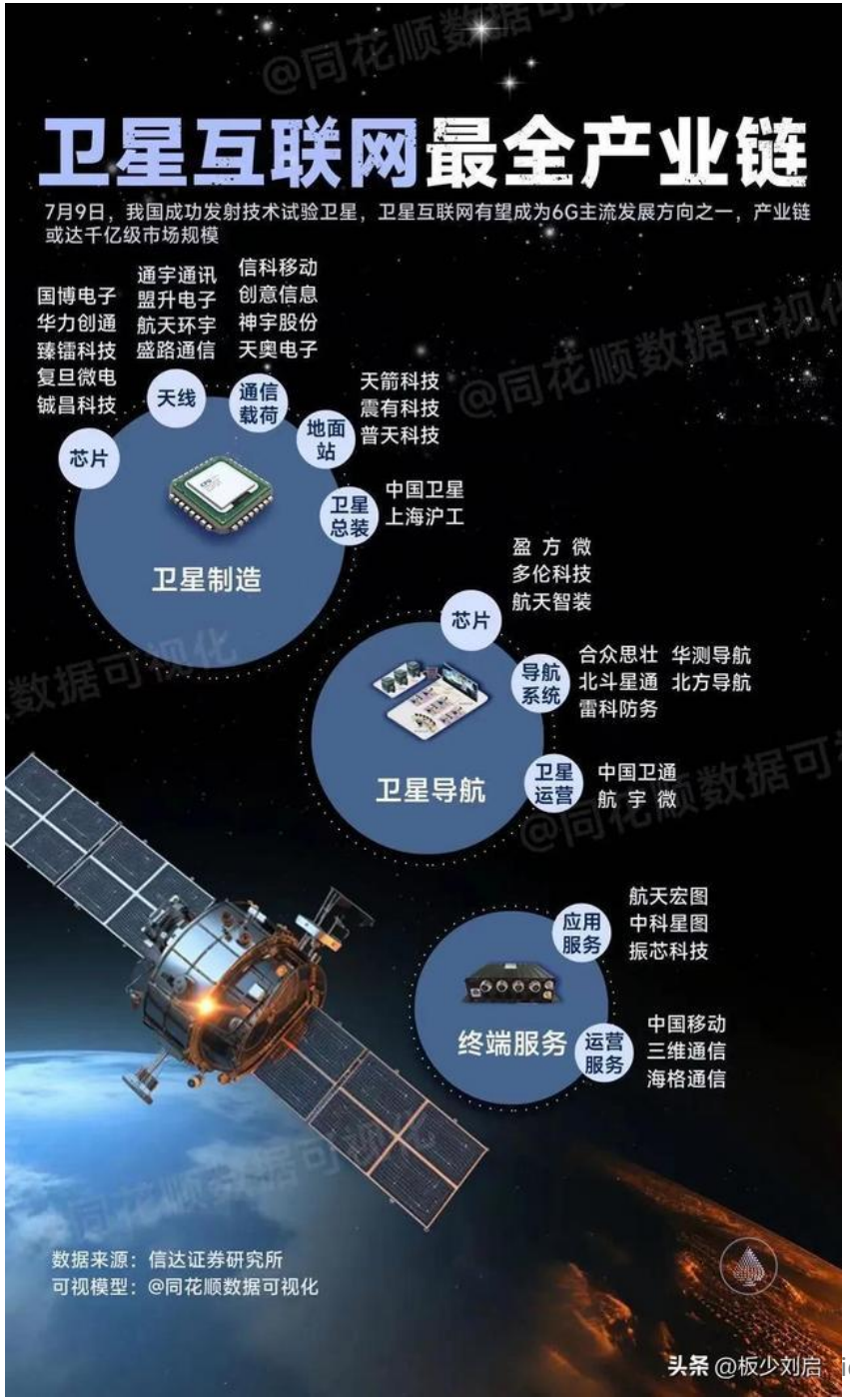




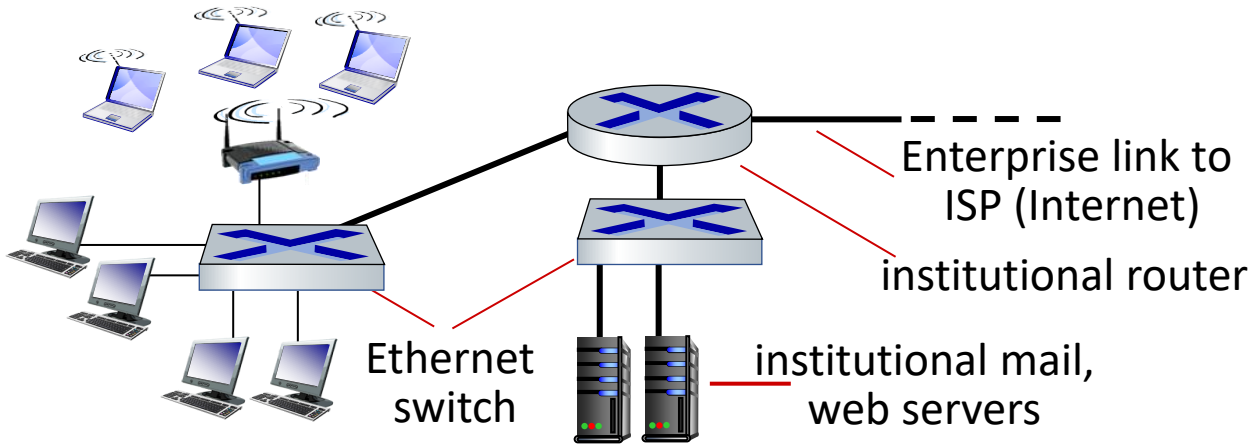
Satellite Internet providers

Satellite Internet is a good option for people who have no other Internet options (e.g., in a rural or remote location).

Company	Features
HughesNet	Best for affordable plans
T-Mobile	Best cheapest satellite alternative
Viasat	Best for faster speeds and more data
Starlink	Best potential (limited availability)
...	

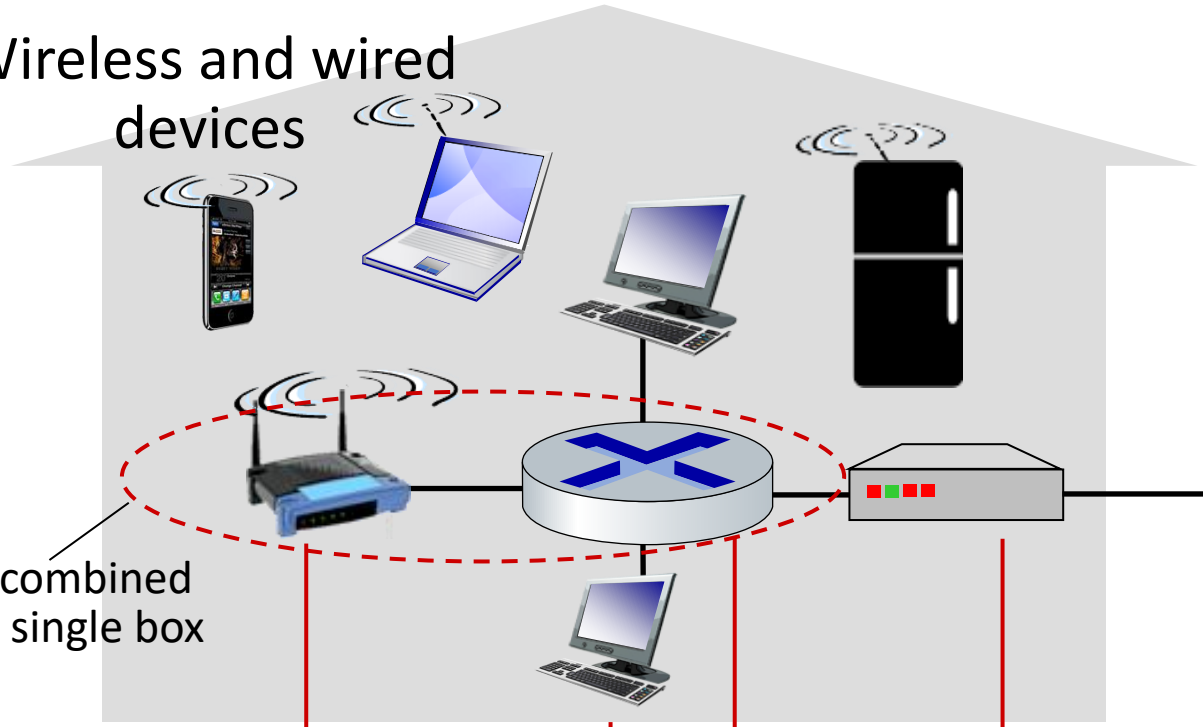


Access in the Enterprise (and the Home): Ethernet and WiFi



- companies, universities, and home
- mix of wired, wireless link technologies, connecting a mix of switches and routers
  - Ethernet: wired access at 100Mbps, 1Gbps, 2.5/5Gbps, 10Gbps …
  - WiFi: wireless access points at 11, 54, 450 Mbps …

Access networks: home networks



MODEM

Optical modem

to/from headend or central office

Cable, DSL or **optical modem**

router, firewall, **NAT**

wired Ethernet (1 Gbps)

WiFi wireless access point (54, 450 Mbps)

核心作用①：信号转换

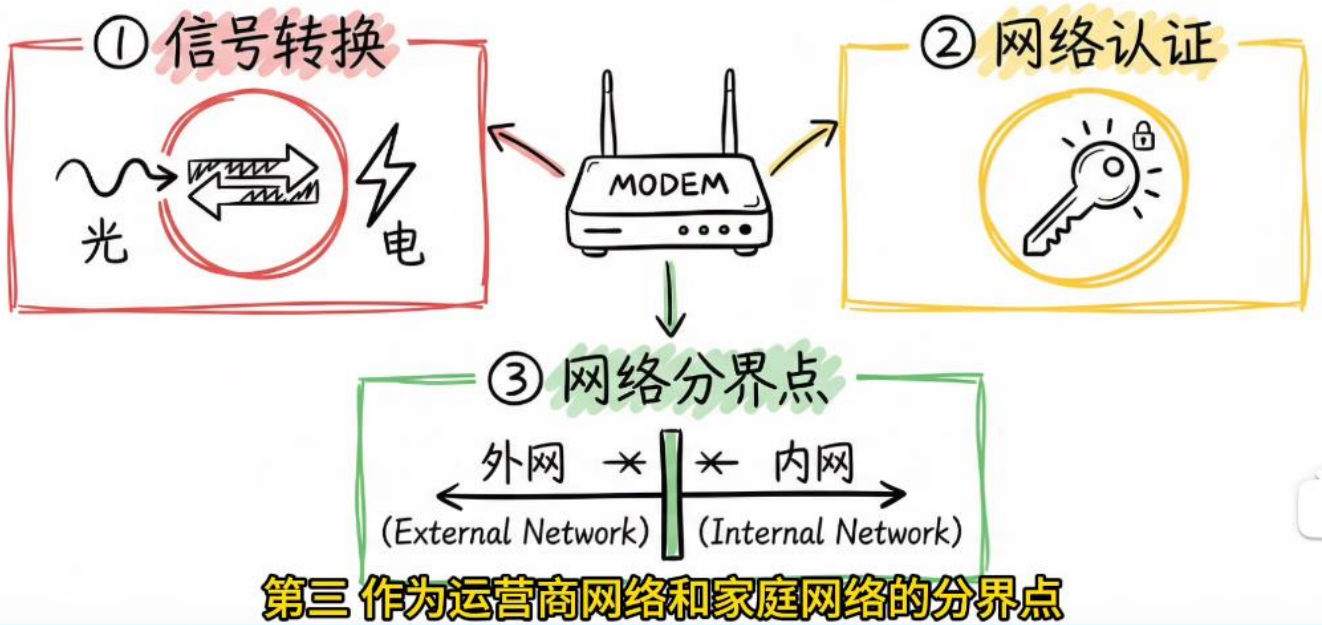


光信号≠电信号



光信号和电信号不一样 不能直接互通

光猫三大作用



第三 作为运营商网络和家庭网络的分界点

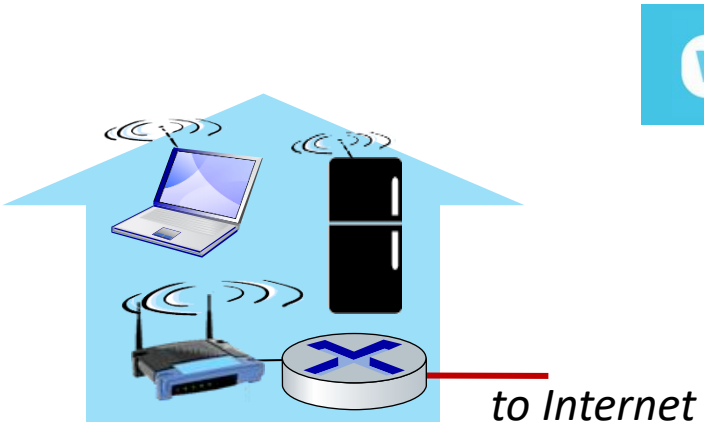
Wireless access networks

Shared *wireless* access network connects end system to router

- via base station aka "access point" (**AP**)

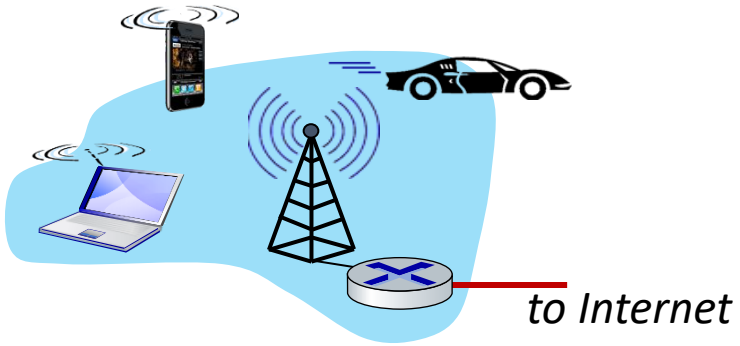
Wireless local area networks (WLANs)

- typically within or around building (~100 ft = 30.48 m)
- 802.11b/g/n/ac/ax/be (WiFi): 11, 54, 450 Mbps, 6.9, 9.6, 46 Gbps transmission rate

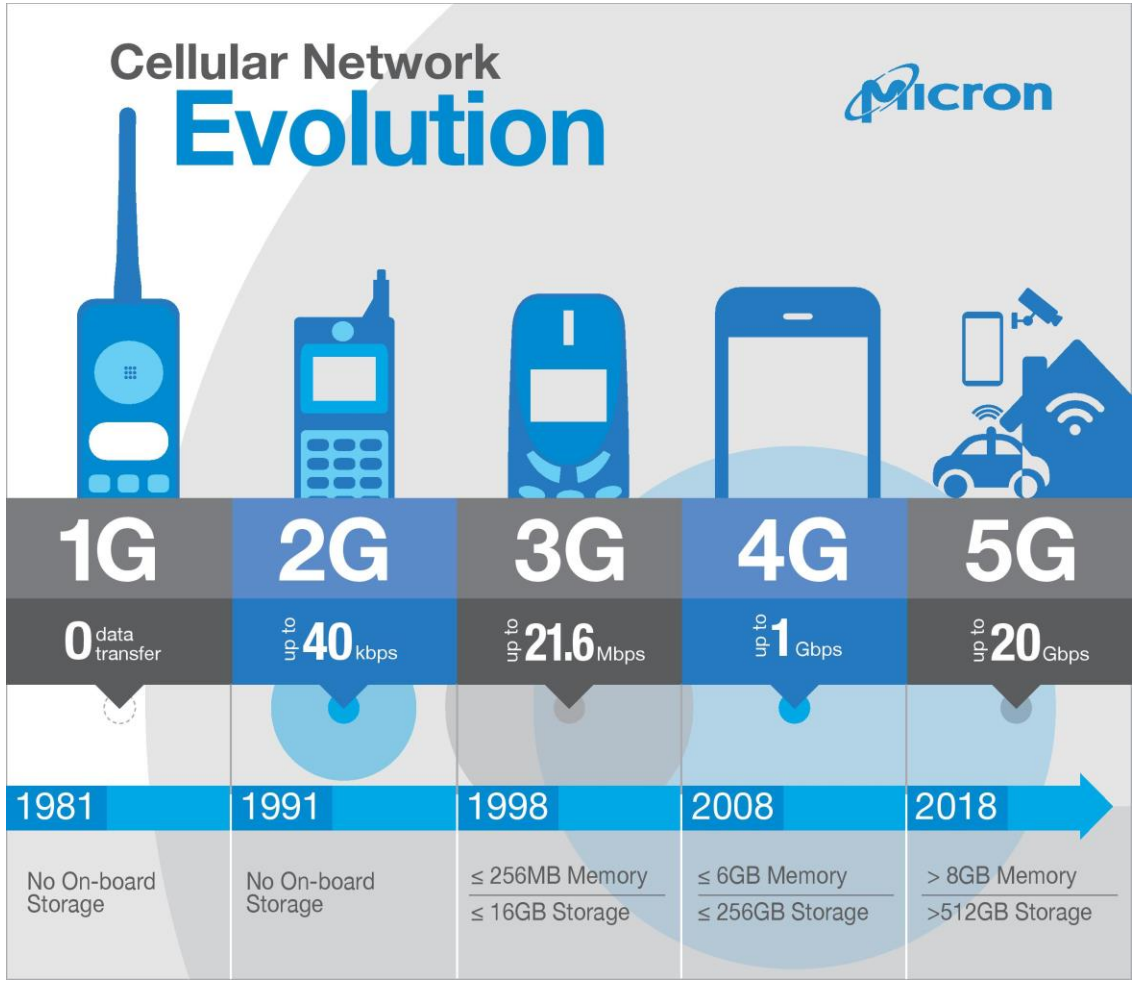


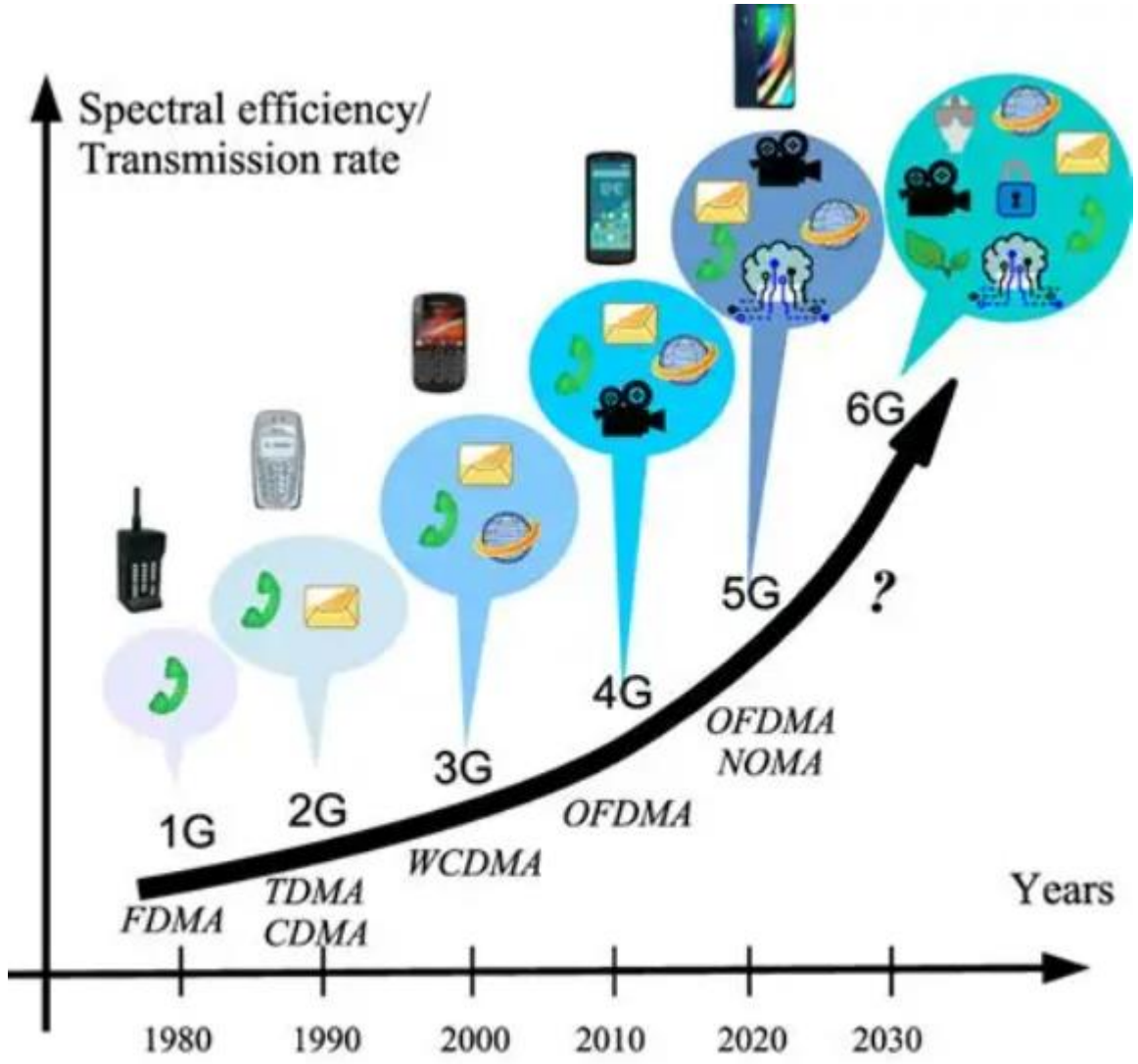
Wide-area cellular access networks

- provided by mobile, cellular network operator (10's km)
- 10's ~ 100's Mbps
- 4G/5G cellular networks

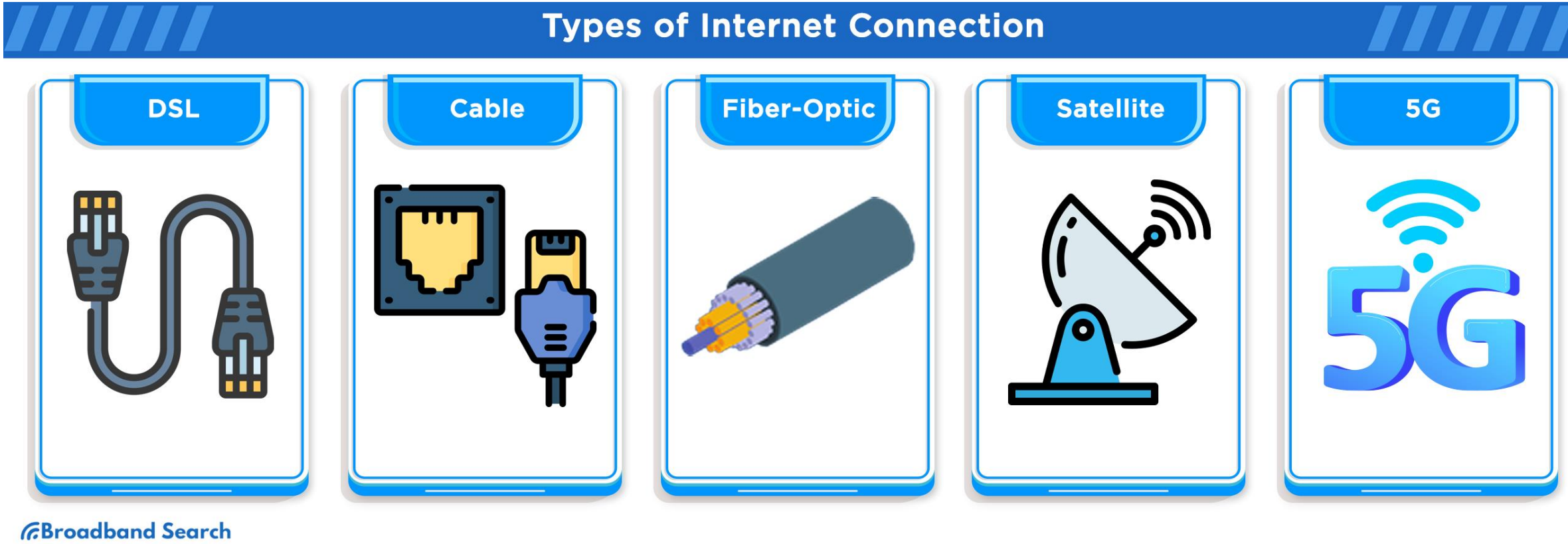


Mobile network evolution



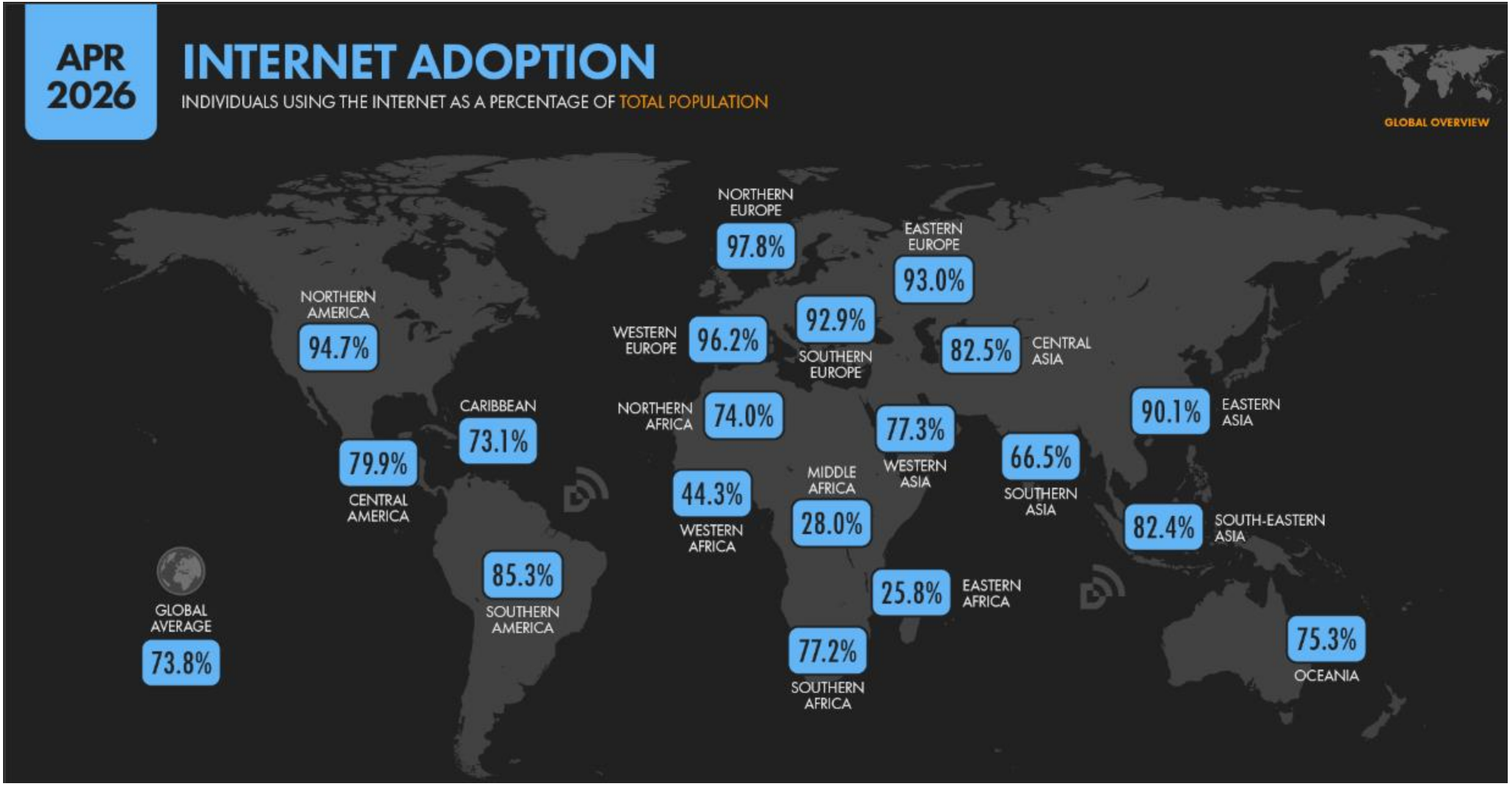


Types of Internet Connection



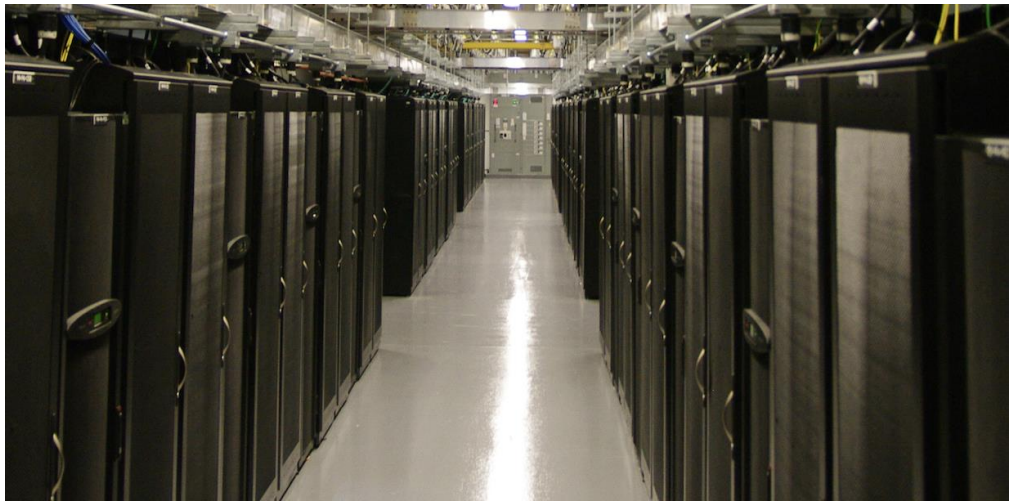
2026 Internet Statistics and Digital Trends

https://www.broadbandsearch.net/blog/internet-statistics

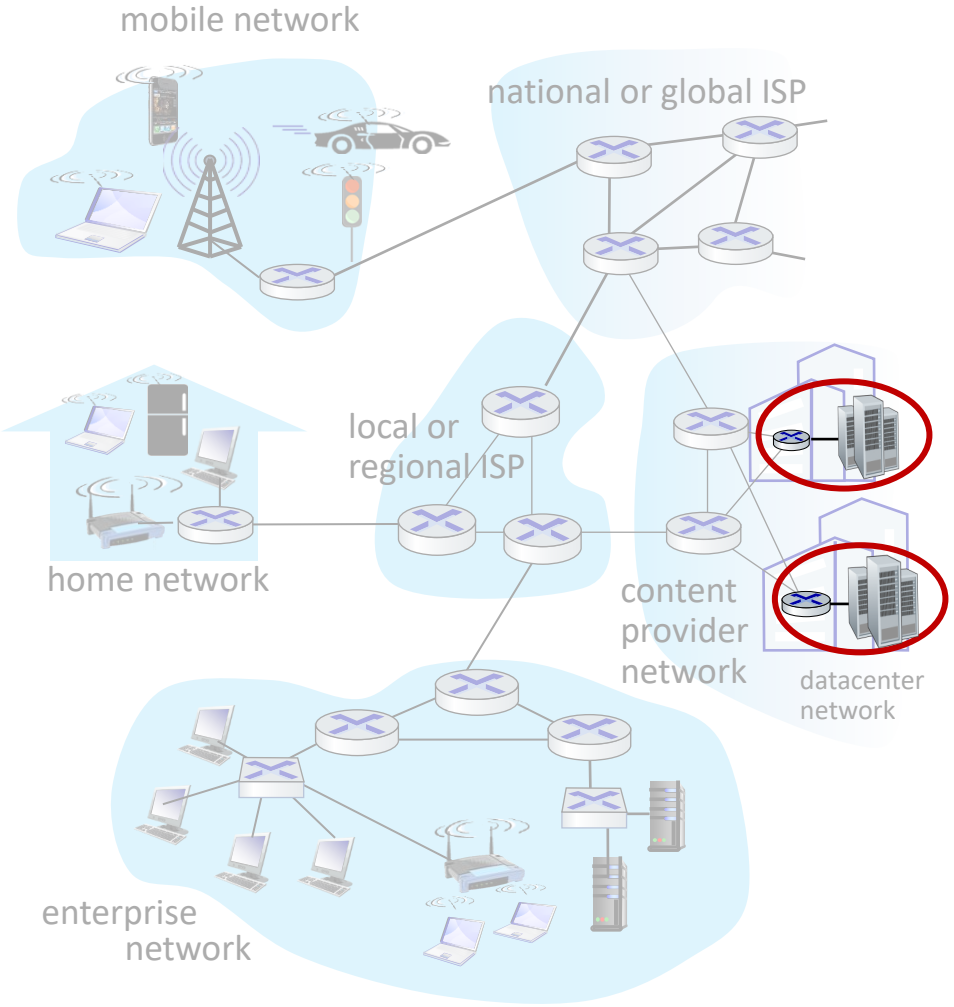

APR 2026
INTERNET ADOPTION
INDIVIDUALS USING THE INTERNET AS A PERCENTAGE OF TOTAL POPULATION
GLOBAL OVERVIEW
NORTHERN EUROPE
97.8%
EASTERN EUROPE
93.0%
NORTHERN AMERICA
94.7%
WESTERN EUROPE
96.2%
92.9%
SOUTHERN EUROPE
82.5%
CENTRAL ASIA
CARIBBEAN
73.1%
NORTHERN AFRICA
74.0%
77.3%
WESTERN ASIA
90.1%
EASTERN ASIA
79.9%
CENTRAL AMERICA
66.5%
SOUTHERN ASIA
44.3%
WESTERN AFRICA
MIDDLE AFRICA
28.0%
82.4%
SOUTH-EASTERN ASIA
85.3%
SOUTHERN AMERICA
25.8%
EASTERN AFRICA
GLOBAL AVERAGE
73.8%
77.2%
SOUTHERN AFRICA
75.3%
OCEANIA

Access networks: data center networks

- high-bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet



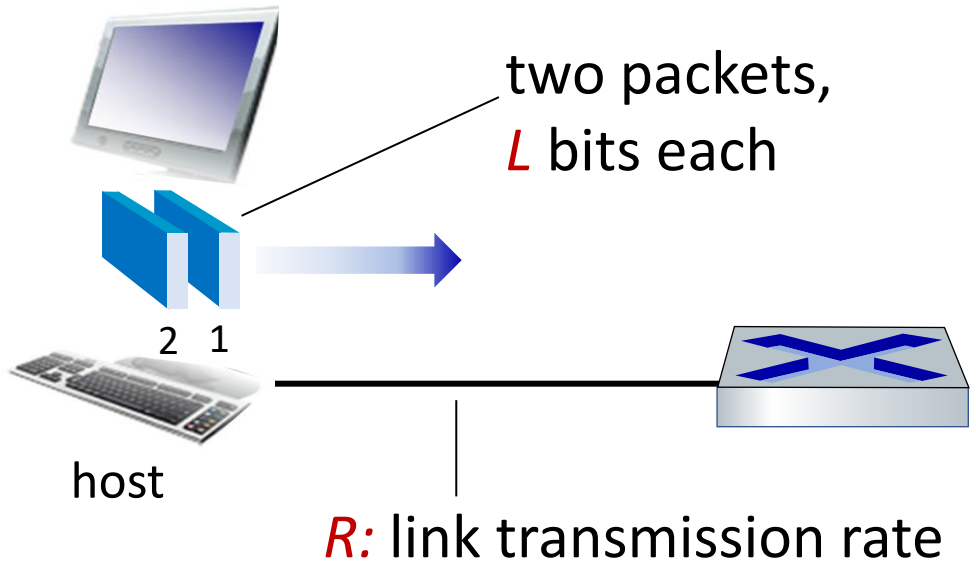
Courtesy: Massachusetts Green High Performance Computing Center (mghpcc.org)



Host: sends *packets* of data

host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length *L* bits
- transmits packet into access network at *transmission rate R*
  - link transmission rate, aka link *capacity, aka link bandwidth*

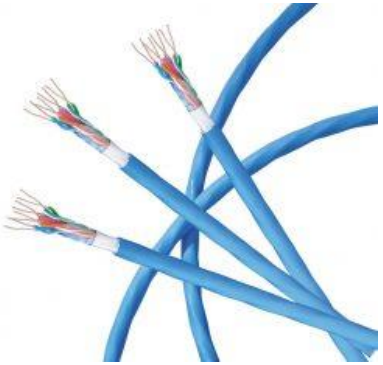


$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Links: physical media (1)

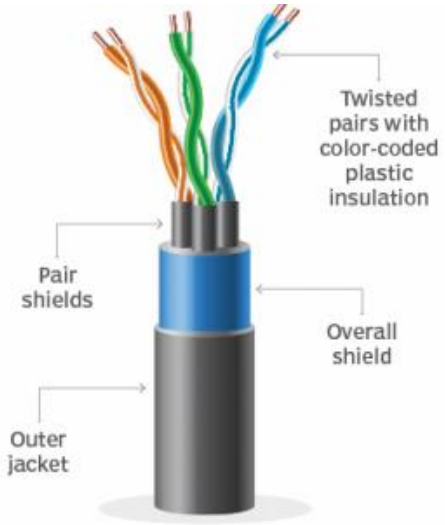


- bit: propagates between transmitter/receiver pairs
- physical link: what lies between transmitter & receiver
- guided media:
  - signals propagate in solid media: copper, fiber, coax
- unguided media:
  - signals propagate freely, e.g., radio

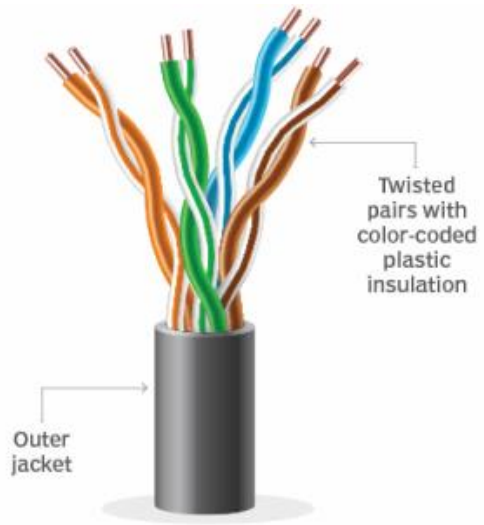


Twisted pair (TP)

- two insulated copper wires
  - Category 5: 100 Mbps, 1 Gbps Ethernet
  - Category 6: 10Gbps Ethernet

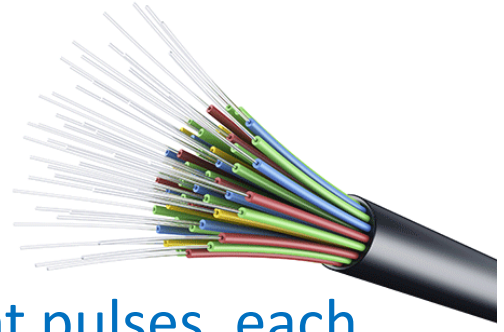


Shielded twisted pair



Unshielded twisted pair

Links: physical media (2)



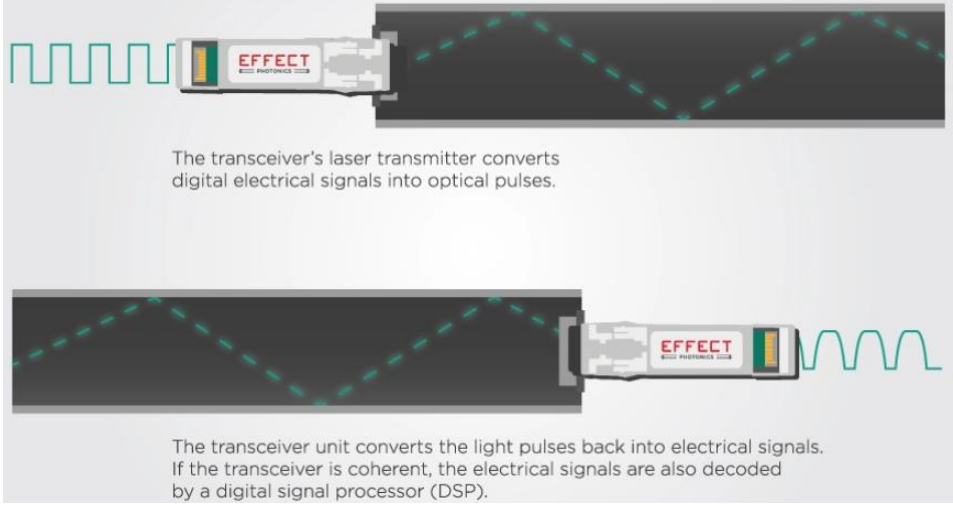
Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
  - multiple frequency channels on cable
  - 100's Mbps per channel

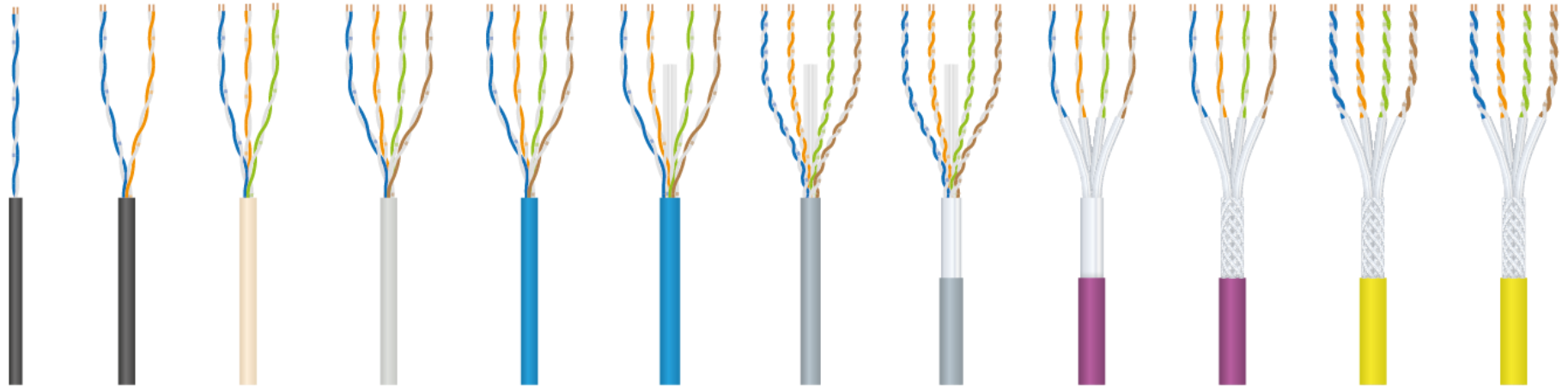


Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
  - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
  - repeaters spaced far apart
  - immune to electromagnetic noise



History of Ethernet LAN Cables' Categories



CAT 1	CAT 2	CAT 3	CAT 4	CAT 5	CAT 5e	CAT 6	CAT 6A	CAT 7	CAT 7A	CAT 8.1	CAT 8.2
1 Mbps	4 Mbps	10 Mbps	16 Mbps	100 Mbps	1 Gbps	1 Gbps	10 Gbps	10 Gbps	10 Gbps	25 Gbps	40 Gbps
400 KHz	4 MHz	16 MHz	20 MHz	100 MHz	100 MHz	250 MHz	500 MHz	600 MHz	1000 MHz	2000 MHz	2000 MHz
1983	1987	1991	1993	1995	2001	2002	2008	2010	2013	2016	2018

bandwidth

Common Name	Speed	Alternative Name	Name of IEEE Standard	Cable Type, Maximum Length
Ethernet	10Mbps	10BASE-T	802.3	Copper, 100 m
Fast Ethernet	100Mbps	100BASE-TX	802.3u	Copper, 100 m
Gigabit Ethernet	1000Mbps	1000BASE-LX	802.3z	Fiber, 550 m
Gigabit Ethernet	1000Mbps	1000BASE-T	802.3ab	Copper, 100 m
10GigE (Gigabit Ethernet)	10Gbps	10GBASE-T	802.3an	Copper, 100 m

Standard	Speed	Distance	MHz	Category
802.3	10Mbps	100 meters	3 MHz	3
802.3u	100Mbps	100 meters	100 MHz	5
802.3ab	1Gbps	100 meters	100 MHz	5e*
802.3an	10Gbps	100 meters	500 MHz	6a
802.3bq	20 or 40Gbps	30 meters	2000 MHz	8**

*Category 6 Ethernet is the same as Cat 5e, except it operates at 250 MHz.
**IEEE did not release a Category 7 standard, although the International Standards Organization (ISO) did.

Links: physical media (3)

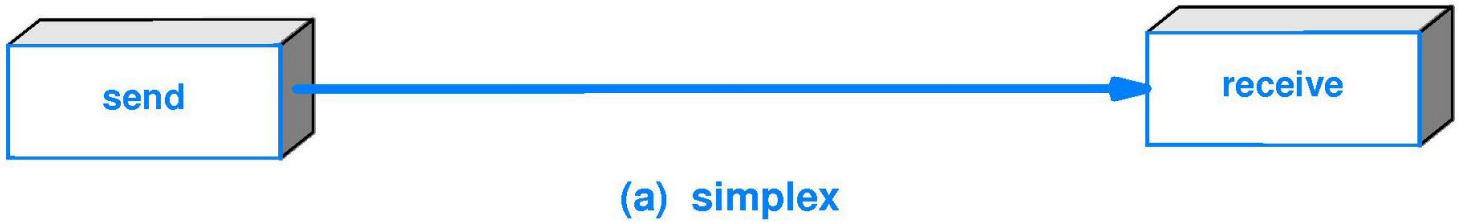
Wireless radio

- signal carried in various "bands" in electromagnetic spectrum
- no physical "wire"
- broadcast, "half-duplex" (sender to receiver)
- propagation environment effects:
  - reflection
  - obstruction by objects
  - Interference/noise

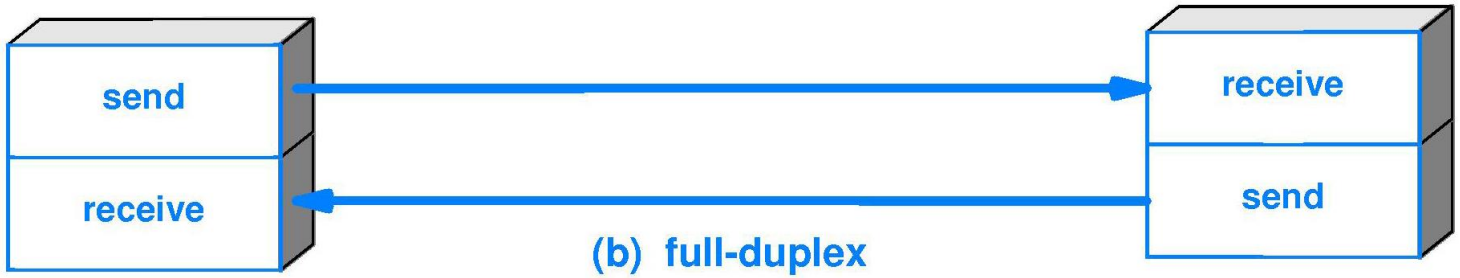
Radio link types:

- Wireless LAN (WiFi, 802.11)
  - 10-100's Mbps; 10's of meters
- wide-area (e.g., 4G/5G cellular)
  - 100's Mbps (4G/5G) over ~10 Km
- Bluetooth: cable replacement
  - short distances, limited rates
  - IEEE 802.15.1
- terrestrial microwave (US; Europe)
  - point-to-point;
  - FWA (Fixed Wireless Access); 100+Mbps
  - WISP (Wireless ISP)
- satellite
  - up to < 100 Mbps (Starlink) downlink
  - 270 msec end-end delay (geostationary)

Simplex, Half-duplex, Full-duplex



(a) simplex



(b) full-duplex



(c) half-duplex

Dedicated channels for send and receive.
No collision

Labor Costs vs. Material Costs

- The actual cost of the physical link (copper wire, fiber optic cable, and so on) is often relatively minor compared with other networking costs.



Labor costs account for up to **40% of a project's budget.**

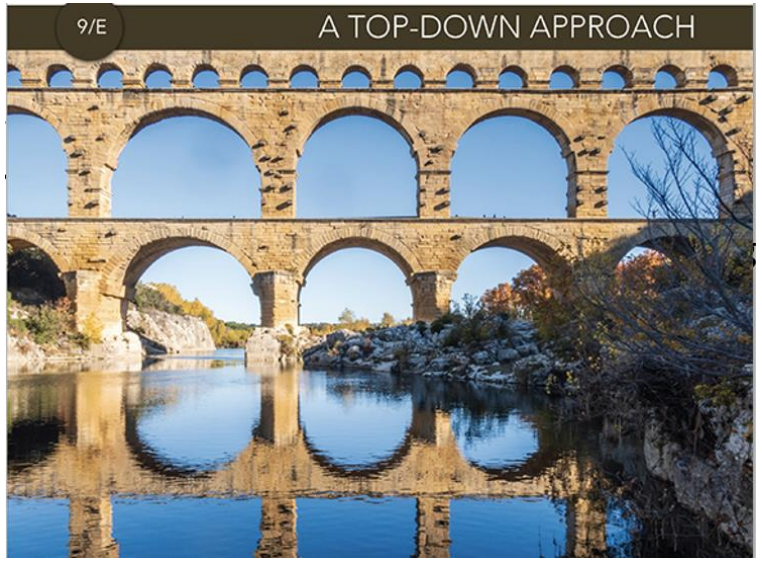
the labor cost (installation of the physical link) >> the cost of the material

can be **orders of magnitude higher than the cost of the material**

- For this reason, many builders install twisted pair, optical fiber, and coaxial cable in every room in a building.
- Even if only one medium is initially used, there is a good chance that another medium could be used in the near future, and so money is saved by not having to lay additional wires in the future.

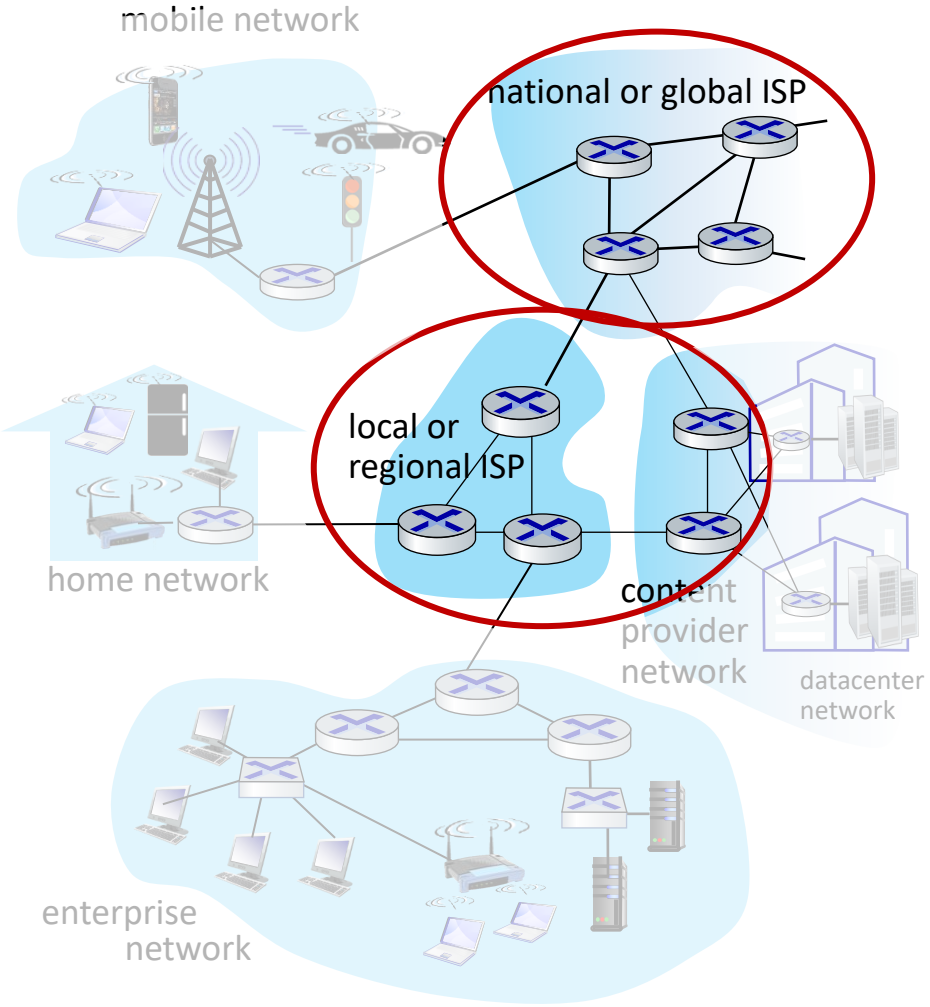
Chapter 1: roadmap

- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- **Network core:** packet/circuit switching, Internet structure
- Performance: loss, delay, throughput
- Protocol layers, service models
- Security
- History



The network core

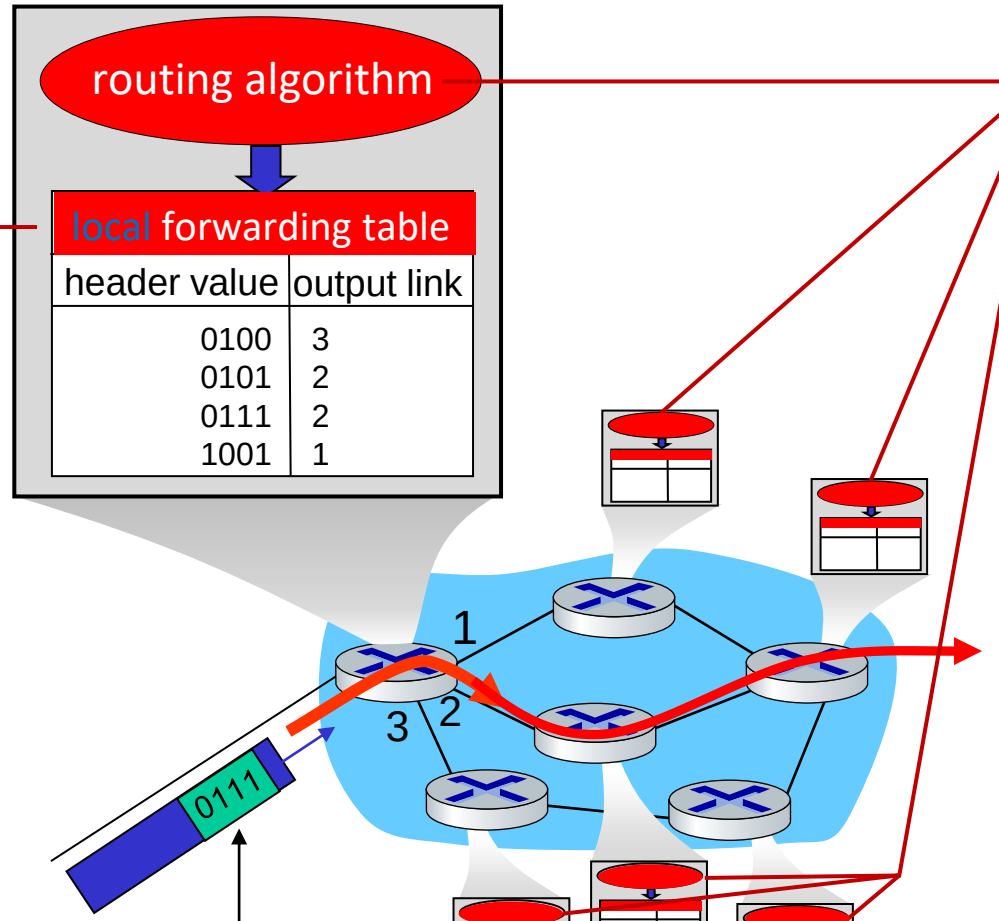
- mesh of interconnected routers
- packet-switching: hosts break application-layer messages into *packets*
  - network forwards packets from one router to the next, across links on path from source to destination



Two key network-core functions

Forwarding:

- aka "switching"
- *local* action: move arriving packets from router's input link to appropriate router output link



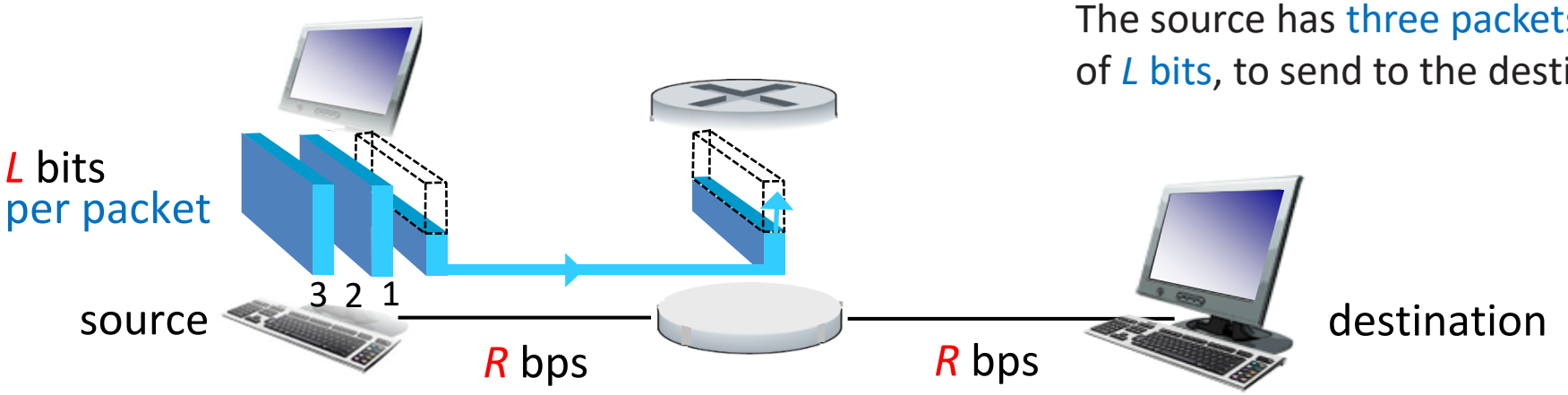
Routing:

- *global* action: determine source-destination paths taken by packets
- routing algorithms


routing
45 h
3,053 miles
48 h
3,309 miles
United States
Victoria
Seattle
Portland
San Francisco
San Jose
Sacramento
Las Vegas
Los Angeles
San Diego
Phoenix
Tucson
El Paso
Dallas
Austin
Houston
San Antonio
New Orleans
Kansas City
Minneapolis
Chicago
Detroit
Toronto
Ottawa
Montreal
Quebec City
Northampton
New York
Philadelphia
Indianapolis
Nashville
Charlotte
Atlanta
Jacksonville
Orlando
Tampa
WASHINGTON
OREGON
CALIFORNIA
NEVADA
IDAHO
MONTANA
WYOMING
UTAH
COLORADO
ARIZONA
NEW MEXICO
NORTH DAKOTA
SOUTH DAKOTA
MINNESOTA
IOWA
MISSOURI
WISCONSIN
MICHIGAN
ILLINOIS
INDIANA
OHIO
KENTUCKY
TENNESSEE
MISSISSIPPI
ALABAMA
GEORGIA
SOUTH CAROLINA
NORTH CAROLINA
VIRGINIA
WEST VIRGINIA
PENNSYLVANIA
MARYLAND
DE
NJ
CT
RI
VERMONT
NEW HAMPSHIRE
MAINE
OKLAHOMA
ARKANSAS
LOUISIANA
TEXAS
FLORIDA
BAJA CALIFORNIA
SONORA
CHIHUAHUA
COAHUILA


forwarding
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45 h
3,053 miles
48 h
3,309 miles
United States
Victoria
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WASHINGTON
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Minneapolis
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WISCONSIN
MICHIGAN
Toronto
Quebec City
Montreal
Ottawa
MAINE
VERMONT
NEW HAMPSHIRE
Northampton
Detroit
Chicago
WYOMING
IOWA
ILLINOIS
INDIANA
OHIO
Kansas City
Indianapolis
COLORADO
KANSAS
MISSOURI
KENTUCKY
Nashville
TENNESSEE
San Francisco
San Jose
CALIFORNIA
Las Vegas
OKLAHOMA
ARKANSAS
Atlanta
Los Angeles
ARIZONA
NEW MEXICO
MISSISSIPPI
ALABAMA
San Diego
Phoenix
Dallas
Tucson
El Paso
TEXAS
BAJA CALIFORNIA
SONORA
Austin
Houston
LOUISIANA
New Orleans
CHIHUAHUA
San Antonio
COAHUILA
Tampa
FLORIDA
Cleveland
DOWNTOWN
CENTRAL
OHIO CITY

Packet-switching: store-and-forward



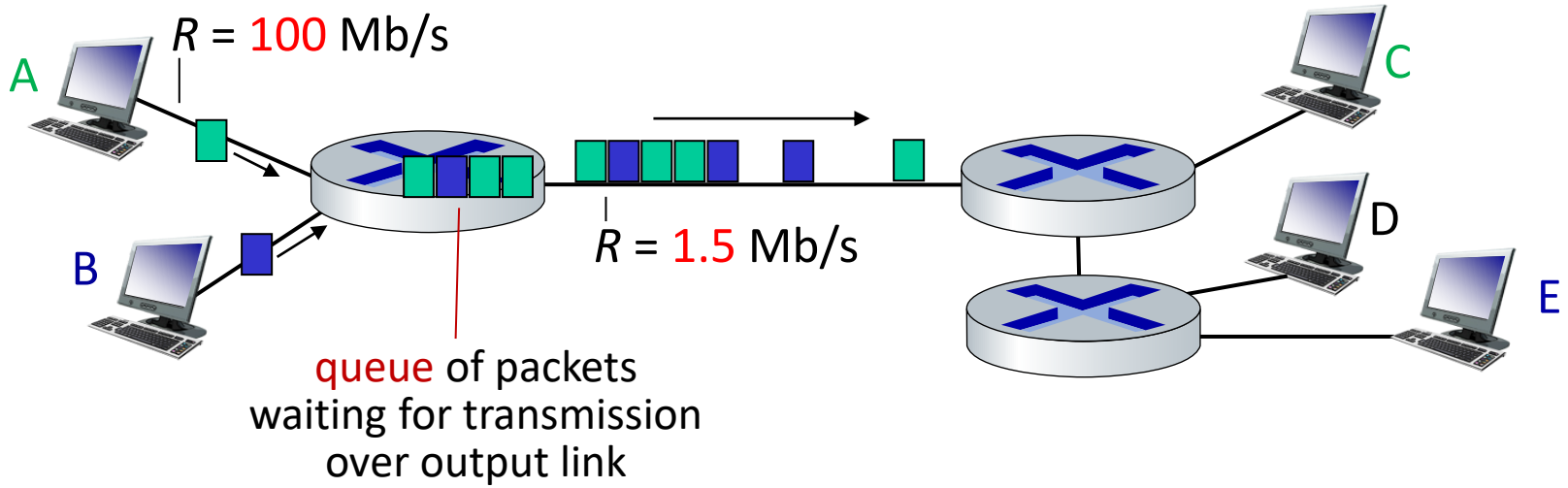
- packet transmission delay: takes L/R seconds to transmit (push out) L-bit packet into link at R bps
- *store and forward: entire* packet must arrive at router before it can be transmitted on next link
- **The general case:** over a path consisting of N links each of rate R (thus, there are N-1 routers between source and destination).

One-hop numerical example:

- L = 10 Kbits
- R = 100 Mbps
- one-hop transmission delay = 0.1 msec

$$d_{\text{end-end}} = N \cdot L/R$$

Packet-switching: queuing



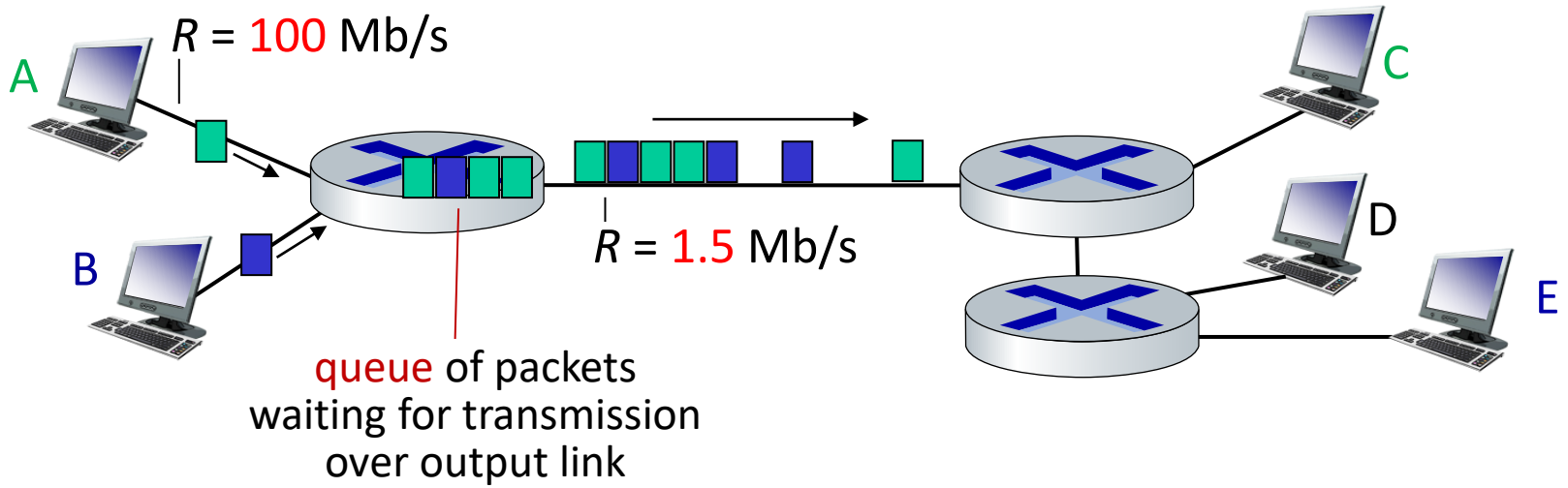
Queuing occurs when work arrives faster than it can be serviced:







Packet-switching: queuing



Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time:

- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

Two fundamental approaches

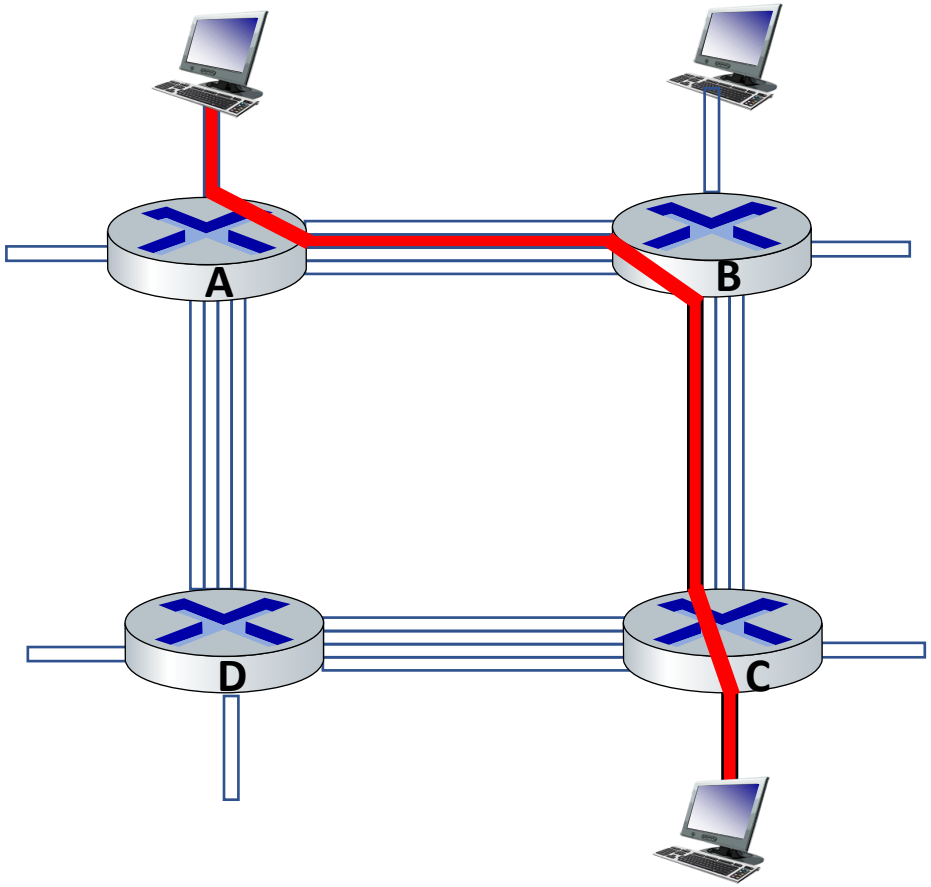
- Two fundamental approaches to moving data through a network of links and switches:
  a) packet-switching: data sent through net in discrete "chunks"
  b) circuit switching: dedicated circuit per call: telephone net
     - In the jargon of telephony, a connection is referred to a circuit



Alternative to packet switching: circuit switching

end-end resources allocated to, reserved for "call" between source and destination

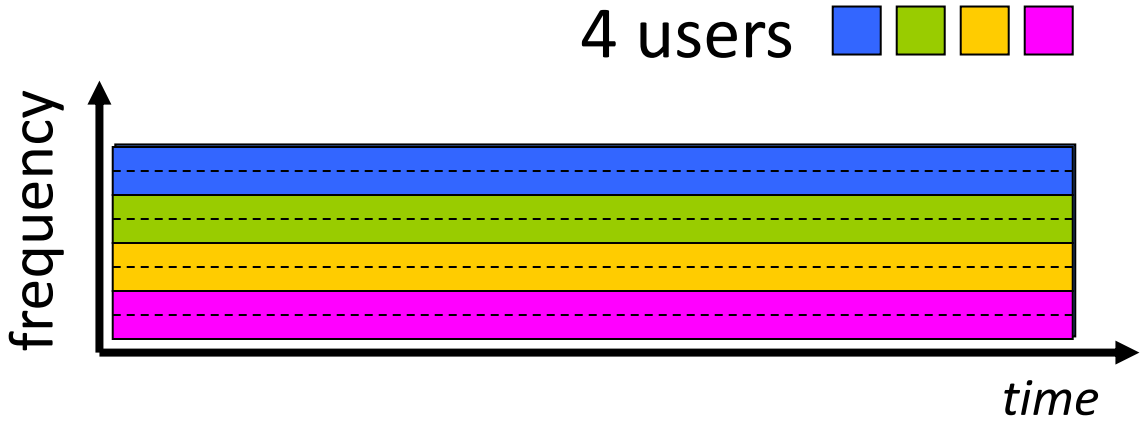
- in diagram, each link has four circuits.
  - call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
  - circuit-like (guaranteed) performance
- circuit segment idle if not used by call (no sharing)
- commonly used in traditional telephone networks



Circuit switching: FDM and TDM

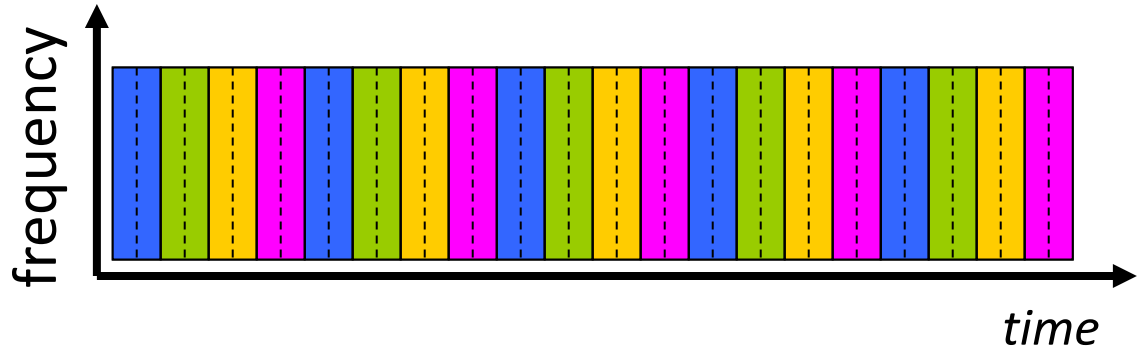
Frequency Division Multiplexing (FDM)

- optical, electromagnetic frequencies divided into (narrow) frequency bands
- each call allocated its own band, can transmit at max rate of that narrow band



Time Division Multiplexing (TDM)

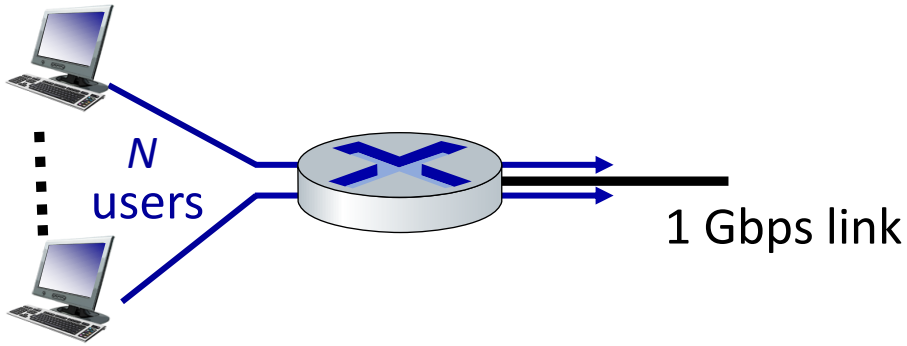
- time divided into slots
- each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band (only) during its time slot(s)



Packet switching versus circuit switching

example:

- 1 Gb/s link
- each user:
  - 100 Mb/s when "active"
  - active 10% of time



Q: how many users can use this network under circuit-switching and packet switching?

- *circuit-switching:* 10 users
- *packet switching:* with 35 users, probability > 10 active at same time is less than .0004 *

Q: how did we get value 0.0004?

Network Core: Packet Switching

each end-end data stream divided into *packets*

- user A, B packets *share* network resources
- each packet uses full link bandwidth
- resources used *as needed*



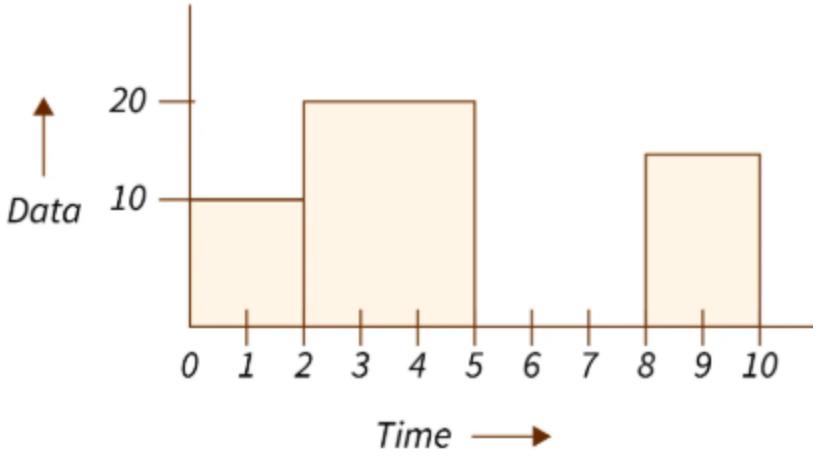
resource contention:

- aggregate resource demand can exceed amount available
- no admission control: circuit switching only allows the number of connections it can afford
- congestion: packets queue, wait for link use
- store and forward: packets move one hop at a time
  - Node receives complete packet before forwarding

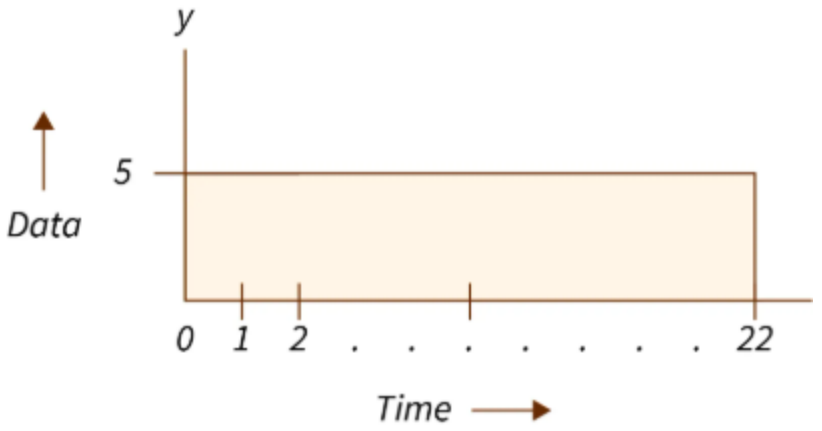
Packet switching versus circuit switching

Is packet switching a “slam dunk winner”?

- great for “bursty” data – sometimes has data to send, but at other times not
  - resource sharing
  - simpler, no call setup
- excessive congestion possible: packet delay and loss due to buffer overflow
  - protocols needed for reliable data transfer, congestion control
- *Q:* How to provide circuit-like behavior with packet-switching?
  - “It’s complicated.” We’ll study various techniques that try to make packet switching as “circuit-like” as possible.



BUSTY



UNIFORM

Restaurants with and without reservations





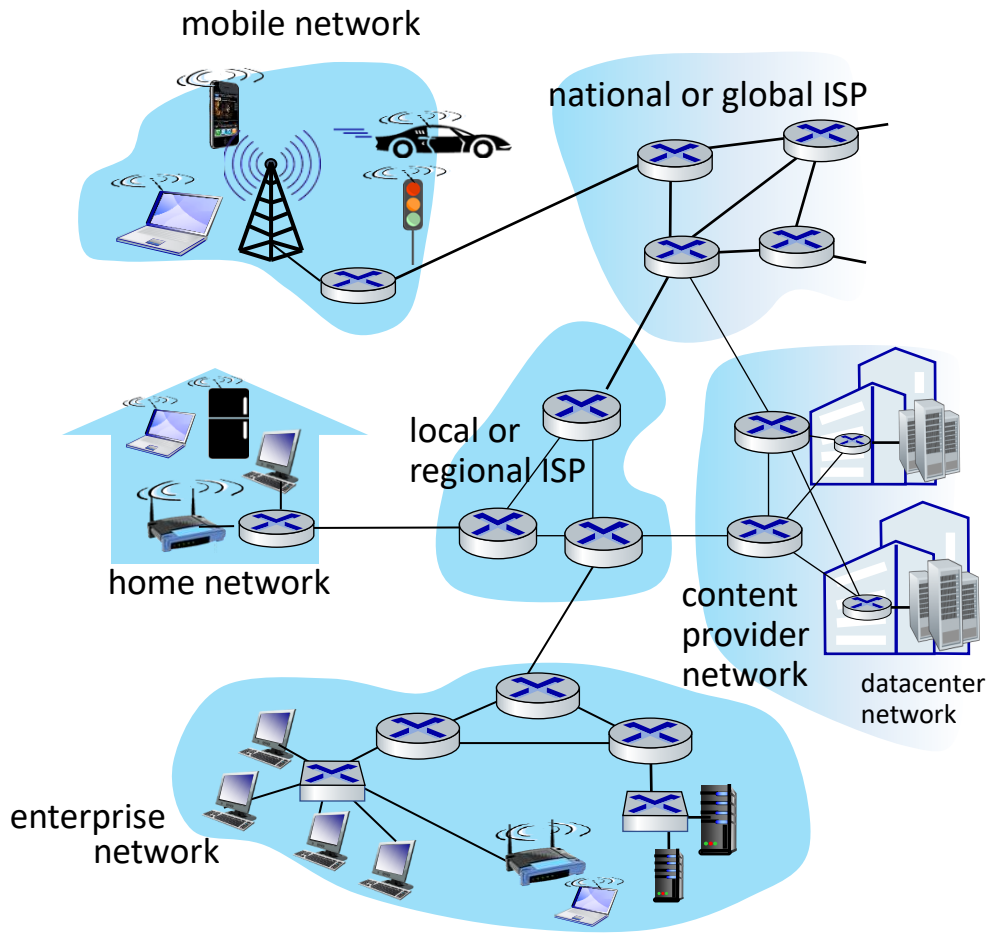


Popular no-booking restaurants

Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet switching)?

Internet structure: a "network of networks"

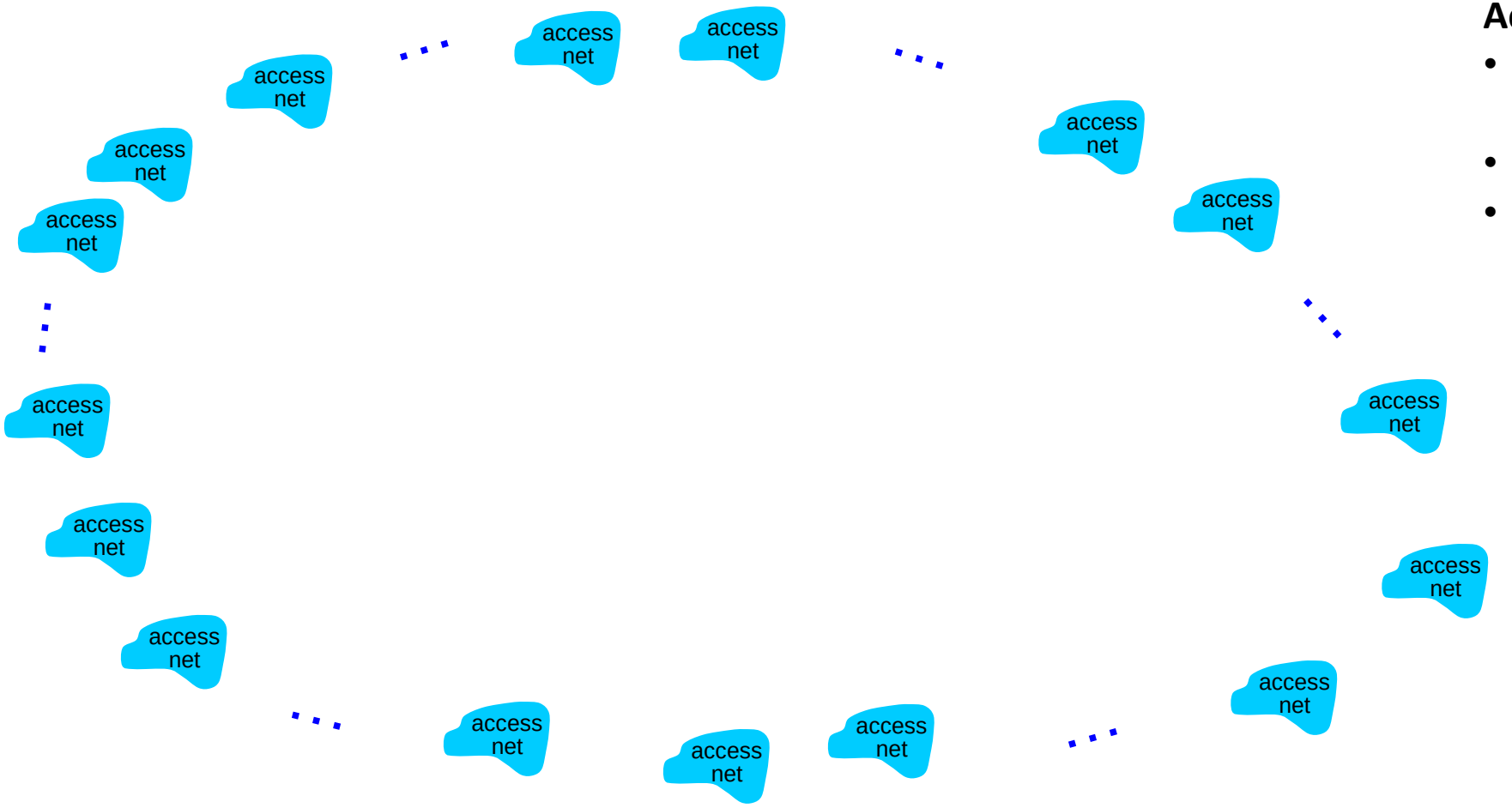
- hosts connect to Internet via access **I**nternet **S**ervice **P**roviders (ISPs)
- access ISPs in turn must be interconnected
  - so that *any* two hosts *(anywhere!)* can send packets to each other
- resulting network of networks is very complex
  - evolution driven by economics, national policies



Let's take a stepwise approach to describe current Internet structure

Internet structure: a "network of networks"

Question: given *millions* of access ISPs, how to connect them together?

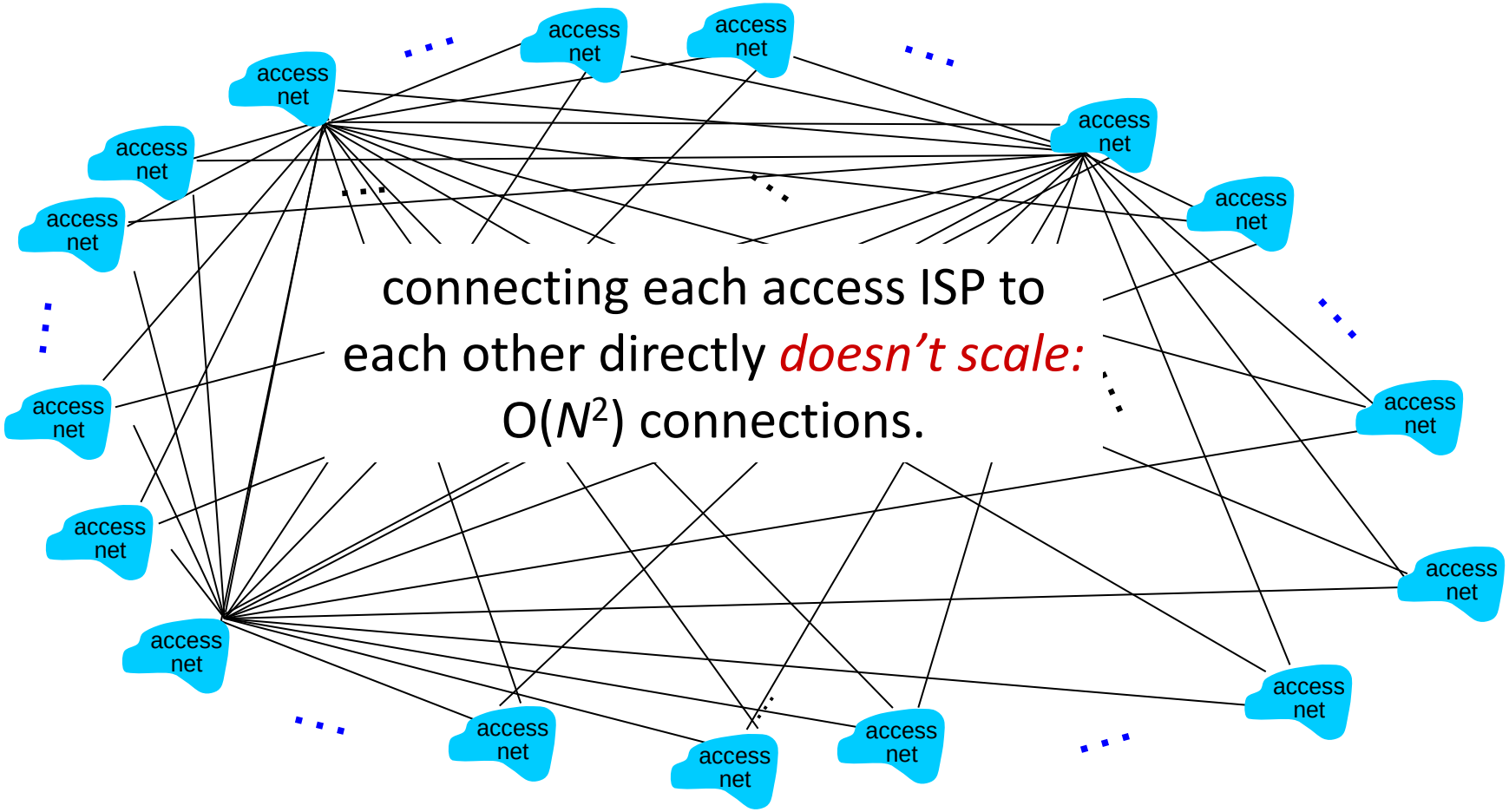


Access ISP:

- a telco/cable company,
- a university,
- a company

Internet structure: a "network of networks"

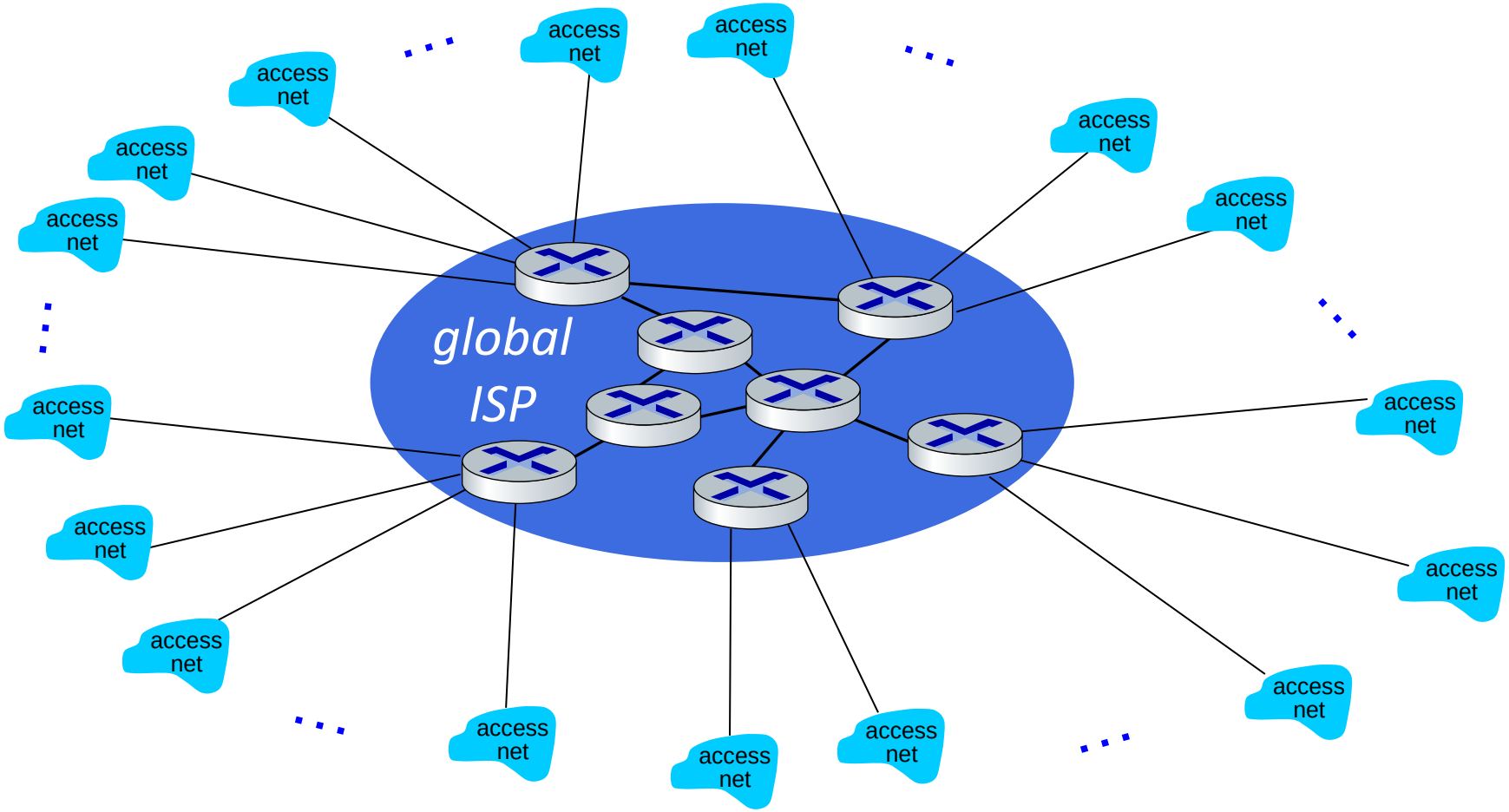
Question: given *millions* of access ISPs, how to connect them together?



Internet structure: a "network of networks"

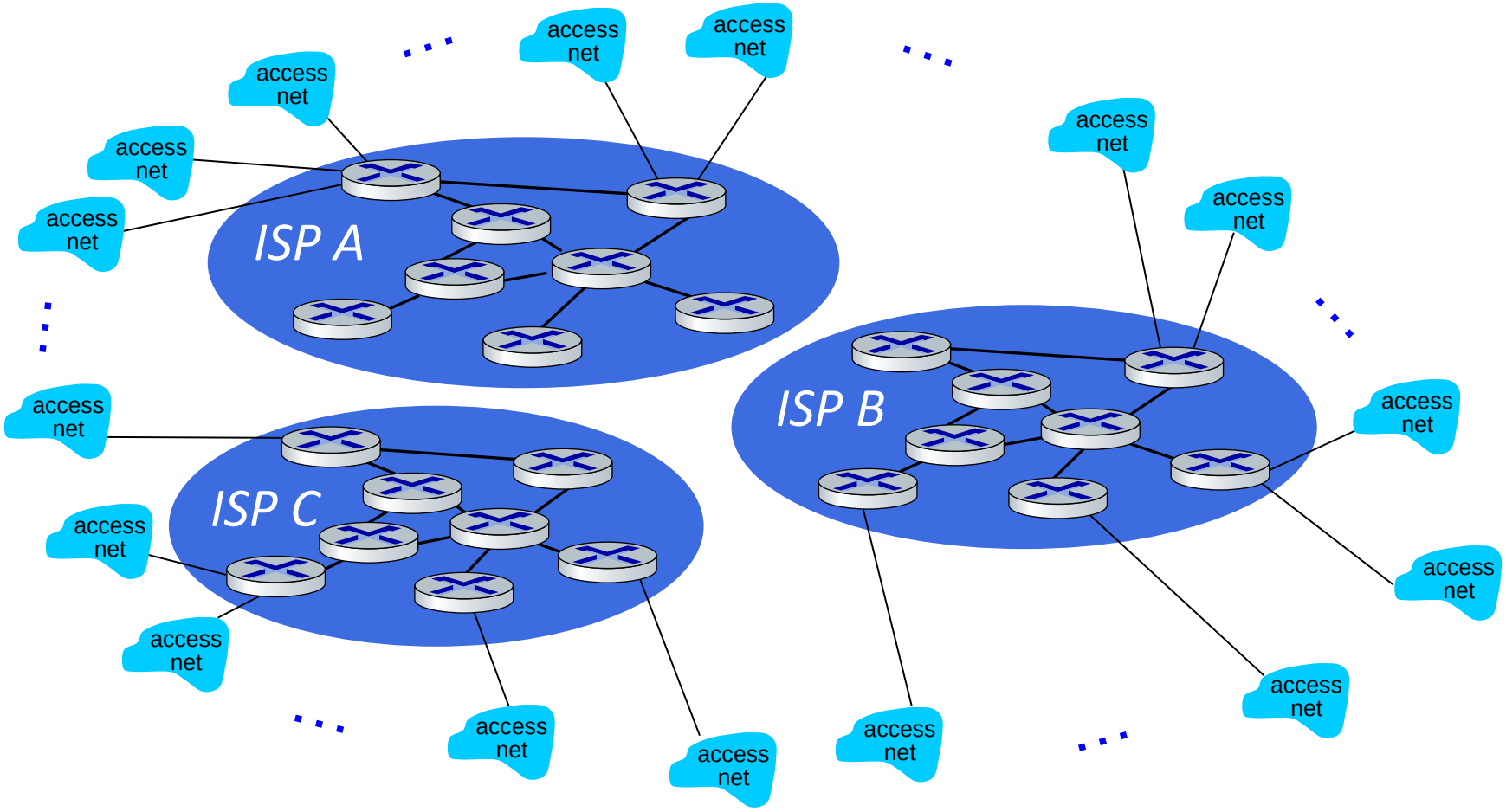
Option: connect each access ISP to one global transit ISP?

Customer and provider ISPs have economic agreement.



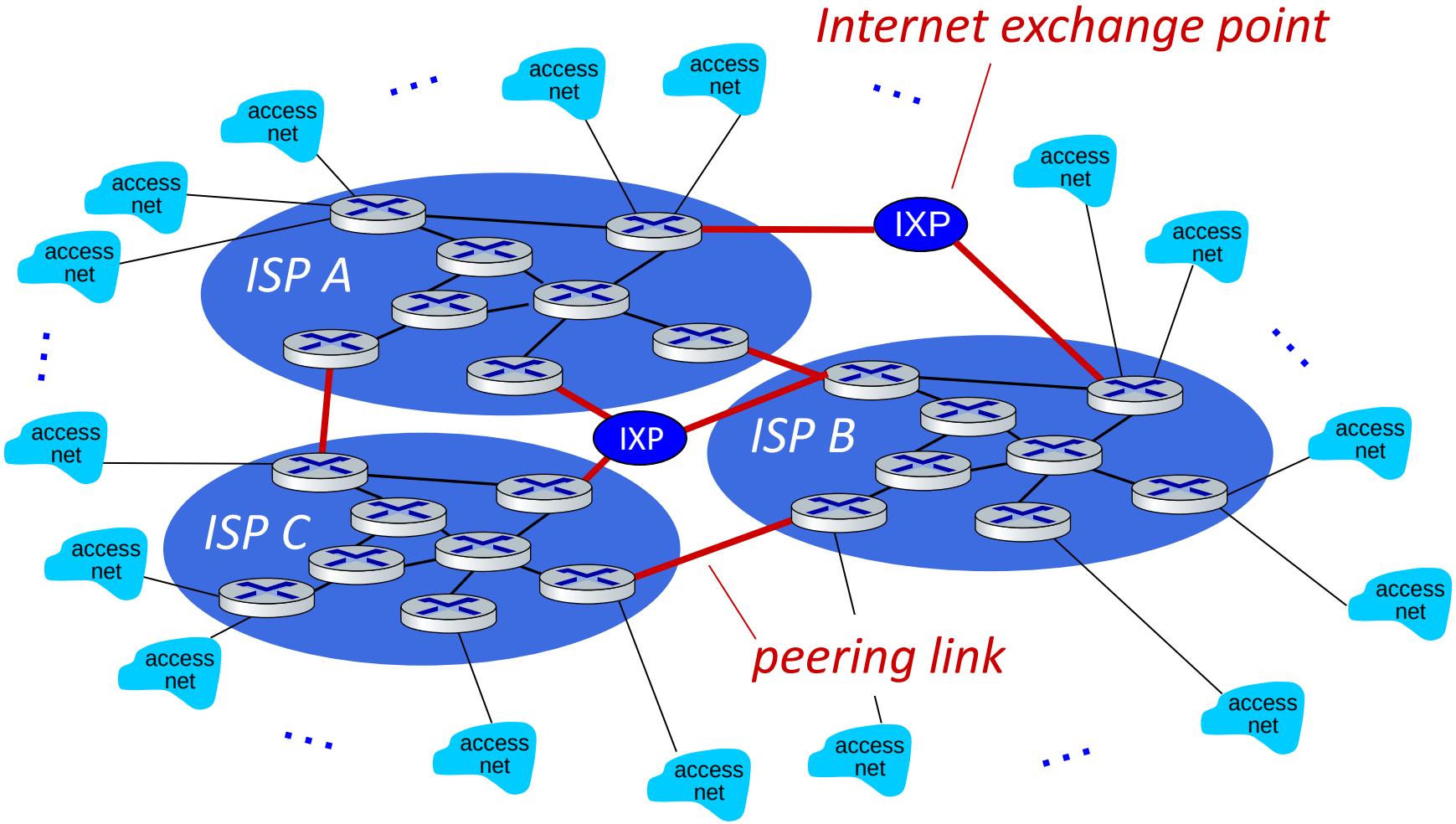
Internet structure: a "network of networks"

But if one global ISP is viable business, there will be competitors ….



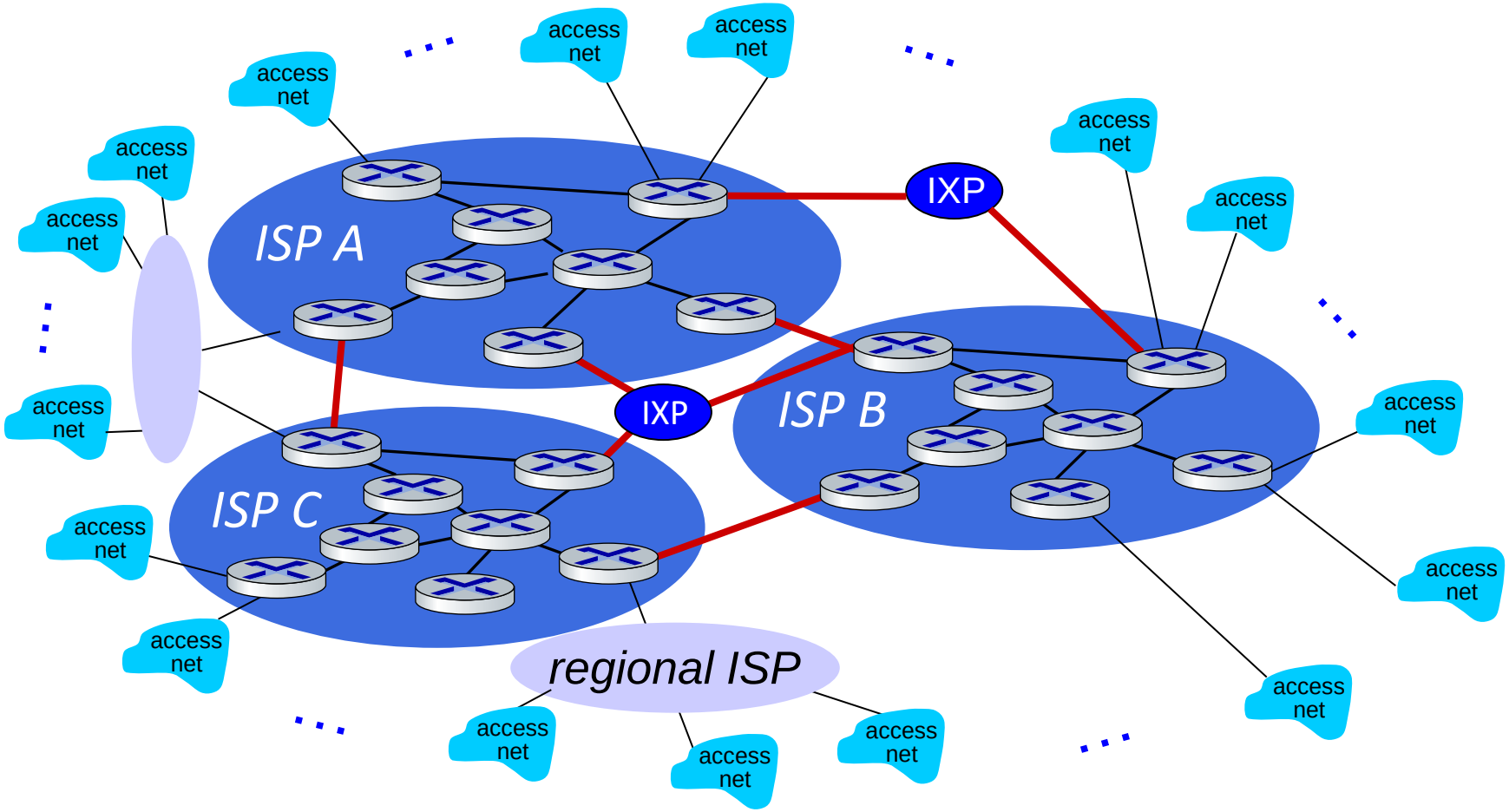
Internet structure: a "network of networks"

But if one global ISP is viable business, there will be competitors …. who will want to be connected



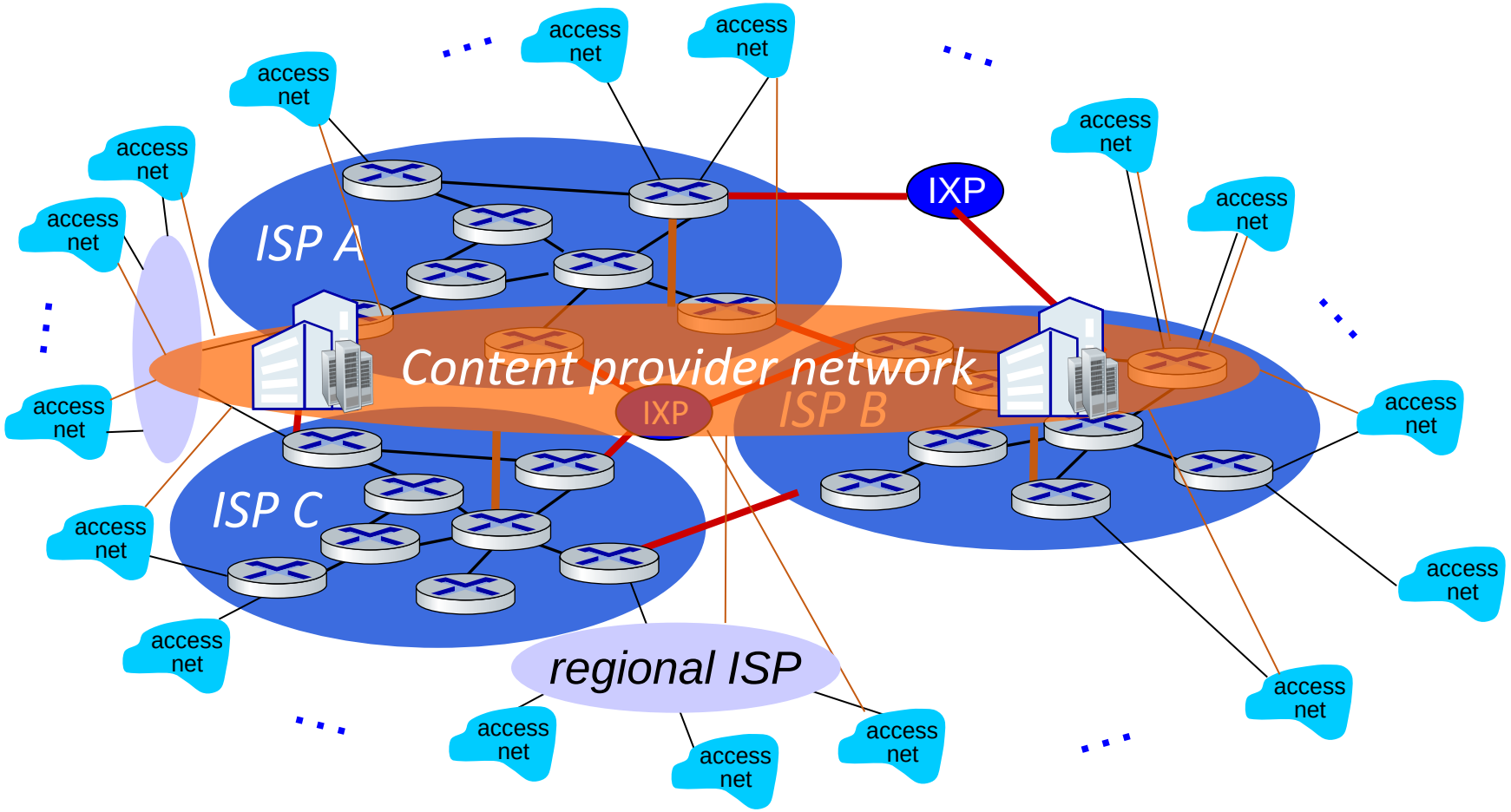
Internet structure: a "network of networks"

… and regional networks may arise to connect access nets to ISPs



Internet structure: a "network of networks"

… and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Google

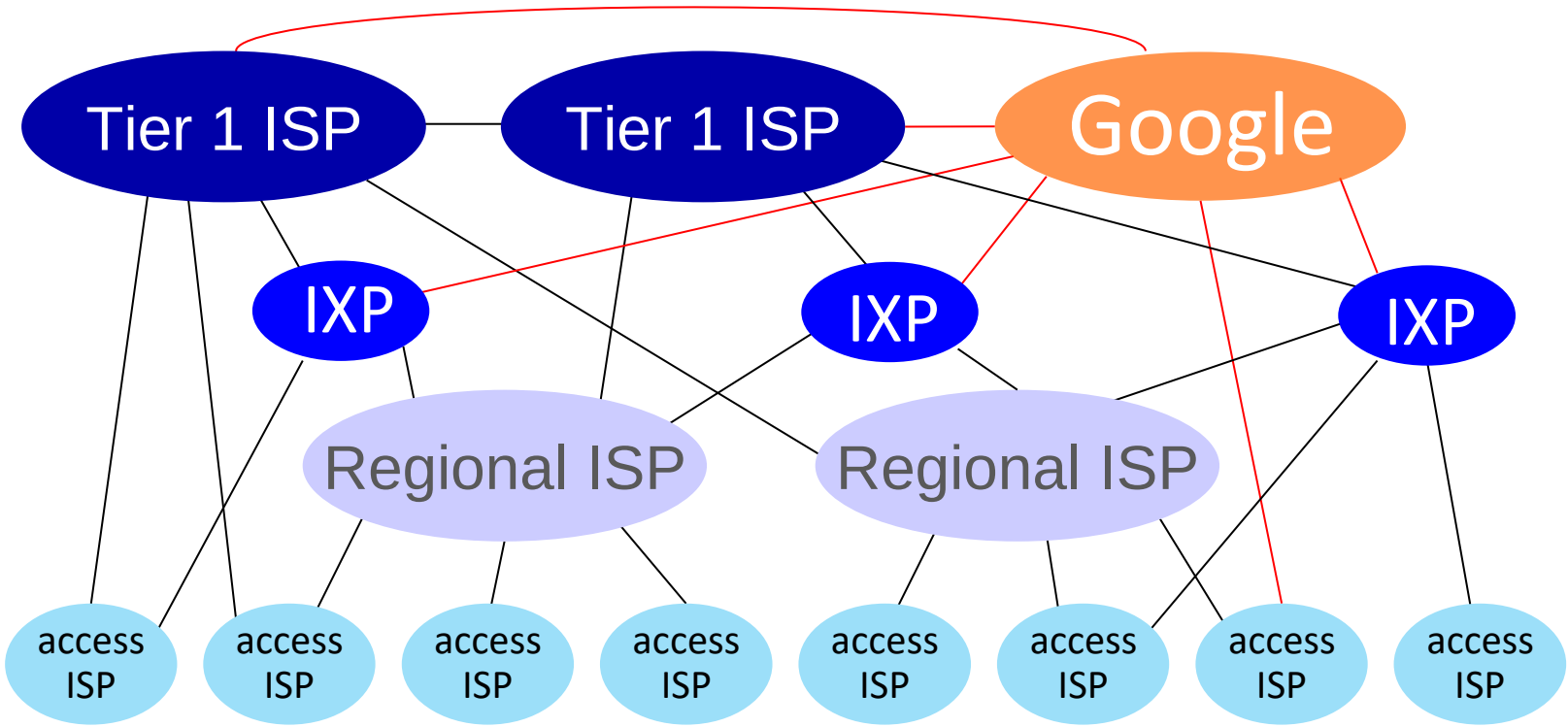
Microsoft

Akamai

NETFLIX

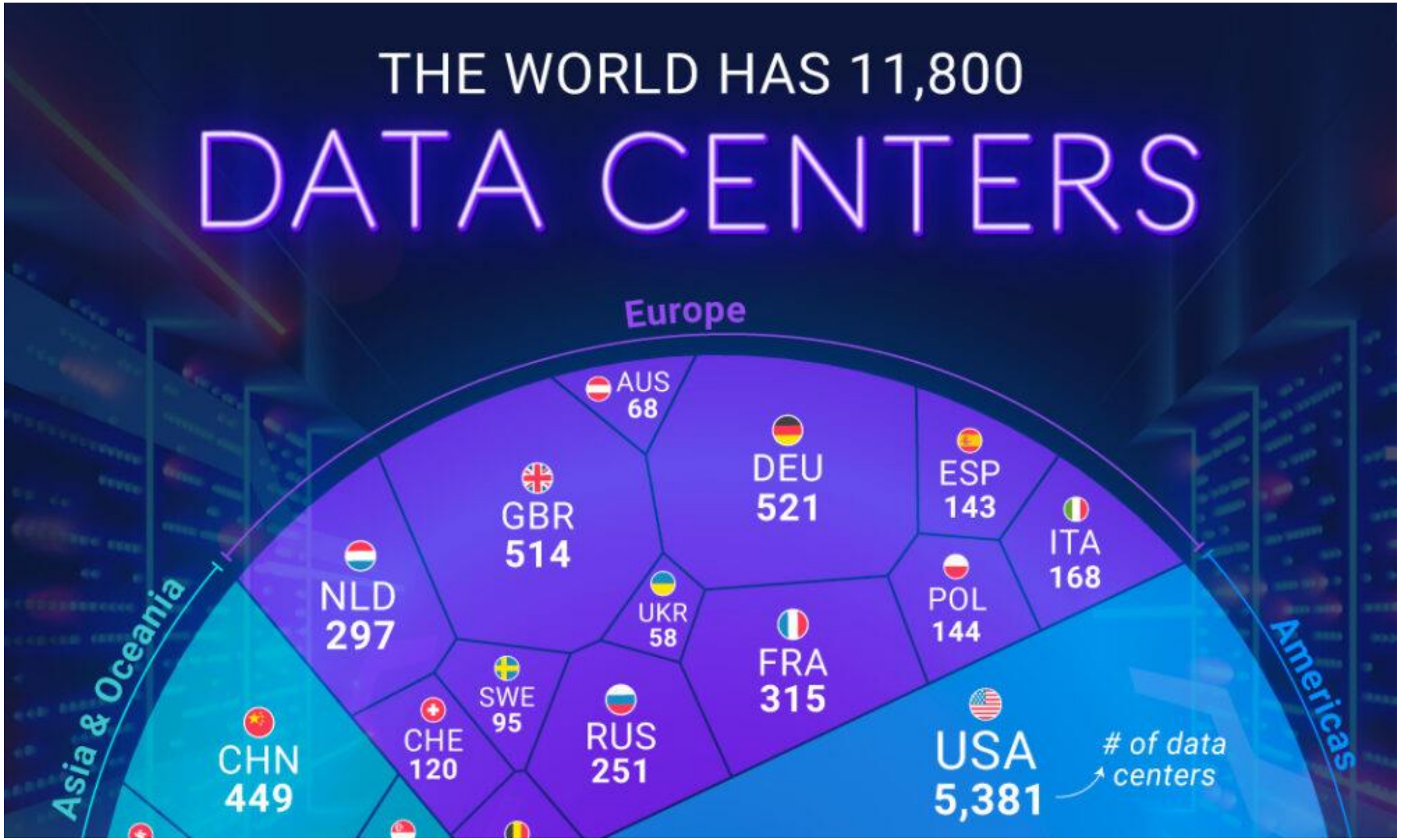
facebook

Internet structure: a "network of networks"



At "center": small # of well-connected large networks

- "tier-1" commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- content provider networks (CDN) (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs


THE WORLD HAS 11,800
DATA CENTERS
Europe
Asia & Oceania
Americas
AUS
68
GBR
514
DEU
521
ESP
143
ITA
168
NLD
297
UKR
58
POL
144
FRA
315
SWE
95
CHE
120
RUS
251
CHN
449
USA
5,381
of data centers

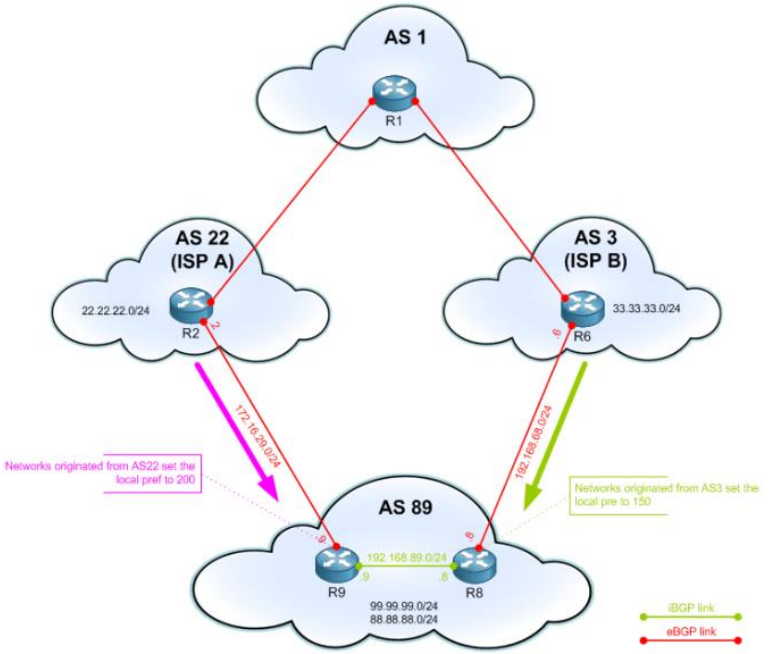
More terms

- PoPs (Points of Presence)
  - A PoP is simply a group of one or more routers (at the same location) in the provider's network where customer ISPs can connect into the provider ISP.
  - PoPs exist in all levels of the hierarchy, except for the bottom (access ISP) level.
- multi-homing
  - to connect to two or more provider ISPs.
- Peering

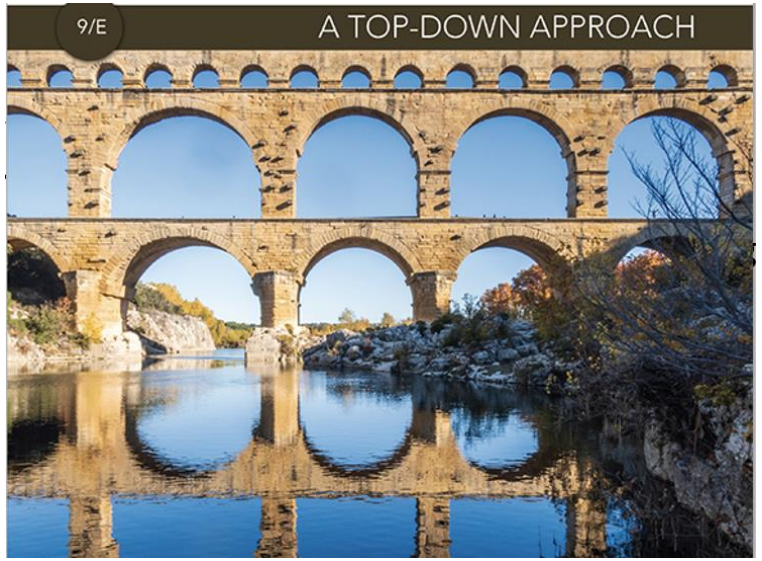


  - To reduce these costs, a pair of nearby ISPs

  at the same level of the hierarchy can peer
- IXPs (Internet Exchange Points)
  - a meeting point where multiple ISPs can peer together
  - typically in a stand-alone building with its own switches



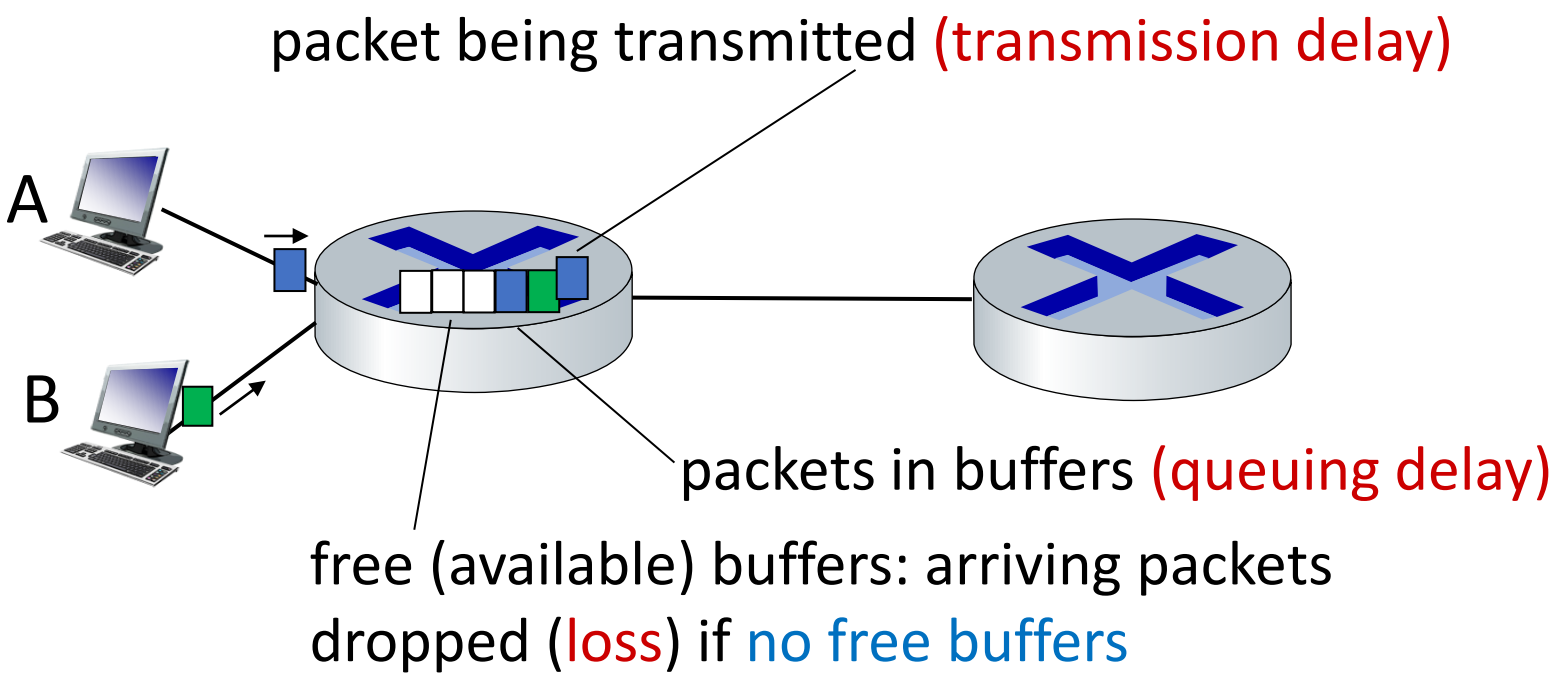
Chapter 1: roadmap



- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, Internet structure
- Performance: loss, delay, throughput
- Protocol layers, service models
- Security
- History

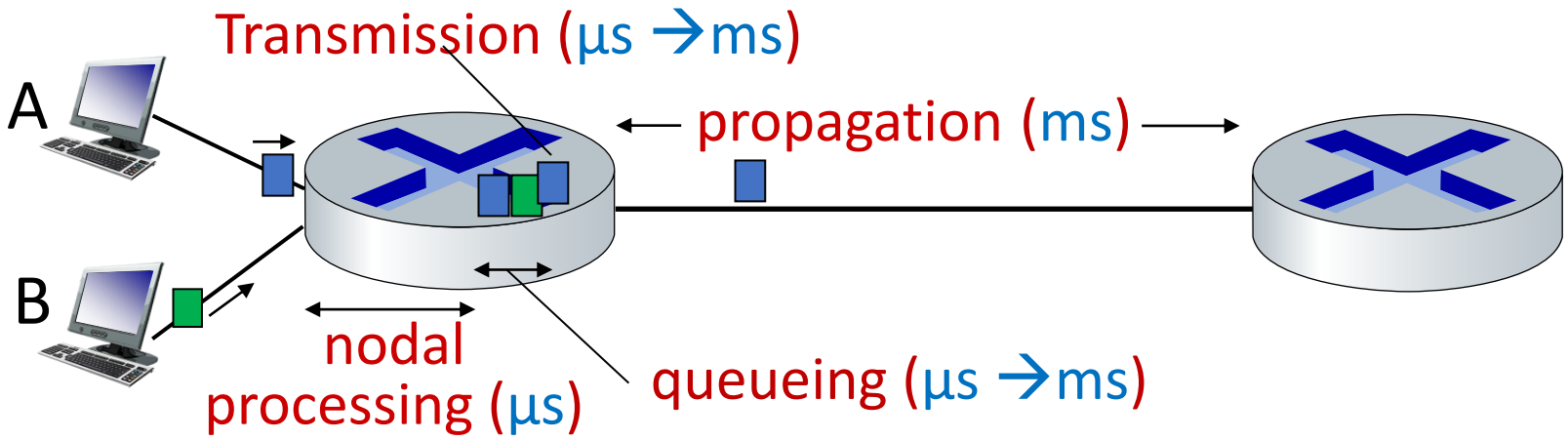
How do packet delay and loss occur?

- packets *queue* in router buffers, waiting for turn for transmission
  - queue length grows when arrival rate to link (temporarily) exceeds output link capacity
- packet *loss* occurs when memory to hold queued packets fills up





Packet delay: four sources



the delay at a single router

Nodal delay: $d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$

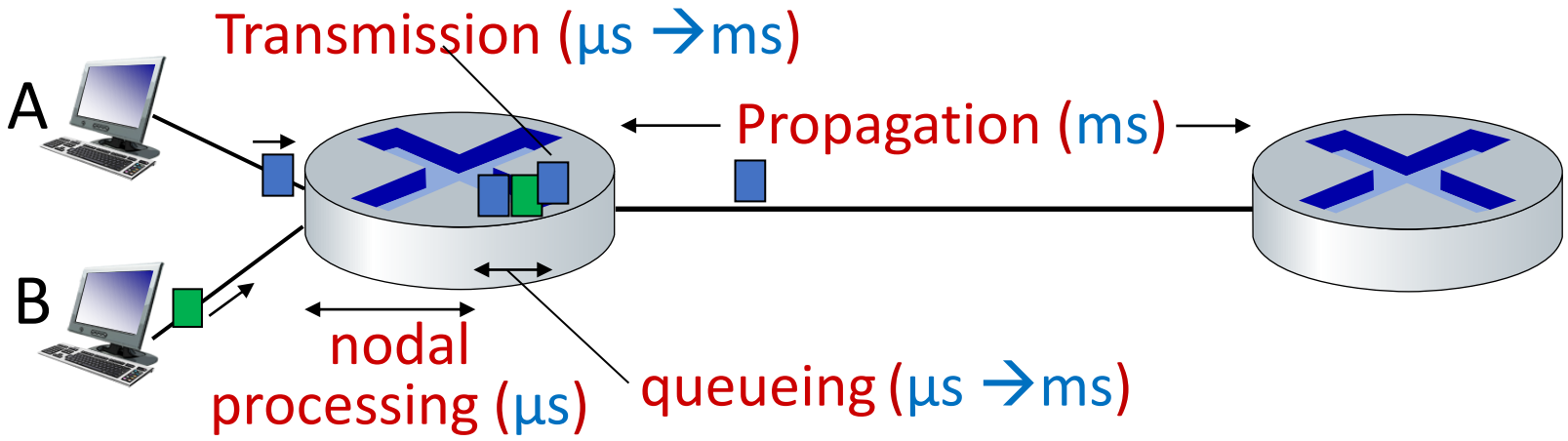
d_{proc}: nodal processing

- check bit errors
- Examine the packet's header and determine output link
- typically negligible < microsecs (µs)

d_{queue}: queuing delay

- time waiting at output link for transmission
- depends on congestion level of router
- The most complicated and interesting component of nodal delay.

Packet delay: four sources



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

d_{trans}: transmission delay:

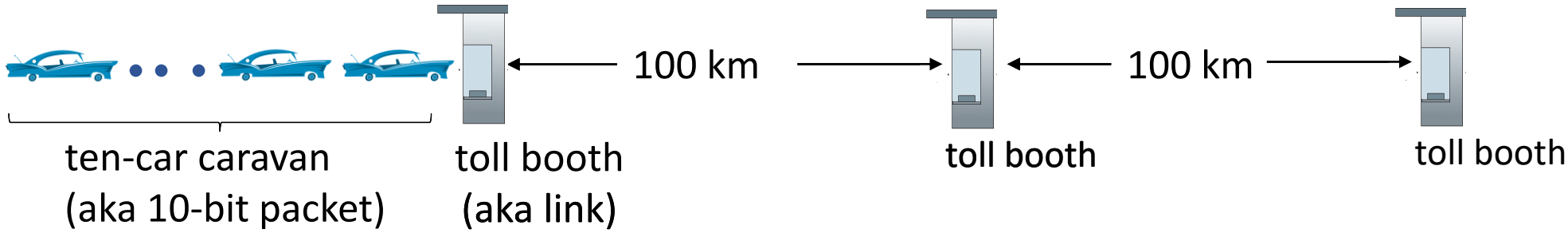
- L: packet length (bits)
- R: link *transmission rate (bps)*
- $d_{trans} = L/R$

d_{prop}: propagation delay:

- d: length of physical link
- s: propagation speed (~$2x10^8$ m/sec)
- $d_{prop} = d/s$

d_{trans} and d_{prop} *very* different

Caravan analogy

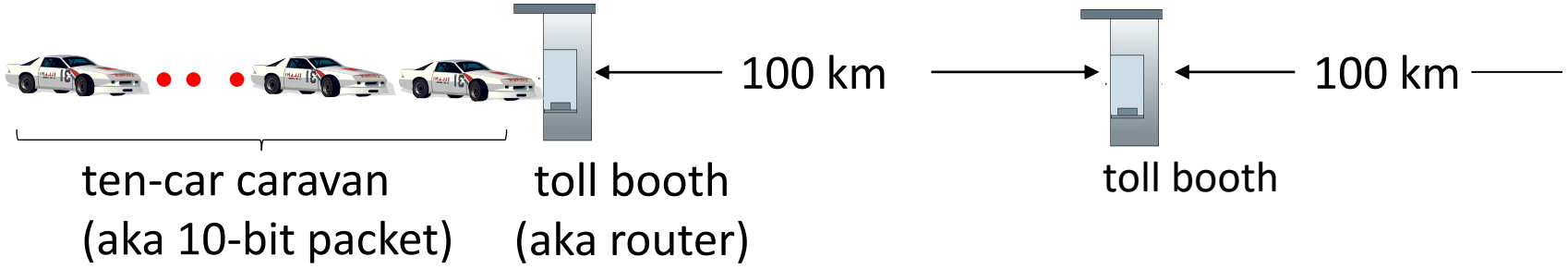


- car ~ bit;
- caravan ~ packet;
- toll service ~ link transmission
- toll booth takes 12 sec to service car (bit transmission time)
- "propagate" at 100 km/hr
- *Q:* How long until caravan is lined up before 2nd toll booth?

Transmission delay

- time to "push" entire caravan through toll booth onto highway = 12*10 = 120 sec
- time for last car to propagate from 1st to 2nd toll both: 100km/(100km/hr) = 1 hr
- *A:* 62 minutes (just like the nodal delay)

Caravan analogy

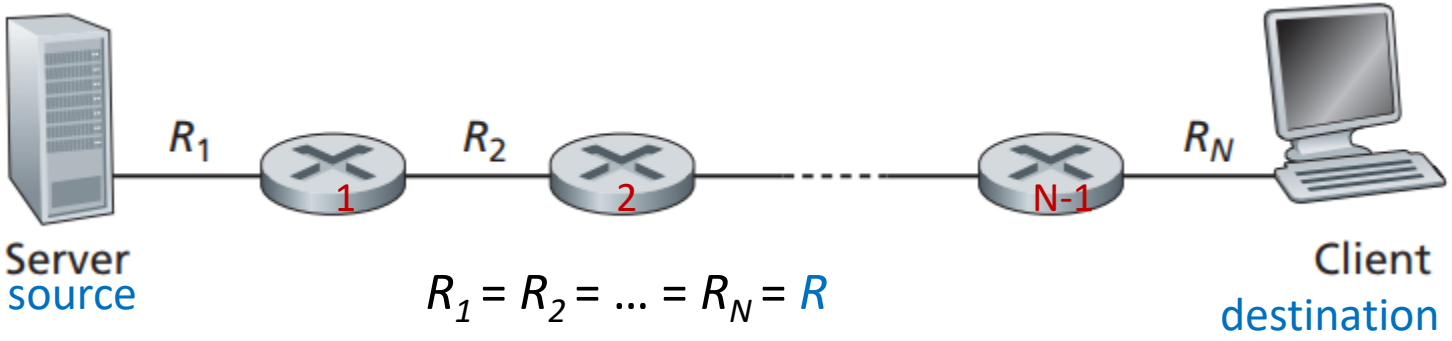


d_{prop} = 100km/(1000km/hr) = 0.1 hr = 6 min

- suppose cars now "propagate" at 1000 km/hr
- and suppose toll booth now takes one min to service a car
- *Q:* Will cars arrive to 2nd booth before all cars serviced at first booth?

  A: Yes! after 7 min, first car arrives at second booth; three cars still at first booth

End-to-End delay



- Suppose: *N* - 1 routers between source and destination; the network is uncongested (so that queuing delays are negligible),
  - at each router and at the source host, the processing delay is d_{proc}, the transmission rate is *R* bits/sec, and the propagation on each link is d_{prop}.
  - The nodal delays accumulate and give an end-to-end delay,

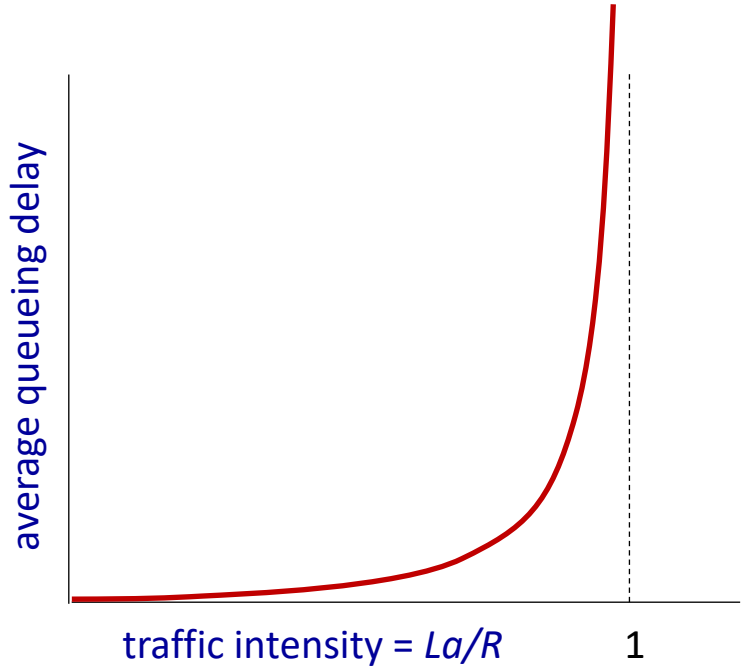
$$d_{\text{end-end}} = N\,(d_{\text{proc}} + d_{\text{trans}} + d_{\text{prop}})$$

Packet queueing delay (revisited)

- *a:* average packet arrival rate
- *L:* packet length (bits)
- *R:* link bandwidth (bit transmission rate)

$$\frac{L \cdot a}{R} = \frac{\text{arrival rate of bits}}{\text{service rate of bits}}$$

"traffic intensity"



- *La/R* ~ 0: avg. **queueing delay** small
- *La/R* → 1: avg. **queueing delay** large
- *La/R* > 1: more "work" arriving is more than can be serviced - average delay infinite!



La/R ~ 0

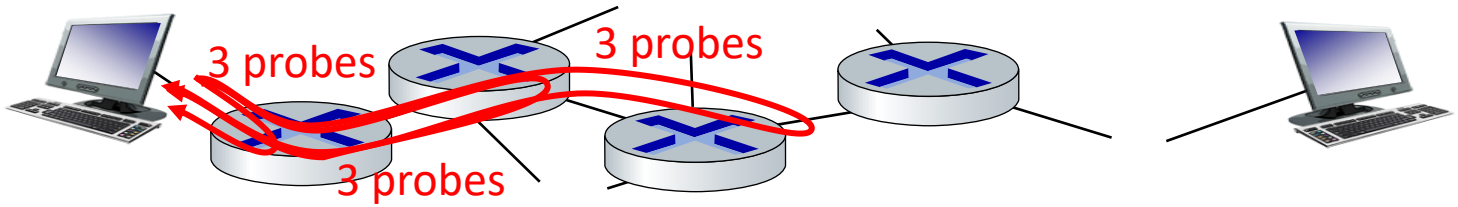


La/R → *1*

“Real” Internet delays and routes

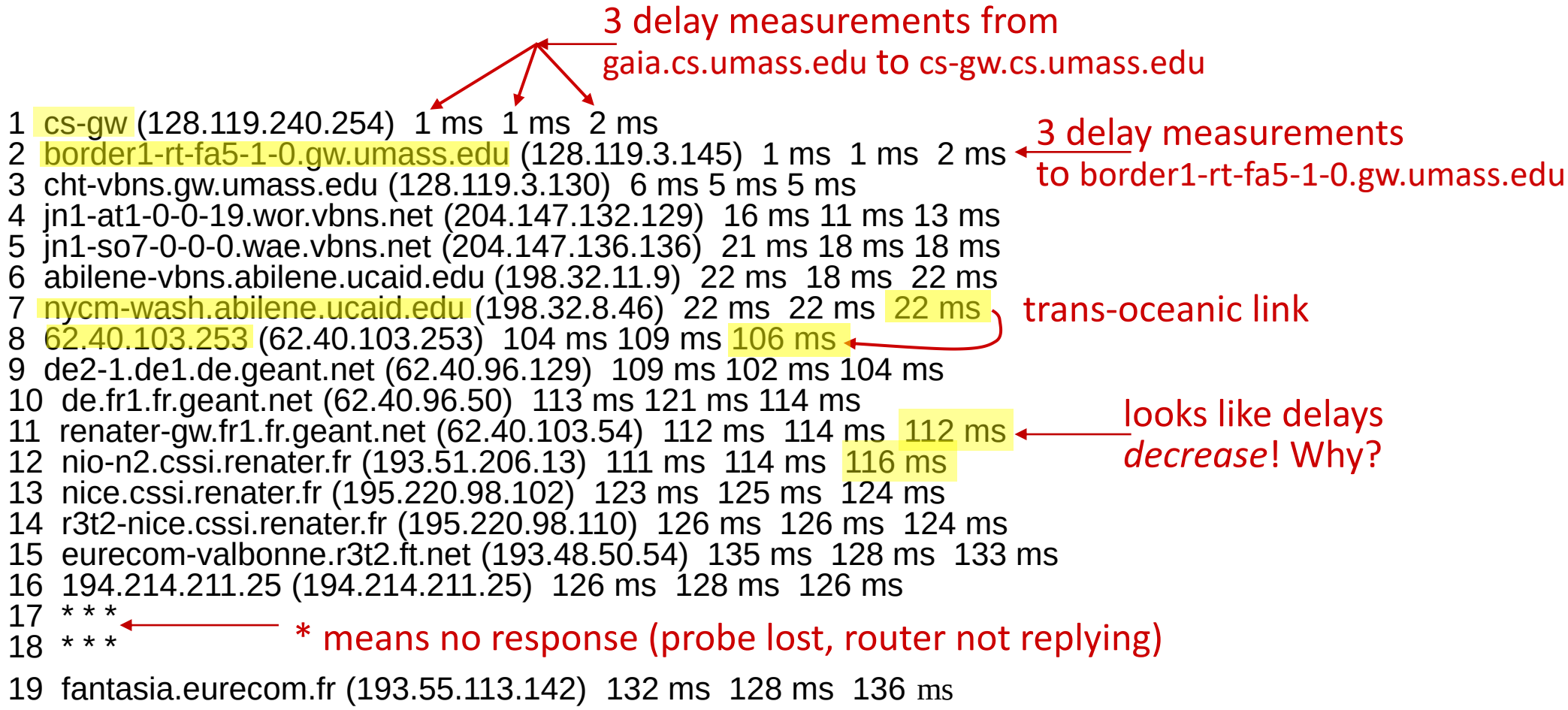
- what do “real” Internet delay & loss look like?
- **traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination. For all *i:*
  - sends three packets that will reach router *i* on path towards destination (with time-to-live field value of *i*)
  - router *i* will return packets to sender
  - sender measures time interval between transmission and reply





Real Internet delays and routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr（in linux）



* Do some traceroutes from exotic countries at www.traceroute.org

In windows, we use “tracert” command

```
C:\Users\Administrator.DESKTOP-EEN44LR>tracert www.baidu.com

通过最多 30 个跃点跟踪
到 www.a.shifen.com [180.101.49.11] 的路由:

  1     3 ms     3 ms     3 ms  192.168.101.1
  2     7 ms    11 ms     8 ms  100.91.128.1
  3    11 ms     7 ms     9 ms  58.217.231.149
  4    28 ms    41 ms    27 ms  58.213.94.54
  5     *        *        *     请求超时。
  6    35 ms    46 ms     *     58.213.96.86
  7    30 ms    29 ms     *     10.166.50.0
  8     *       43 ms     *     10.166.96.38
  9     *        *        *     请求超时。
 10    33 ms    34 ms    32 ms  180.101.49.11

跟踪完成。
```

Local default gateway

```
C:\Users\Administrator.DESKTOP-EEN44LR>tracert news.sina.com.cn

通过最多 30 个跃点跟踪
到 ww1.sinaimg.cn.w.alikunlun.com [240e:978:1219:0:3::3fe] 的路由:

  1    10 ms     3 ms     4 ms  240e:3a1:d11:8175:d2d7:83ff:feb9:613d
  2    10 ms     8 ms     7 ms  240e:1a:2000::54
  3     9 ms     5 ms     6 ms  240e:1a:2000:5154::2
  4     *        9 ms     9 ms  240e:1a:2000:f112::2
  5    15 ms    20 ms    15 ms  240e:1a:f0:f112::3
  6    14 ms    14 ms    16 ms  240e:1a:f0:1308::3
  7    17 ms    18 ms    17 ms  240e:978:200:aa::3
  8    15 ms    16 ms    17 ms  240e:978:1219:0:3::3fe

跟踪完成。
```

```
C:\Users\Administrator.DESKTOP-EEN44LR>tracert www.ethz.ch

通过最多 30 个跃点跟踪
到 www.ethz.ch [2001:67c:10ec:254::216] 的路由:

  1     5 ms     3 ms     3 ms  240e:3a1:d11:8175:d2d7:83ff:feb9:613d
  2     6 ms    13 ms     6 ms  240e:1a:2000::54
  3     5 ms     5 ms     8 ms  240e:1a:2000:5154::2
  4    16 ms     9 ms     9 ms  240e:1a:2000:f118::2
  5    13 ms    12 ms    14 ms  240e::1:21:61:6d02
  6     *        *       13 ms  240e:0:a::c9:5384
  7    13 ms    21 ms    13 ms  240e:0:a::c9:533d
  8     *        *        *     请求超时。
  9   232 ms   227 ms   224 ms  2001:7f8:1::a502:1320:1
 10   268 ms   272 ms   274 ms  ae7.mx1.fra.de.geant.net [2001:798:cc::1a]
 11   241 ms   240 ms   234 ms  ae3.mx1.gen.ch.geant.net [2001:798:cc::e]
 12   281 ms   287 ms   280 ms  switch-ias-switch-gw.gen.ch.geant.net [2001:798:1::be]
 13   286 ms   295 ms   284 ms  swiCE4-B4.switch.ch [2001:620:0:c00c::2]
 14   215 ms   210 ms   222 ms  swiBF1-B2.switch.ch [2001:620:0:c02d::1]
 15   232 ms   232 ms   236 ms  swiEZ3-B5.switch.ch [2001:620:0:c004::2]
 16   233 ms   236 ms   239 ms  rou-rz-gw-giga-to-switch.ethz.ch [2001:620:0:ffe6::2]
 17   247 ms   233 ms   278 ms  rou-fw-rz-rz-gw.ethz.ch [2001:67c:10ec:201::1]
 18   240 ms   237 ms   235 ms  cms-publish.ethz.ch [2001:67c:10ec:254::216]

跟踪完成。

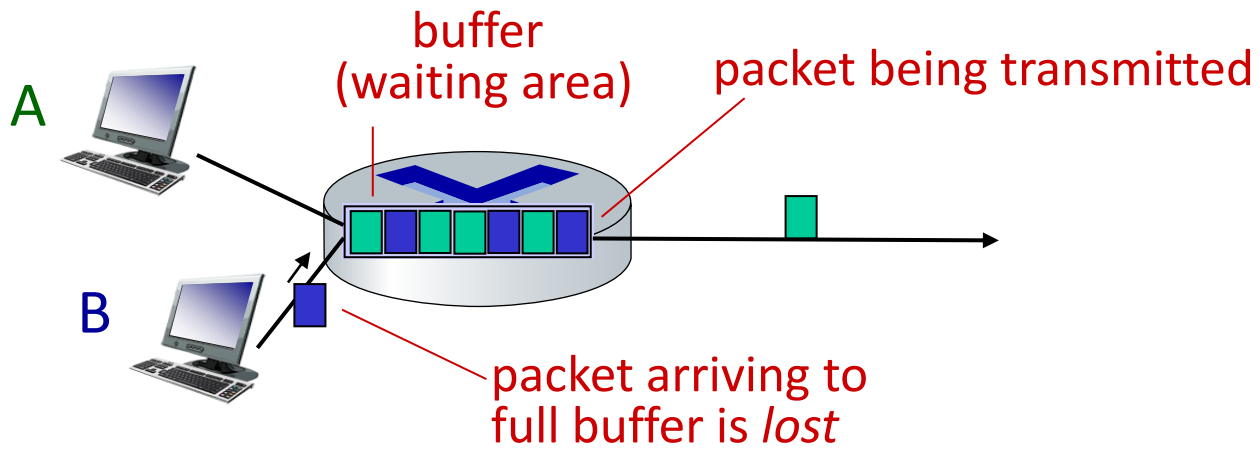
C:\Users\Administrator.DESKTOP-EEN44LR>
```

Command line: (search cmd)

Tracert hostname

Packet loss

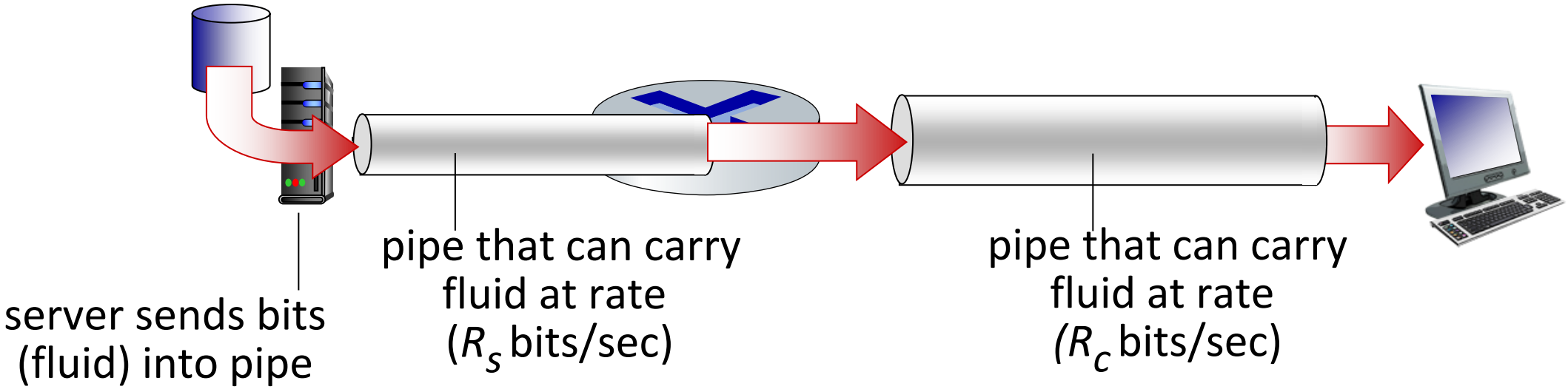
- queue (aka buffer) preceding link in buffer has finite capacity
- packet arriving to full queue dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all



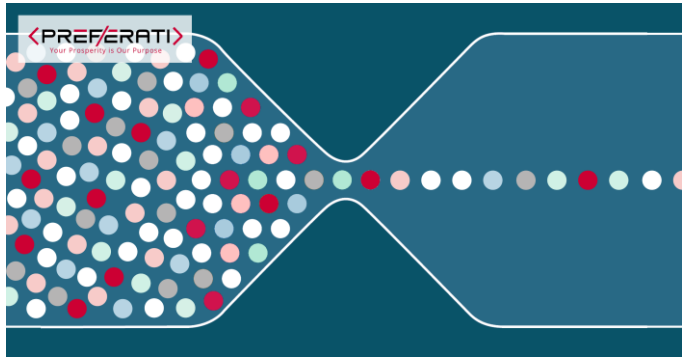
Throughput

- *throughput:* rate (bits/time unit) at which bits are being sent from sender to receiver
  - *instantaneous:* rate at given point in time
  - *average:* rate over longer period of time

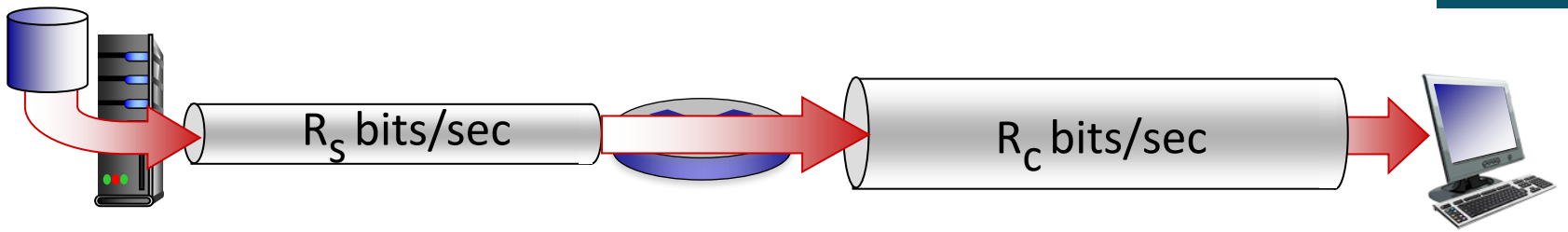
Assume transmission rate:
Server access link: R_s
Client access link: R_c



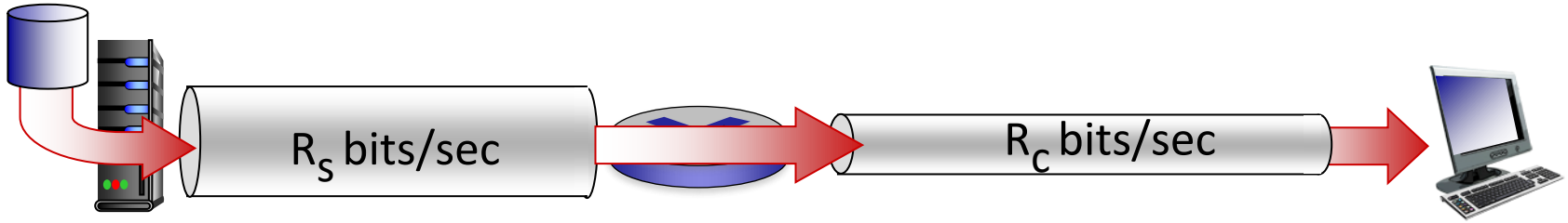
Throughput (for this simple two-link network)



$R_s < R_c$ What is average end-end throughput?



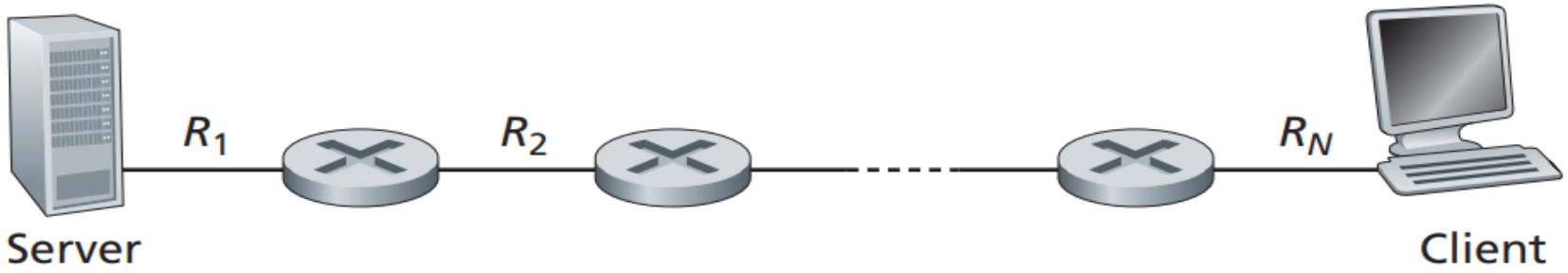
$R_s > R_c$ What is average end-end throughput?



bottleneck link

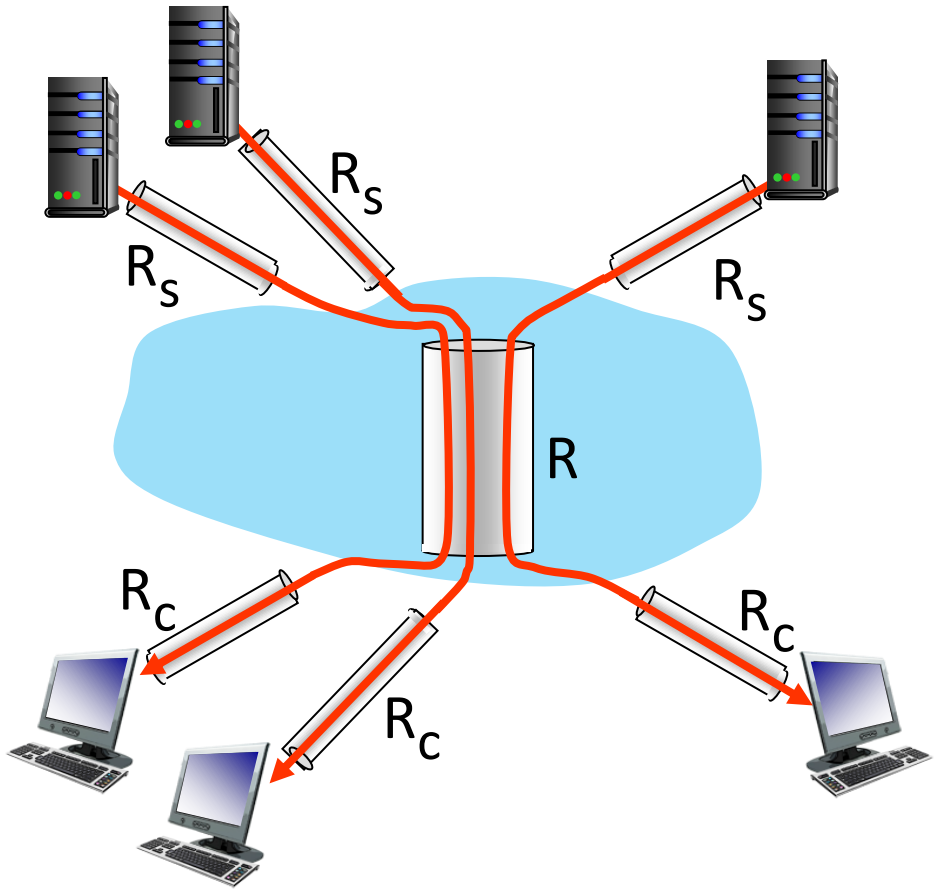
link on end-end path that constrains end-end throughput ($\min\{R_c, R_s\}$)

Throughput for a file transfer from server to client



- A network with *N* links between the server and the client, with the transmission rates of the *N* links being R_1, R_2, …, R_N.
- The throughput for a file transfer from server to client is $\min\{R_1, R_2, \ldots, R_N\}$,
  - i.e., the transmission rate of the bottleneck link along the path between server and client.

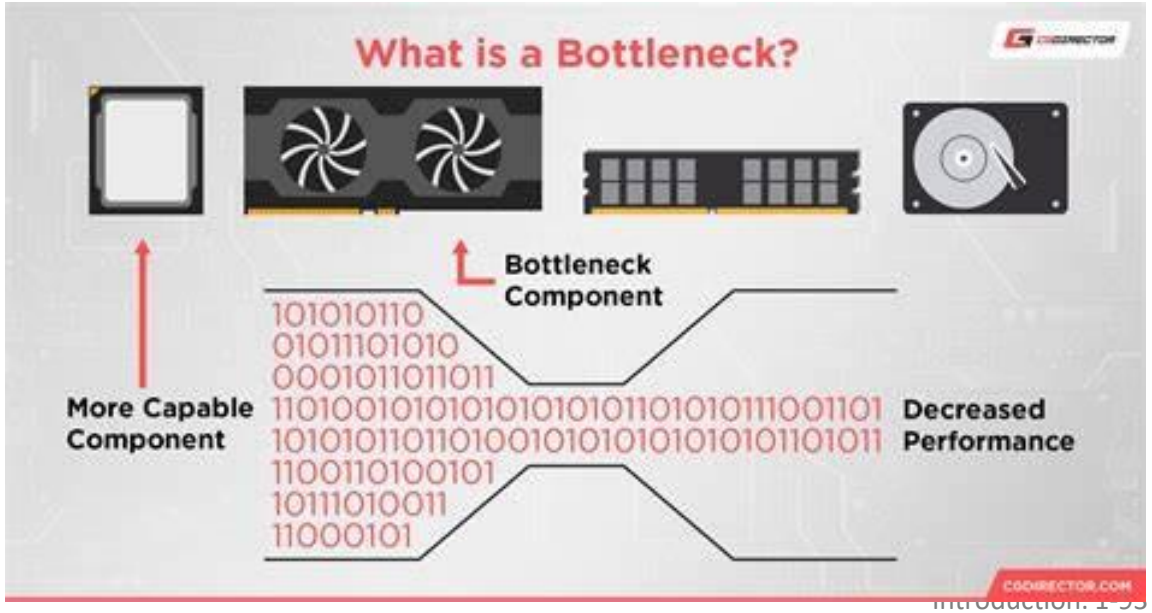
Throughput: network scenario



10 connections (fairly) share backbone bottleneck link *R* bits/sec

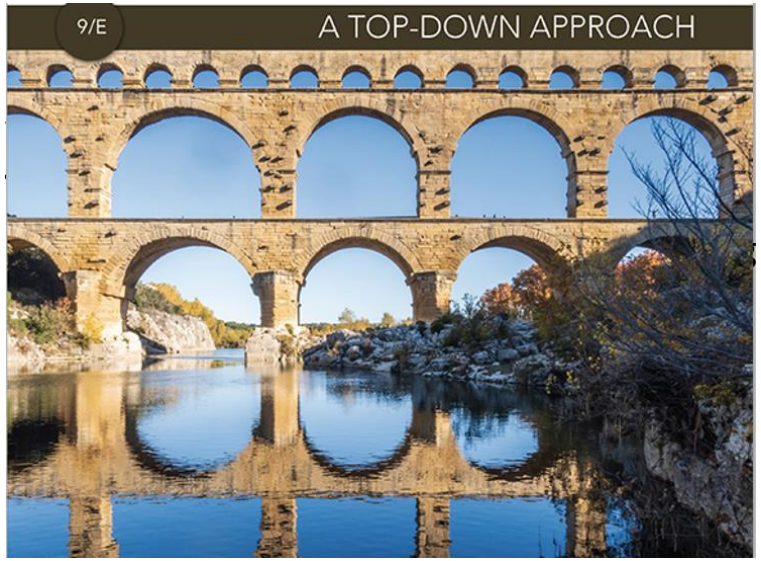
- Only when *R* in the core is of the same order of R_c,R_s
  - per-connection end-end throughput:

    $min(R_c,R_s,R/10)$
- *Generally speaking,* $R >> R_c,R_s$



Chapter 1: roadmap

- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, Internet structure
- Performance: loss, delay, throughput
- Protocol layers, service models
- Security
- History



Protocol “layers” and reference models

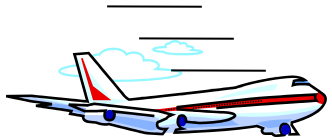
Networks are complex, with many “pieces”:

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question: is there any hope of *organizing* structure of network?

- and/or our *discussion* of networks?

Example: organization of air travel



end-to-end transfer of person plus baggage

ticket (purchase)

baggage (check)

gates (load)

runway takeoff

airplane routing

ticket (complain)

baggage (claim)

gates (unload)

runway landing

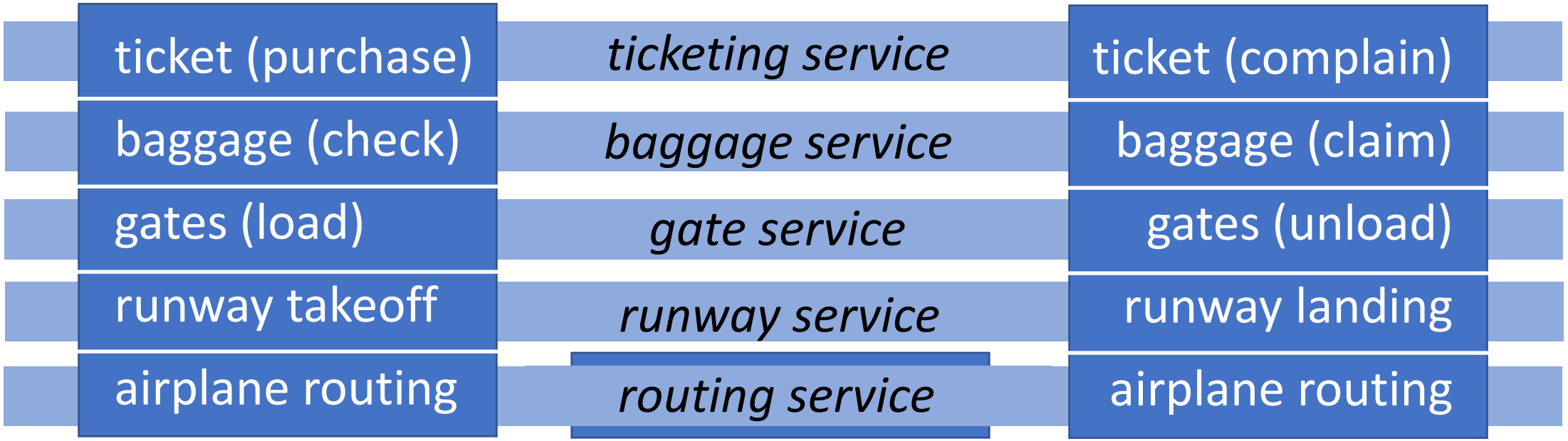
airplane routing

airplane routing

How would you *define/discuss* the *system* of airline travel?

- a series of steps, involving many services

Example: organization of air travel



layers: each layer implements a service

- via its own internal-layer actions
- relying on services provided by layer below

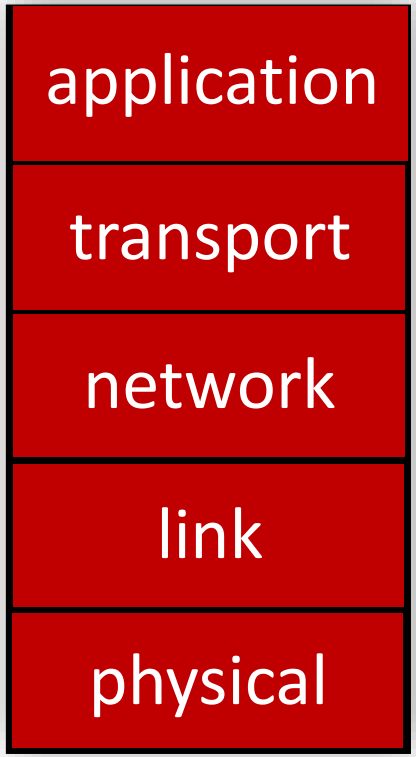
Why layering?

Approach to designing/discussing complex systems:

- explicit structure allows identification, relationship of system's pieces
  - layered *reference model* for discussion
- modularization eases maintenance, updating of system
  - change in layer's service *implementation*: transparent to rest of system
  - e.g., change in gate procedure doesn't affect rest of system

Layered Internet protocol stack (most often used)

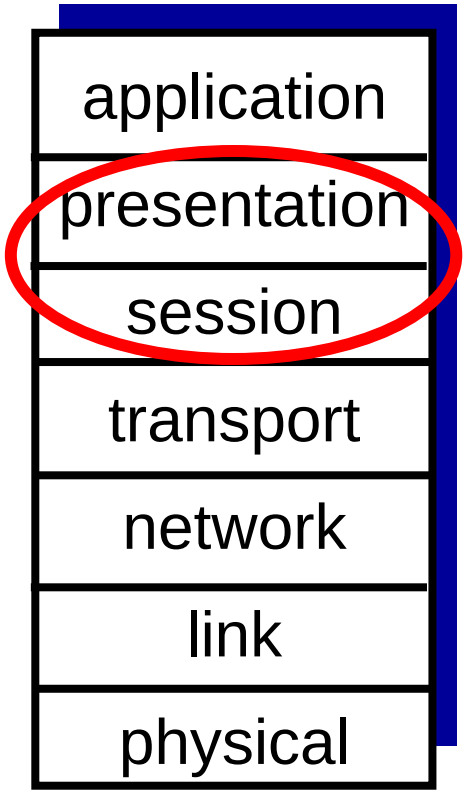
- *application:* supporting network applications
  - HTTP, IMAP, SMTP, DNS
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.11 (WiFi), PPP
- *physical:* bits "on the wire"



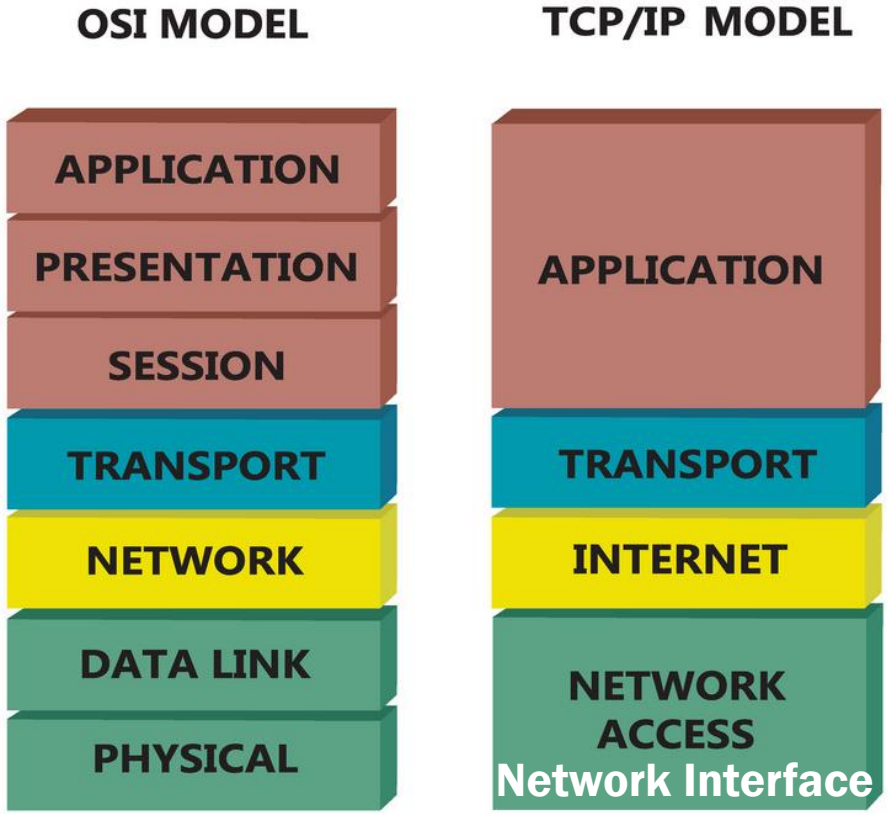
ISO/OSI reference model

Two layers not found in Internet protocol stack!

- *presentation:* allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- *session:* synchronization, checkpointing, recovery of data exchange
- Internet stack "missing" these layers!
  - these services, *if needed,* must be implemented in application
  - needed?

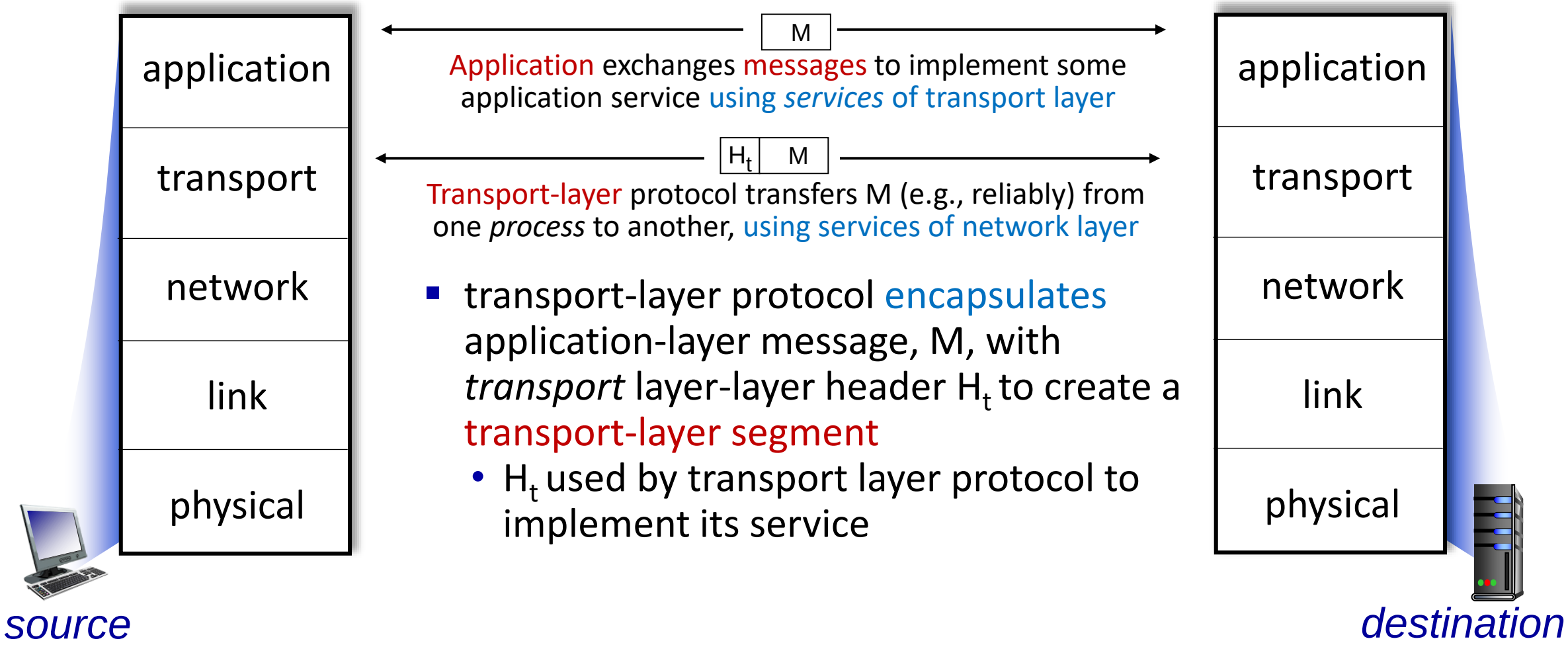


The seven layer ISO/OSI reference model



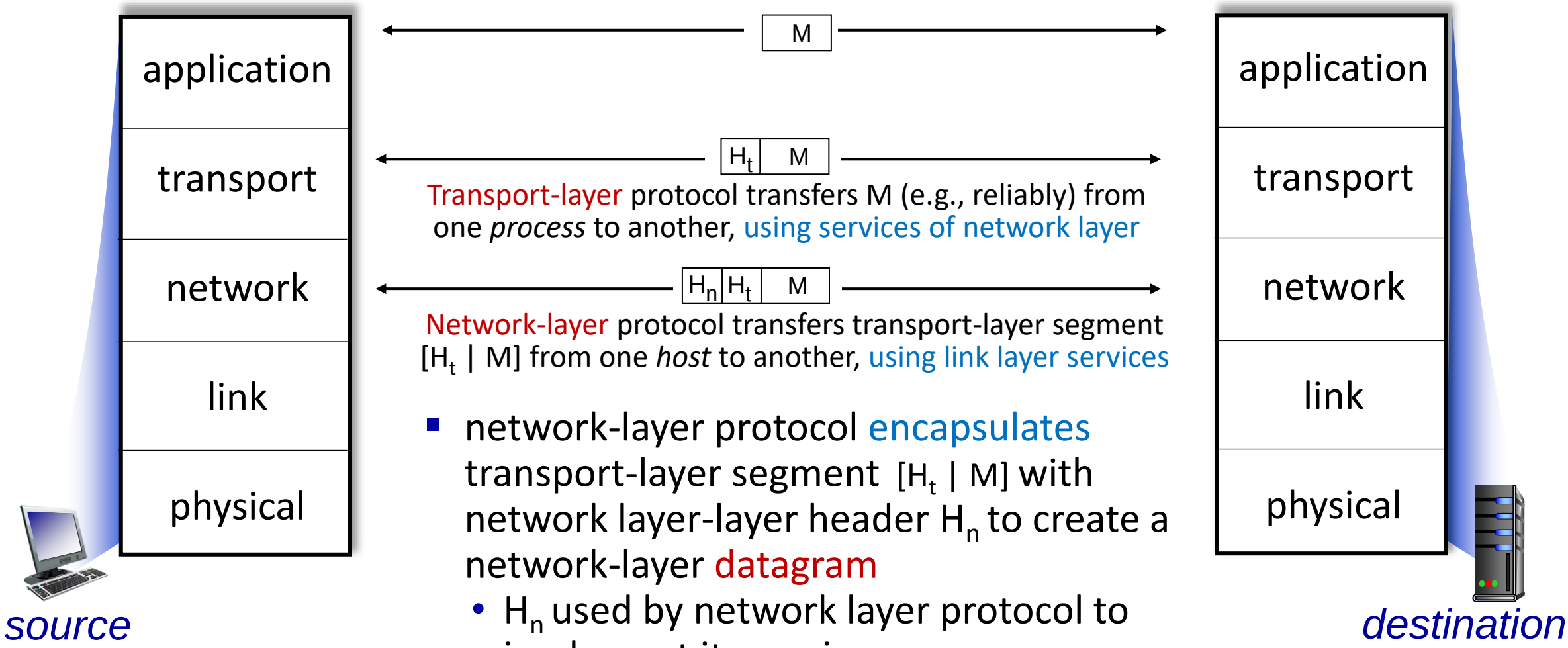
ISO-OSI: International organization of Standardization - Open System Interconnection

Services, Layering and Encapsulation



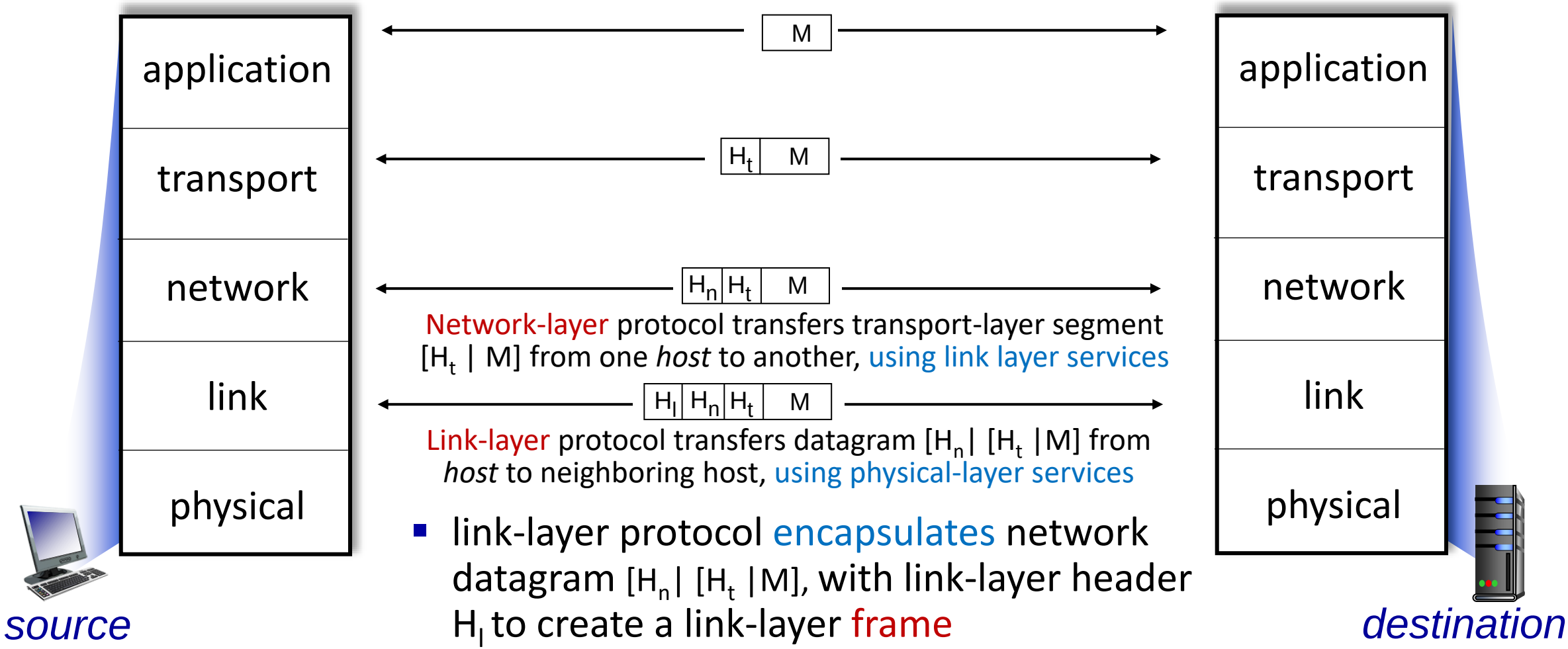
- transport-layer protocol encapsulates application-layer message, M, with *transport* layer-layer header H_t to create a transport-layer segment
  - H_t used by transport layer protocol to implement its service

Services, Layering and Encapsulation



- network-layer protocol encapsulates transport-layer segment [H_t | M] with network layer-layer header H_n to create a network-layer datagram
  - H_n used by network layer protocol to implement its service

Services, Layering and Encapsulation

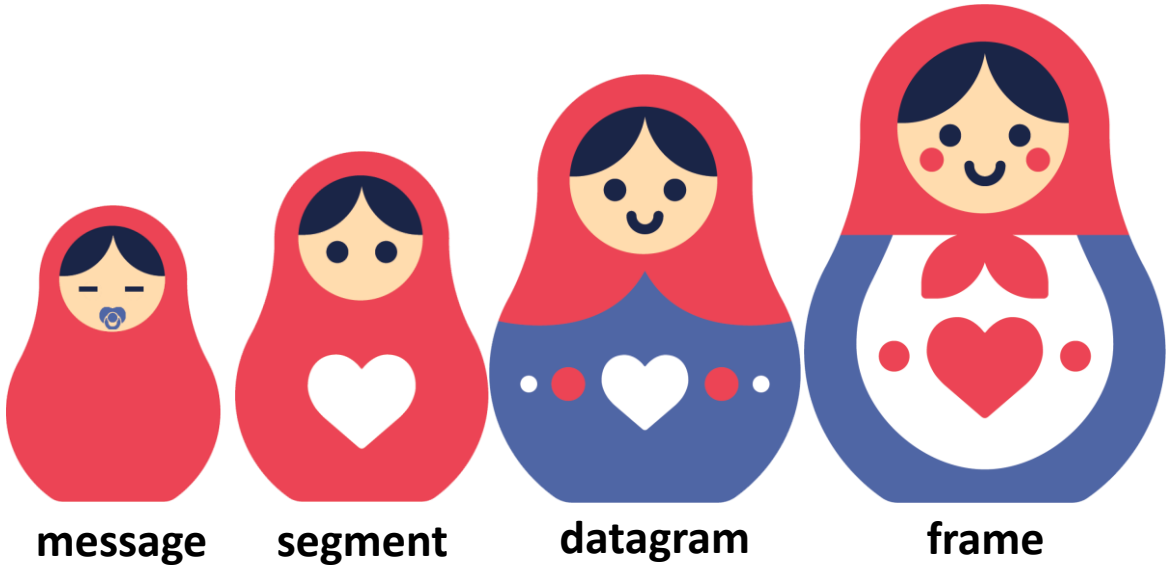


- link-layer protocol encapsulates network datagram [H_n| [H_t |M], with link-layer header H_l to create a link-layer frame

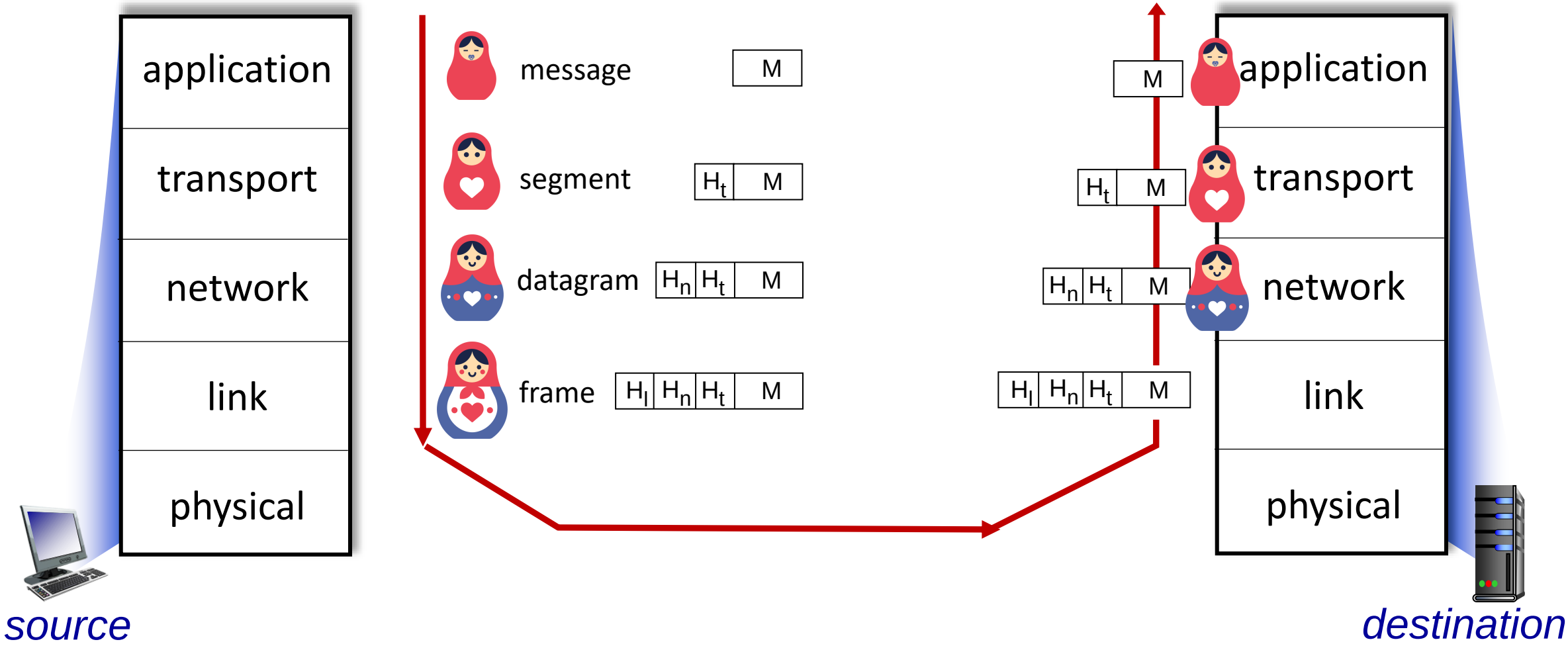
Encapsulation

Matryoshka dolls (stacking dolls)

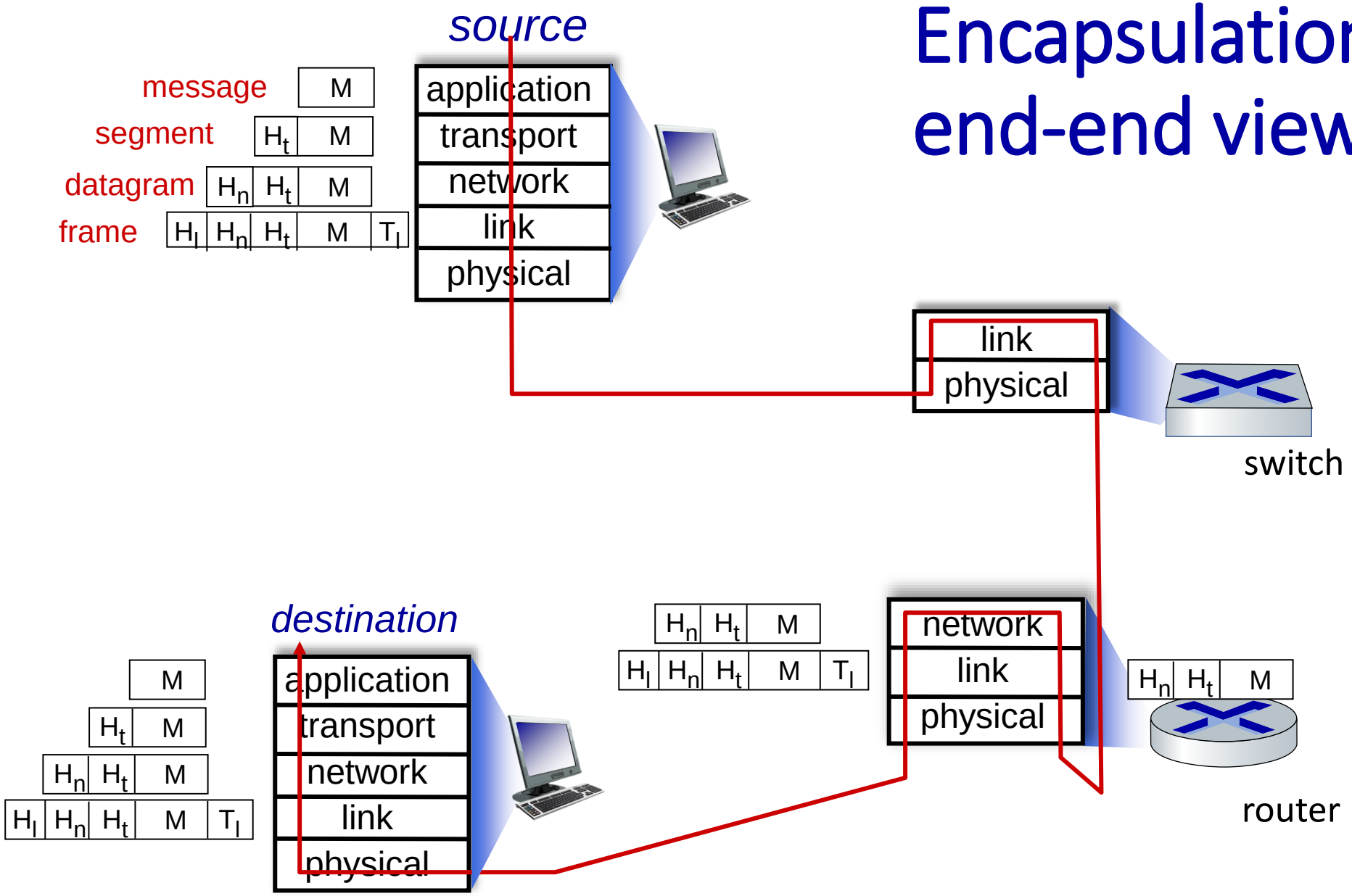




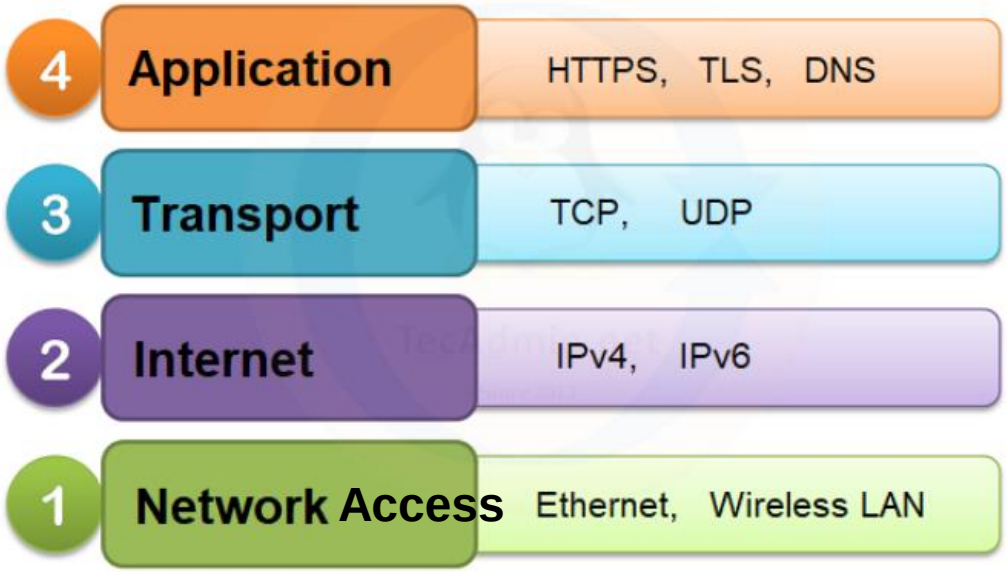
Services, Layering and Encapsulation

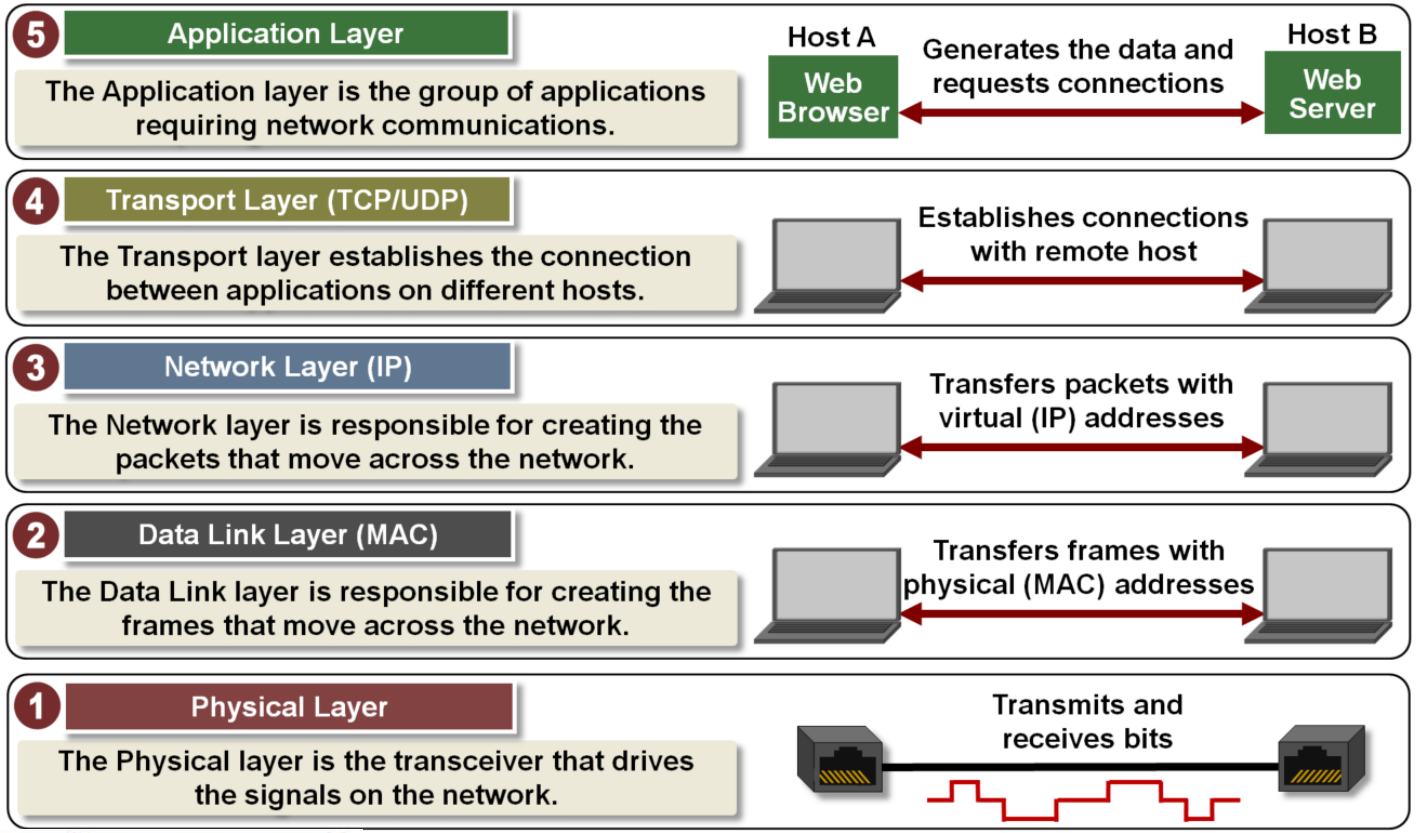


Encapsulation: an end-end view



TCP/IP Model

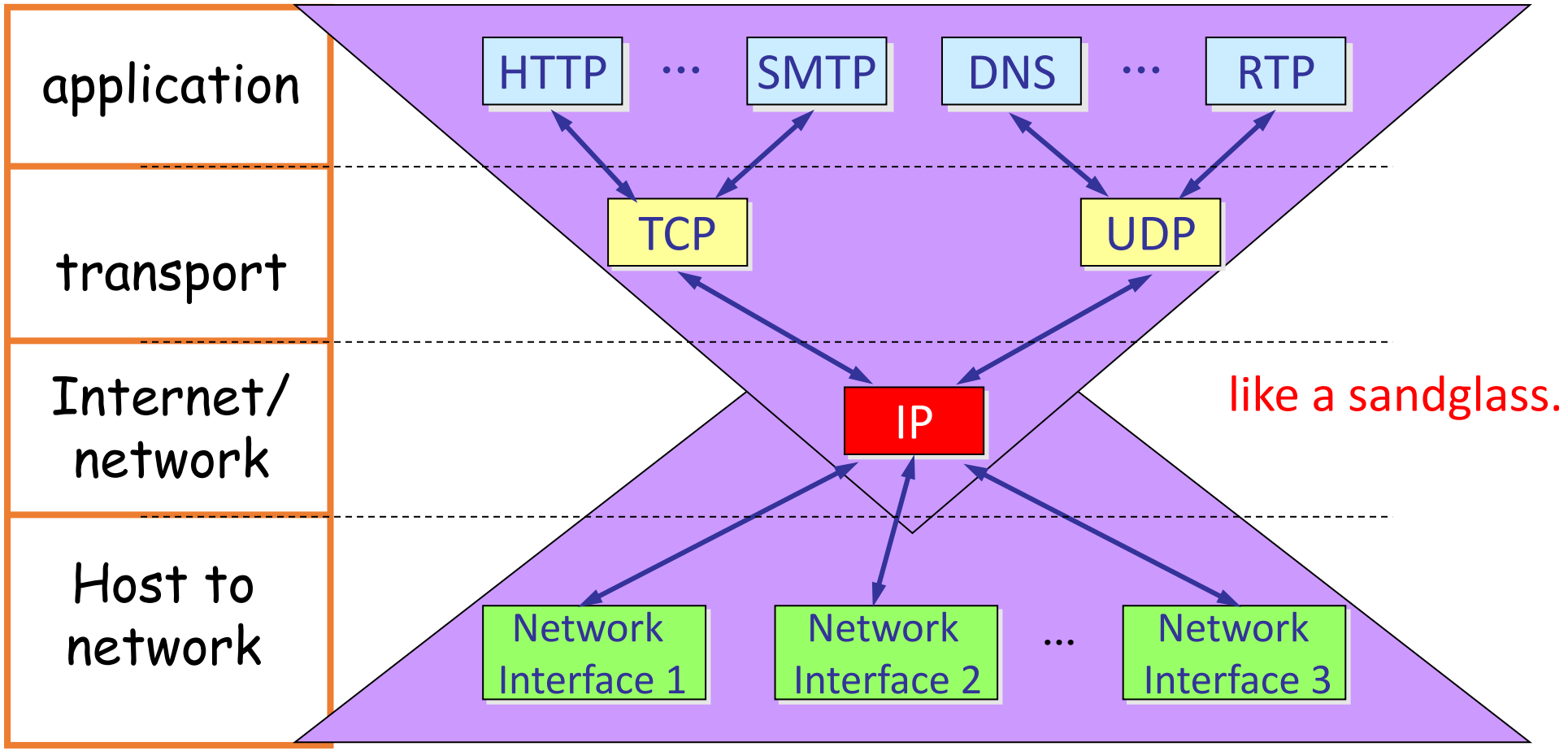




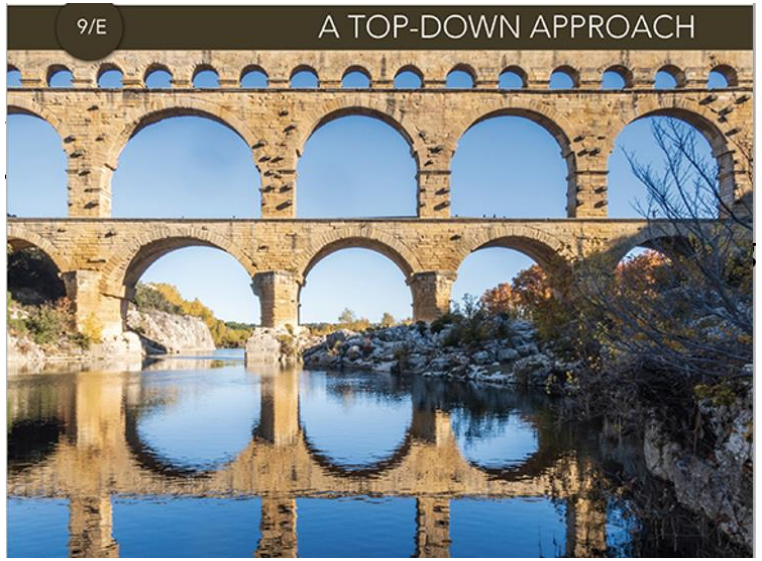
Layer #	Layer Name	Protocol	Protocol Data Unit	Addressing
5	Application	HTTP, SMTP, etc…	Messages	n/a
4	Transport	TCP/UDP	Segments	Port #s
3	Network or Internet	IP	Datagrams	IP Address
2	Data Link	Ethernet, Wi-Fi	Frames	MAC Address
1	Physical	10 Base T, 802.11	Bits	n/a

TCP/IP Model or Internet Protocol Suite

- TCP/IP protocols can serve different applications, so called "everything over IP"
- TCP/IP protocols allow IP protocol runs on different networks, i.e., "IP over everything".



Chapter 1: roadmap



- What *is* the Internet?
- What *is* a protocol?
- Network edge: hosts, access network, physical media
- Network core: packet/circuit switching, Internet structure
- Performance: loss, delay, throughput
- Protocol layers, service models
- Security
- History

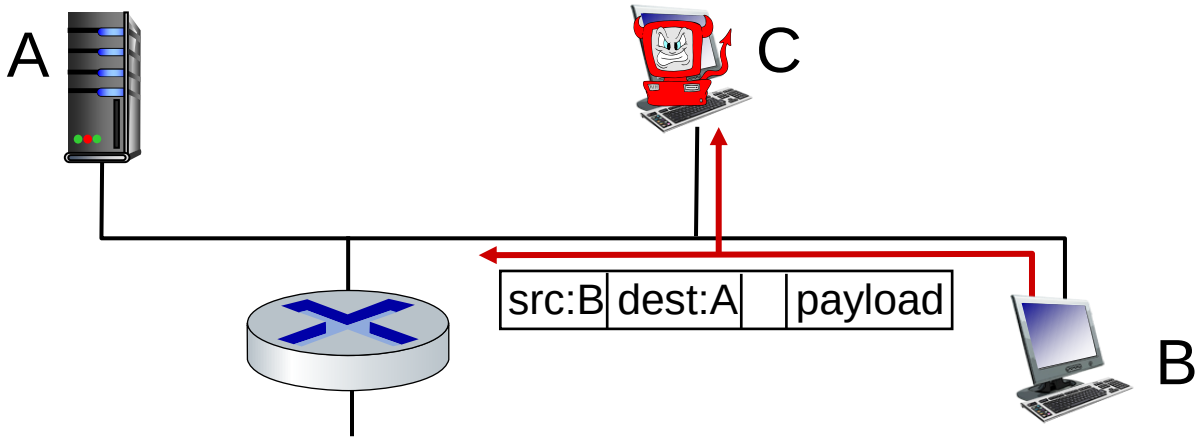
Network security

- Internet not originally designed with (much) security in mind
  - *original vision:* "a group of mutually trusting users attached to a transparent network" ☺
  - Internet protocol designers playing "catch-up"
  - security considerations in all layers!
- We now need to think about:
  - how bad guys can attack computer networks
  - how we can defend networks against attacks
  - how to design architectures that are immune to attacks

Bad guys: packet interception

packet "sniffing":

- broadcast media (shared Ethernet, wireless)
- promiscuous network interface reads/records all packets (e.g., including passwords!) passing by

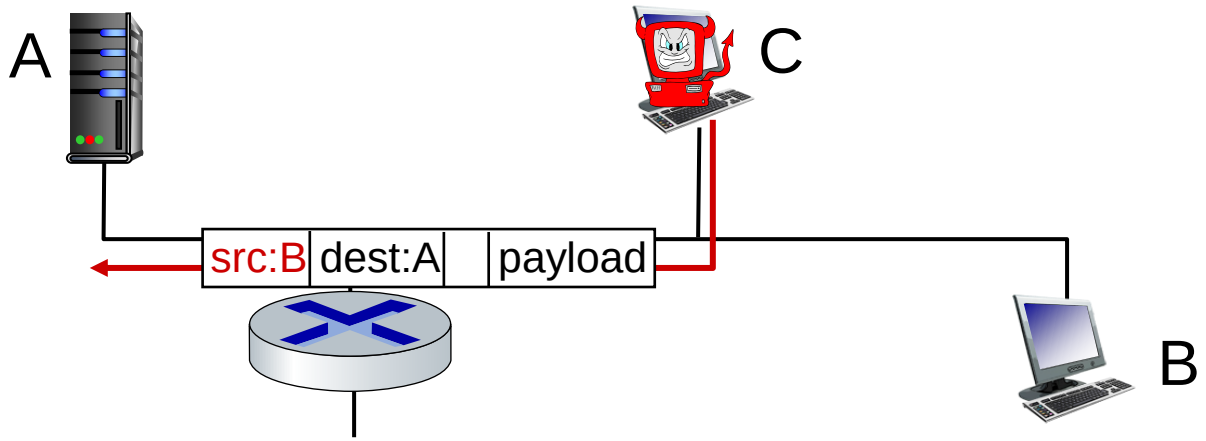




Wireshark software is a (free) packet-sniffer

Bad guys: fake identity

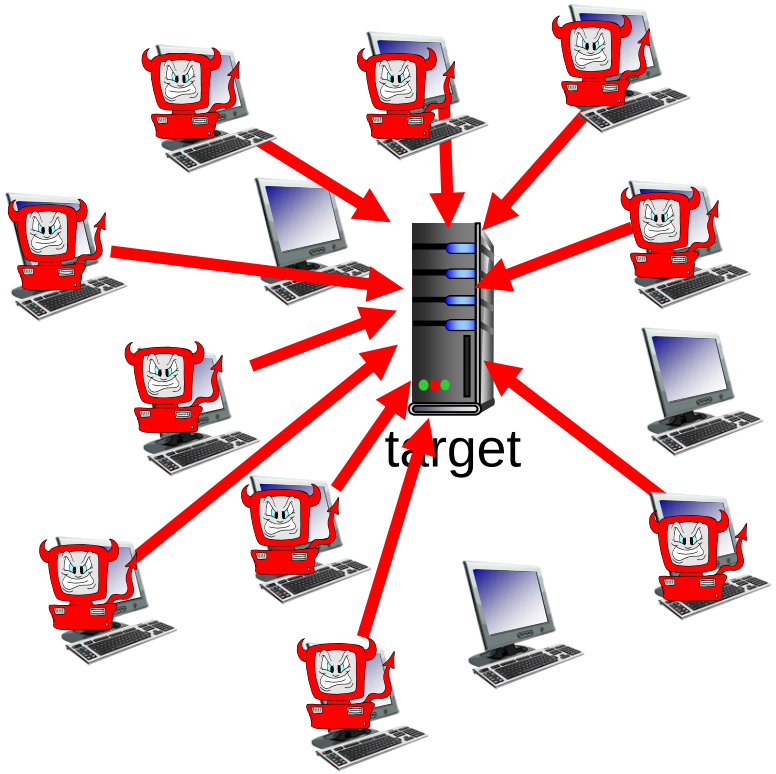
IP spoofing: injection of packet with false source address



Bad guys: denial of service

Denial of Service (DoS): attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic

1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts



Lines of defense:

- authentication: proving you are who you say you are
  - cellular networks provides hardware identity via SIM card; no such hardware assist in traditional Internet
- confidentiality: via encryption
- integrity checks: digital signatures prevent/detect tampering
- access restrictions: password-protected VPNs
- firewalls: specialized "middleboxes" in access and core networks:
  - off-by-default: filter incoming packets to restrict senders, receivers, applications
  - detecting/reacting to DOS attacks

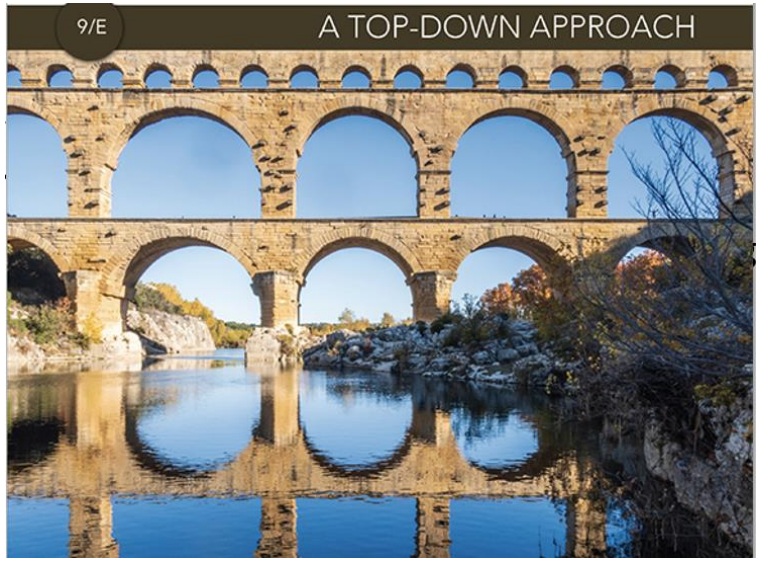








Chapter 1: roadmap

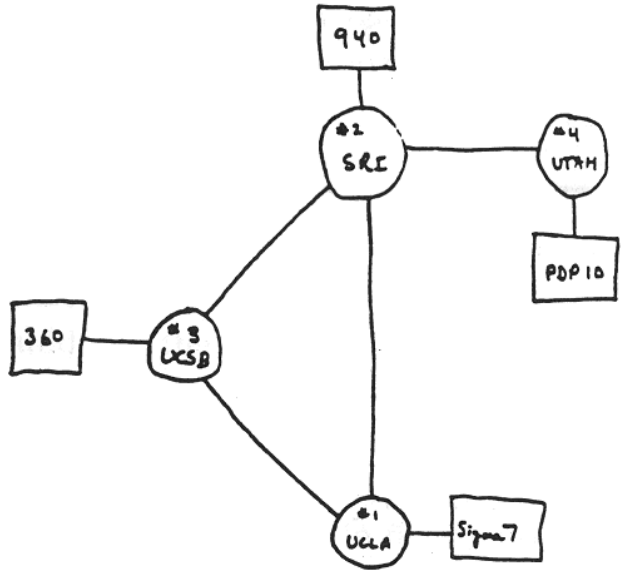


- What *is* the Internet?
- What *is* a protocol?
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- Protocol layers, service models
- Security
- History

Internet history

1961-1972: Early packet-switching principles

- 1961: Kleinrock - queueing theory shows effectiveness of packet-switching
- 1964: Baran - packet-switching in military nets
- 1967: ARPAnet conceived by Advanced Research Projects Agency
- 1969: first ARPAnet node operational



THE ARPA NETWORK

- 1972:
  - ARPAnet public demo
  - NCP (Network Control Protocol) first host-host protocol
  - first e-mail program
  - ARPAnet has 15 nodes

Internet history

1972-1980: Internetworking, new and proprietary networks

- 1970: ALOHAnet satellite network in Hawaii
- 1974: Cerf and Kahn - architecture for interconnecting networks
- 1976: Ethernet at Xerox PARC
- late70's: proprietary architectures: DECnet, SNA, XNA
- 1979: ARPAnet has 200 nodes

Cerf and Kahn's internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best-effort service model
- stateless routing
- decentralized control

define today's Internet architecture

The Principal Architects of the Internet



Robert E. Kahn
(born December 23, 1938, Brooklyn, New York, U.S.)



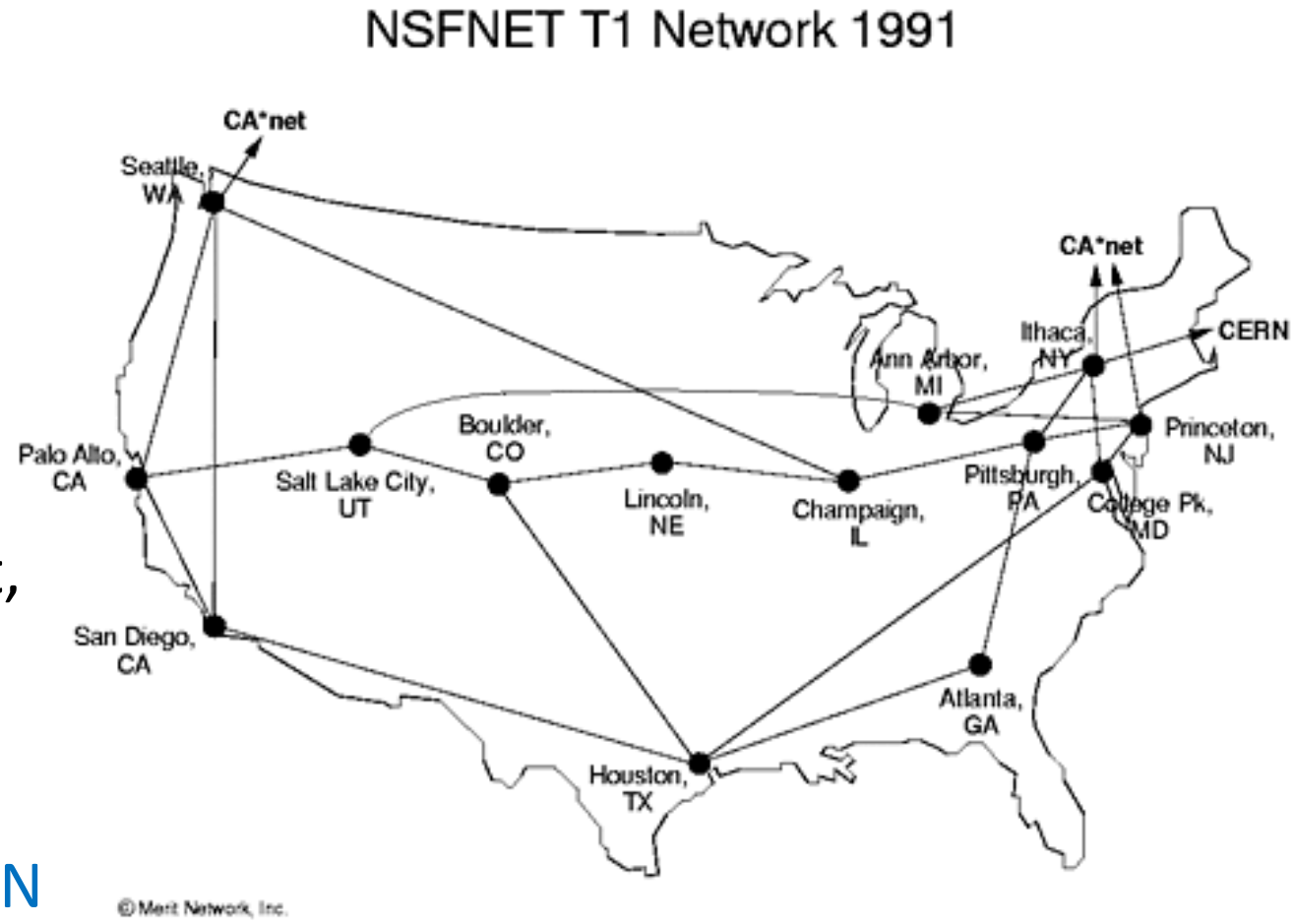


- In 2004 both Kahn and Cerf won the A.M. Turing Award

Internet history

1980-1990: new protocols, a proliferation of networks

- 1983: deployment of TCP/IP
- 1982: smtp e-mail protocol defined
- 1983: DNS defined for name-to-IP-address translation
- 1985: ftp protocol defined
- 1988: TCP congestion control
- new national networks: CSnet, BITnet, NSFnet, Minitel
- 100,000 hosts connected to confederation of networks
- 1989-1991: Web was invented at CERN



Internet history

1990, 2000s: commercialization, the Web, new applications

- early 1990s: ARPAnet decommissioned
- 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- early 1990s: Web
  - hypertext [Bush 1945, Nelson 1960's]
  - HTML, HTTP: Berners-Lee
  - 1994: Mosaic, later Netscape
  - late 1990s: commercialization of the Web

late 1990s – 2000s:

- more killer apps: instant messaging, P2P file sharing
- network security to forefront
- est. 50 million host, 100 million+ users
- backbone links running at Gbps

Internet history

2005-present: scale, SDN, mobility, cloud

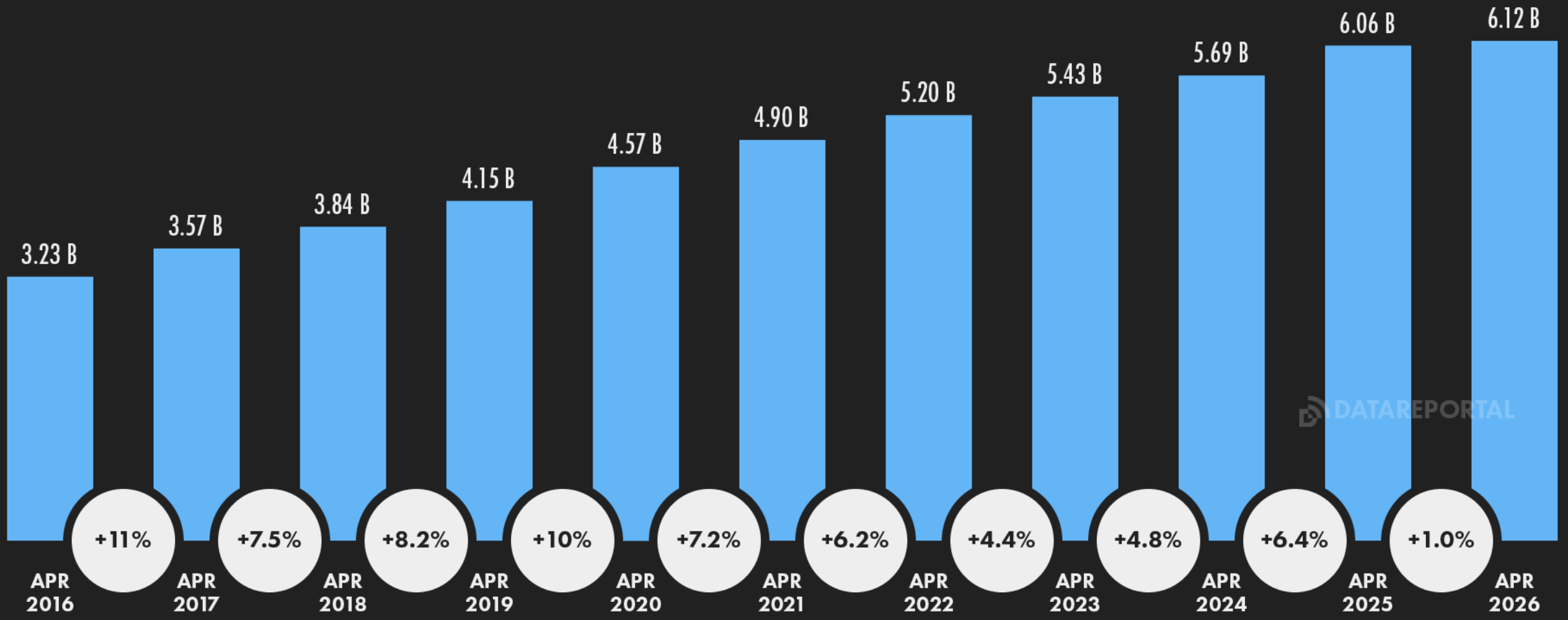
- aggressive deployment of broadband home access (10-100's Mbps)
- 2008: software-defined networking (SDN)
- increasing ubiquity of high-speed wireless access: 4G/5G, WiFi
- service providers (Google, FB, Microsoft) create their own networks
  - bypass commercial Internet to connect "close" to end user, providing "instantaneous" access to social media, search, video content, …
- enterprises run their services in "cloud" (e.g., Amazon Web Services, Microsoft Azure)
- rise of smartphones: more mobile than fixed devices on Internet (2017)
- More than 6.12 billion people use the Internet as of April 2026, representing 73.8% of the world's population

APR 2026

INTERNET USE OVER TIME (YOY)

NUMBER OF INDIVIDUALS USING THE INTERNET, AND YEAR-ON-YEAR CHANGE

GLOBAL OVERVIEW



SOURCES: MANOCHI ANALYSIS; ITU; GSMA INTELLIGENCE; EUROSTAT; GOOGLE'S ADVERTISING RESOURCES; CNNIC; KANTAR & IAMAI; GOVERNMENT RESOURCES; UNITED NATIONS. **NOTE:** WHERE LETTERS ARE SHOWN NEXT TO FIGURES ABOVE BARS, "K" DENOTES THOUSANDS (E.G. "123 K" = 123,000), "M" DENOTES MILLIONS (E.G. "1.23 M" = 1,230,000), AND "B" DENOTES BILLIONS (E.G. "1.23 B" = 1,230,000,000). WHERE NO LETTER IS PRESENT, VALUES ARE SHOWN AS IS. **COMPARABILITY:** SOURCE AND BASE CHANGES. ALL FIGURES USE THE LATEST AVAILABLE DATA, BUT SOME SOURCES DO NOT PUBLISH REGULAR UPDATES, SO FIGURES FOR RECENT PERIODS MAY UNDER-REPRESENT ACTUAL USE. SEE NOTES ON DATA.

APR 2026

ESSENTIAL DIGITAL HEADLINES

OVERVIEW OF THE ADOPTION AND USE OF CONNECTED DEVICES AND SERVICES

GLOBAL OVERVIEW

TOTAL POPULATION	UNIQUE MOBILE PHONE SUBSCRIBERS	INDIVIDUALS USING THE INTERNET	SOCIAL MEDIA USER IDENTITIES	MONTHLY ACTIVE GENERATIVE AI USERS
				
8.28 BILLION	**5.83 BILLION**	**6.12 BILLION**	**5.79 BILLION**	**2.42 BILLION**
URBANISATION	vs. POPULATION	vs. POPULATION	vs. POPULATION	vs. POPULATION
80.5%	**70.4%**	**73.8%**	**69.9%**	**29.2%**

SOURCES: MANOCHI ANALYSIS; UNITED NATIONS; GOVERNMENT RESOURCES; GSMA INTELLIGENCE; ITU; EUROSTAT; CNNIC; SIMILARWEB; PLATFORM RESOURCES; COMPANY EARNINGS REPORTS; OCDH; BETA RESEARCH CENTER. **ADVISORY:** SOCIAL MEDIA USER IDENTITIES AND GENERATIVE AI USERS MAY NOT REPRESENT UNIQUE INDIVIDUALS. **NOTES:** WHERE AVAILABLE, THE FIGURE FOR URBANISATION REPRESENTS THE U.N.'S REPORTED VALUE FOR THE PERCENTAGE OF THE TOTAL POPULATION LIVING IN TOWNS AND CITIES. FIGURES FOR GENERATIVE AI USE ONLY INCLUDE THE USE OF STANDALONE GENERATIVE AI PLATFORMS, AND DO NOT FACTOR THE USE OF AI FUNCTIONALITY WITHIN OTHER SOFTWARE OR APPLICATIONS. **COMPARABILITY:** BASE REVISIONS; SOURCE CHANGES. SEE NOTES ON DATA.

we are social

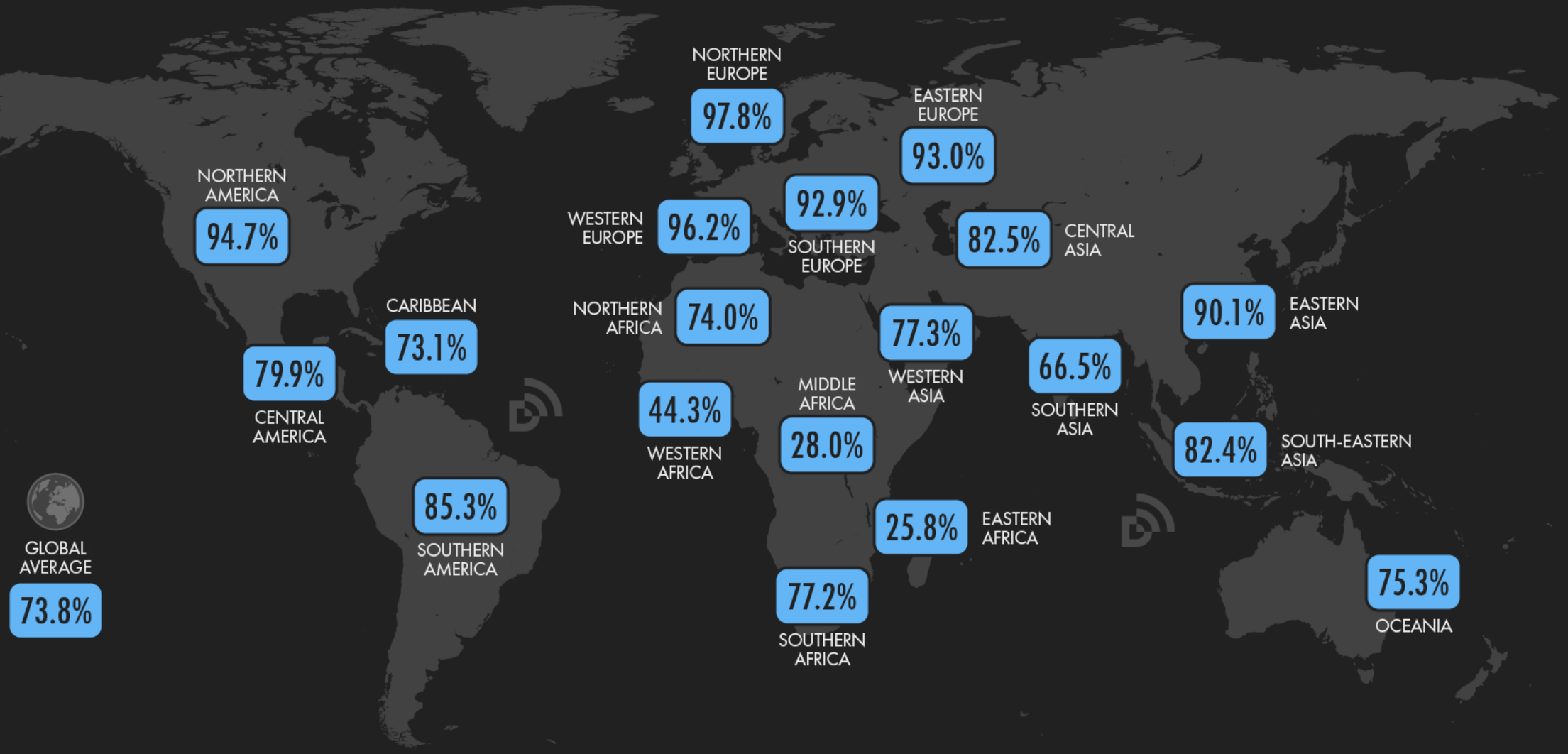
MANOCHI

APR 2026

INTERNET ADOPTION

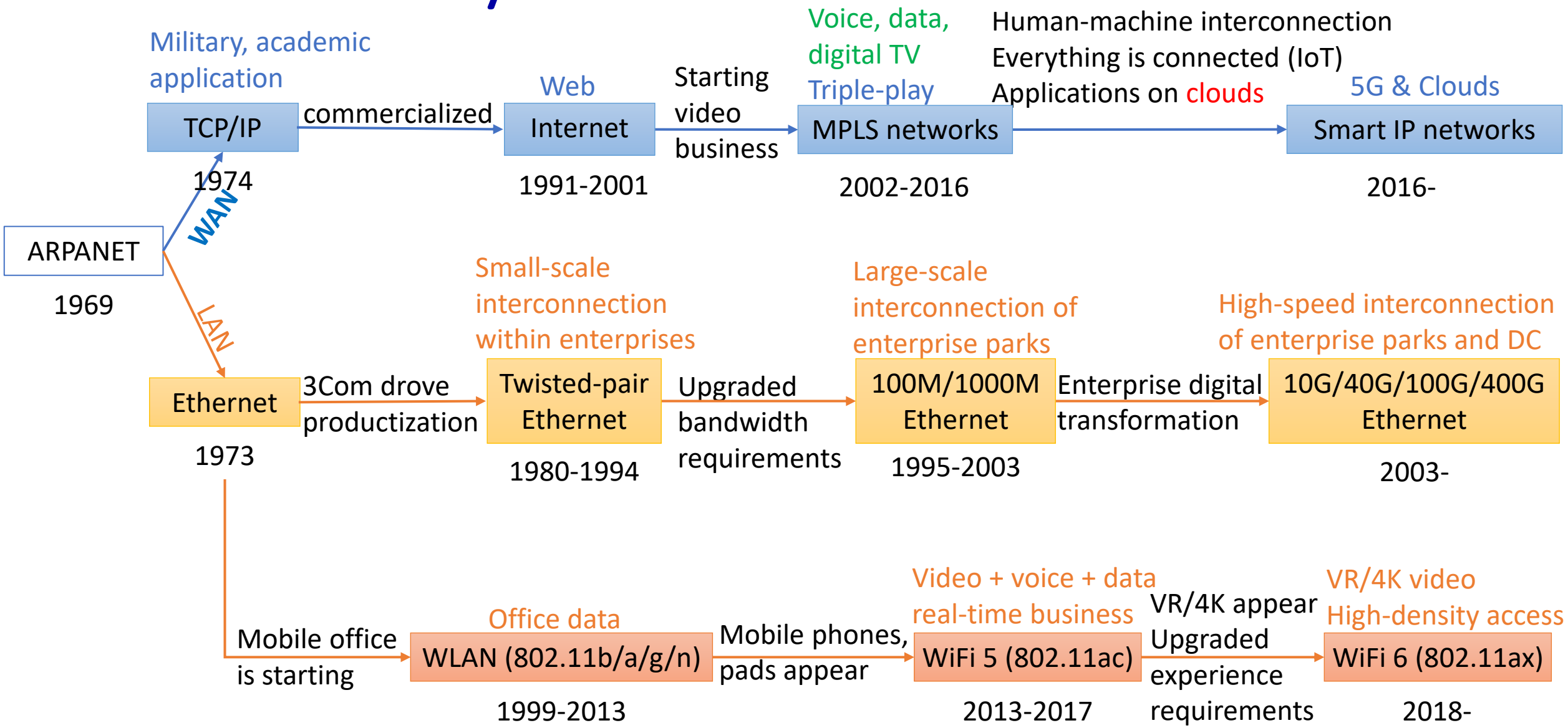
INDIVIDUALS USING THE INTERNET AS A PERCENTAGE OF TOTAL POPULATION

GLOBAL OVERVIEW

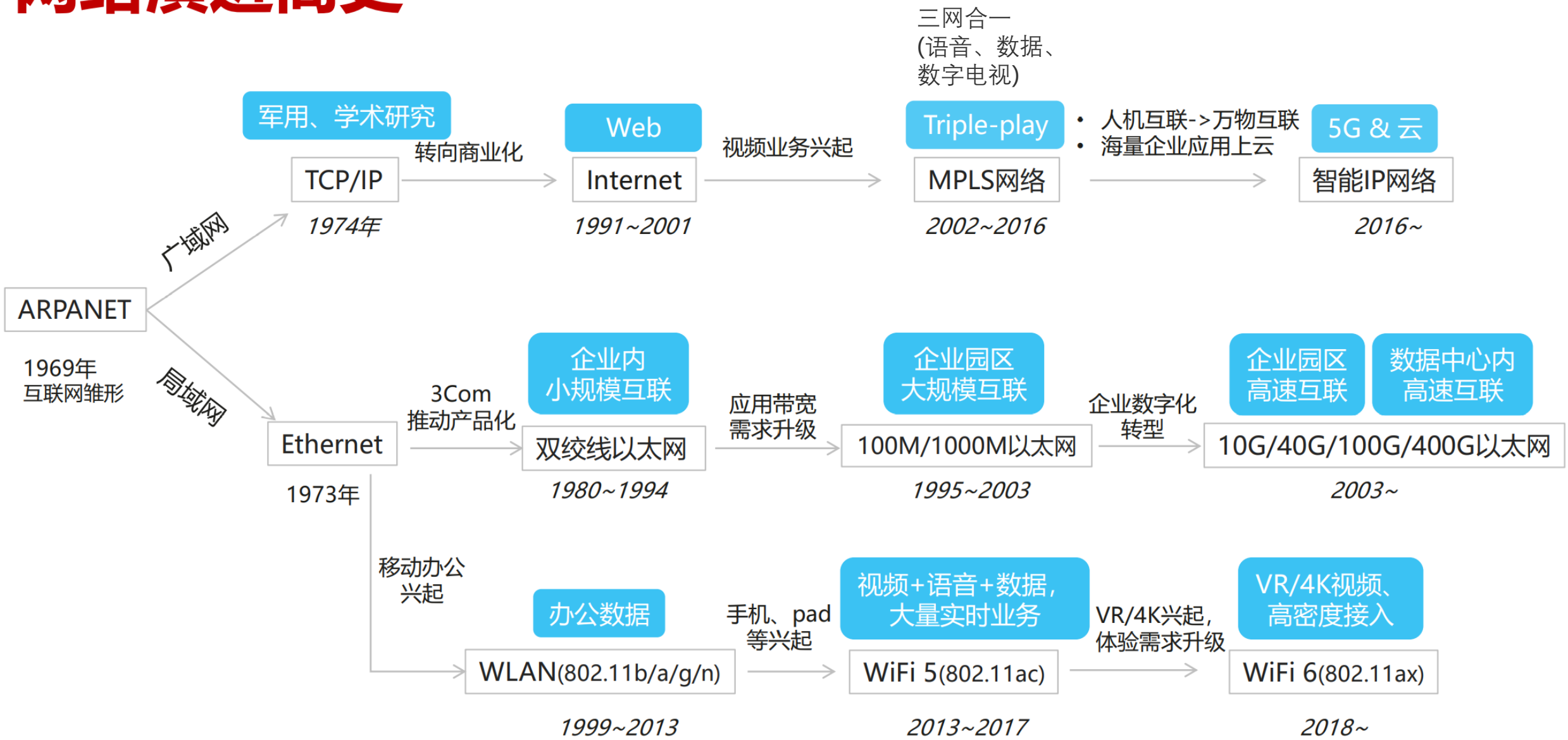


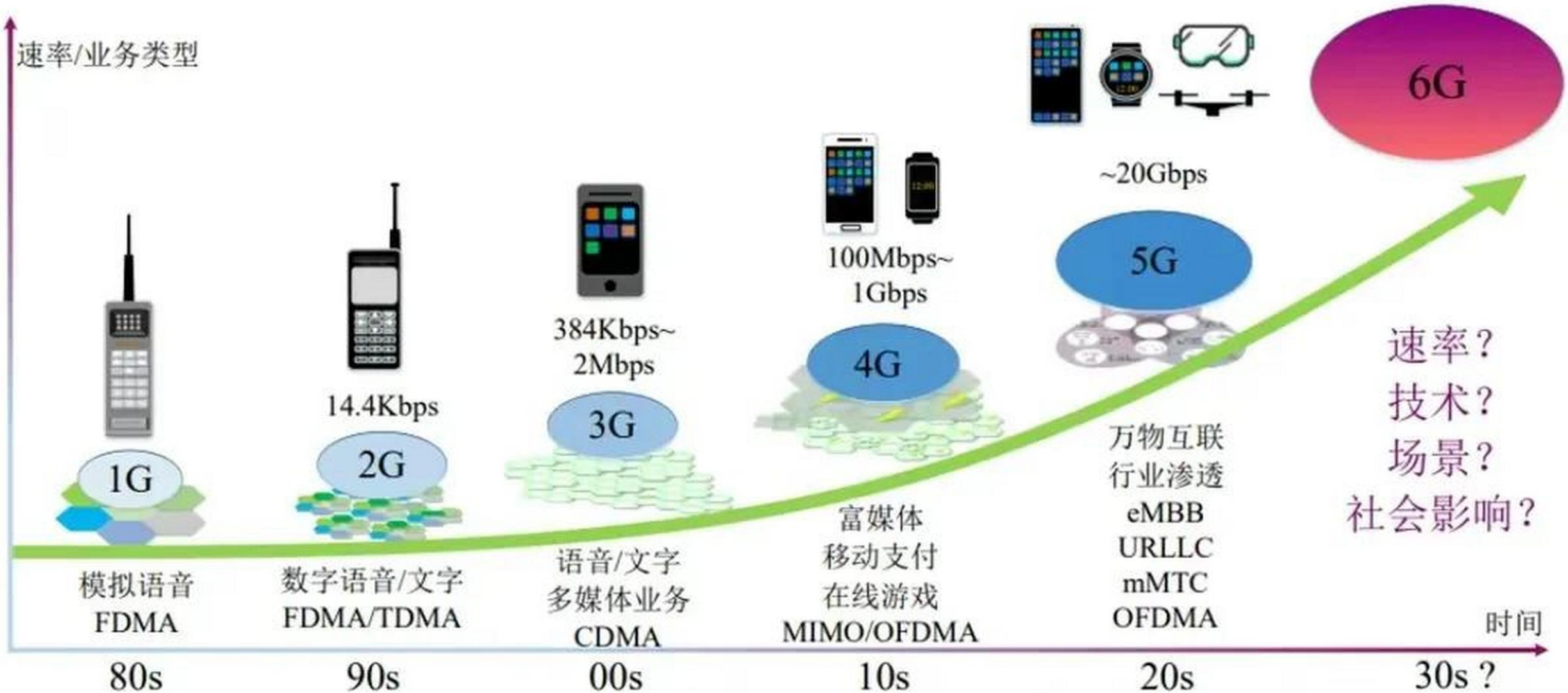
SOURCES: MANOCHI ANALYSIS; ITU; GSMA INTELLIGENCE; EUROSTAT; GOOGLE'S ADVERTISING RESOURCES; CNNIC; GOVERNMENT RESOURCES; UNITED NATIONS. **NOTE:** REGIONS BASED ON THE UNITED NATIONS GEOSCHEME. LOCAL COUNTRY VALUES HAVE BEEN CAPPED AT 99% OF THE POPULATION. **COMPARABILITY:** SOURCE AND BASE CHANGES. ALL FIGURES USE THE LATEST AVAILABLE DATA, BUT SOME SOURCES DO NOT PUBLISH REGULAR UPDATES, SO FIGURES MAY UNDER-REPRESENT ACTUAL USE. SEE NOTES ON DATA.

A brief history of network evolution

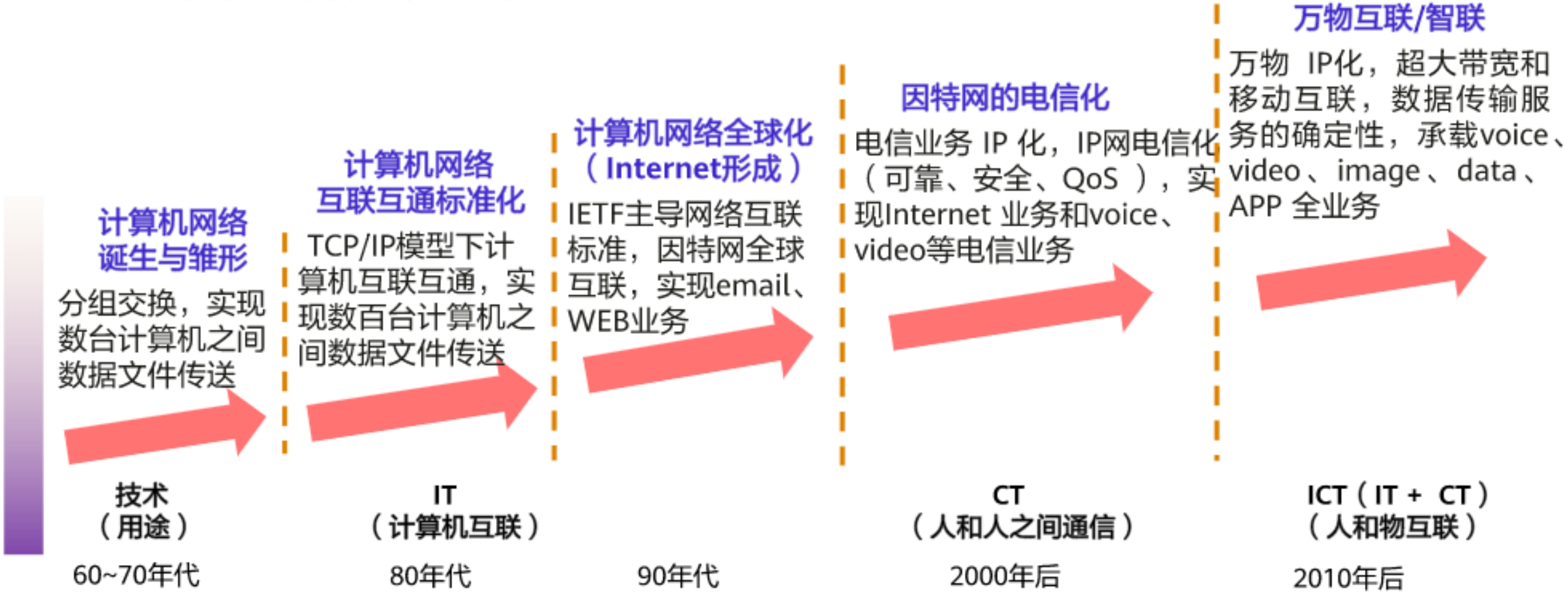


网络演进简史


三网合一
(语音、数据、
数字电视)
军用、学术研究
TCP/IP
1974年
转向商业化
Web
Internet
1991~2001
视频业务兴起
Triple-play
MPLS网络
2002~2016
人机互联->万物互联
海量企业应用上云
5G & 云
智能IP网络
2016~
广域网
ARPANET
1969年
互联网雏形
局域网
Ethernet
1973年
3Com
推动产品化
企业内
小规模互联
双绞线以太网
1980~1994
应用带宽
需求升级
企业园区
大规模互联
100M/1000M以太网
1995~2003
企业数字化
转型
企业园区
高速互联
数据中心内
高速互联
10G/40G/100G/400G以太网
2003~
移动办公
兴起
办公数据
WLAN(802.11b/a/g/n)
1999~2013
手机、pad
等兴起
视频+语音+数据,
大量实时业务
WiFi 5(802.11ac)
2013~2017
VR/4K兴起,
体验需求升级
VR/4K视频、
高密度接入
WiFi 6(802.11ax)
2018~

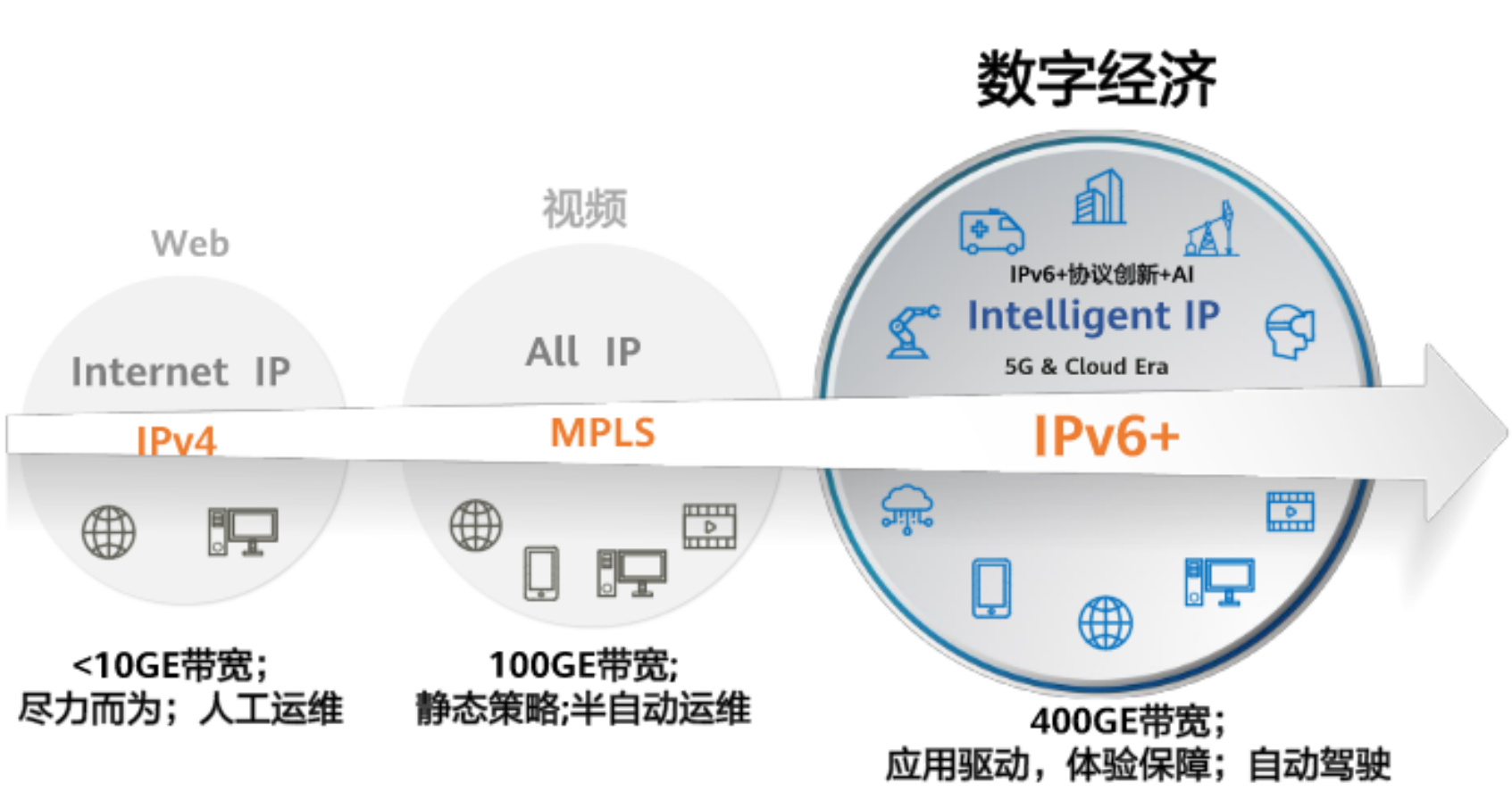

速率/业务类型
1G
模拟语音
FDMA
80s
14.4Kbps
2G
数字语音/文字
FDMA/TDMA
90s
384Kbps~
2Mbps
3G
语音/文字
多媒体业务
CDMA
00s
100Mbps~
1Gbps
4G
富媒体
移动支付
在线游戏
MIMO/OFDMA
10s
~20Gbps
5G
万物互联
行业渗透
eMBB
URLLC
mMTC
OFDMA
20s
6G
速率？
技术？
场景？
社会影响？
30s？
时间

数据通信网络发展历程



- **是一个泛在连接网**：数据通信网（IP网）无处不在，已经是网络的代名词
- **是一个“种豆得瓜”的网络**：种的“豆”是用于文件传送的计算机网，得的“瓜”是用于万物智联的数字经济底座
- **是一个不断发展的网络**：一直在追求更快、更安全、更可靠、更智能

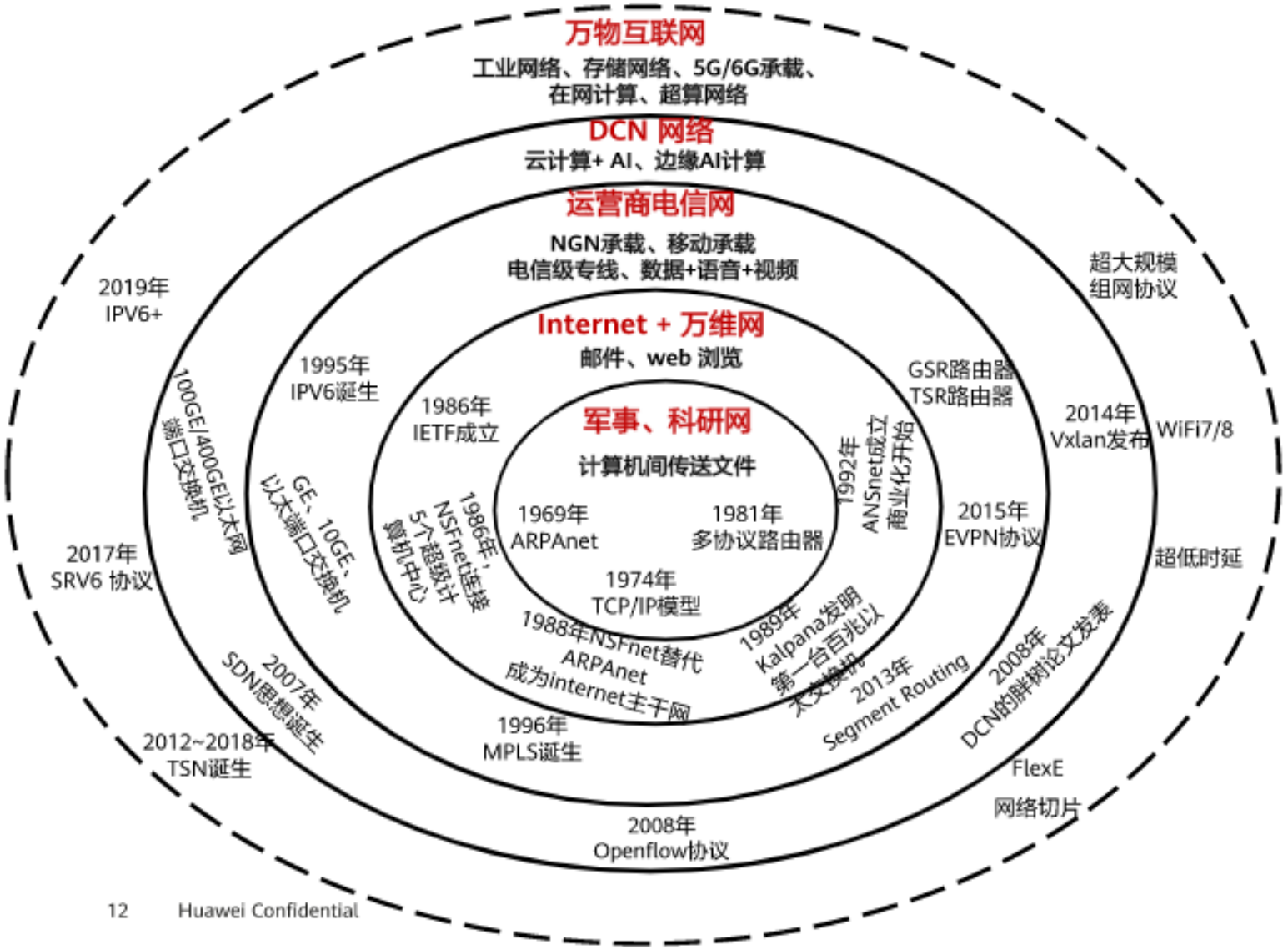
明天的网络：打造数字经济的基建底座





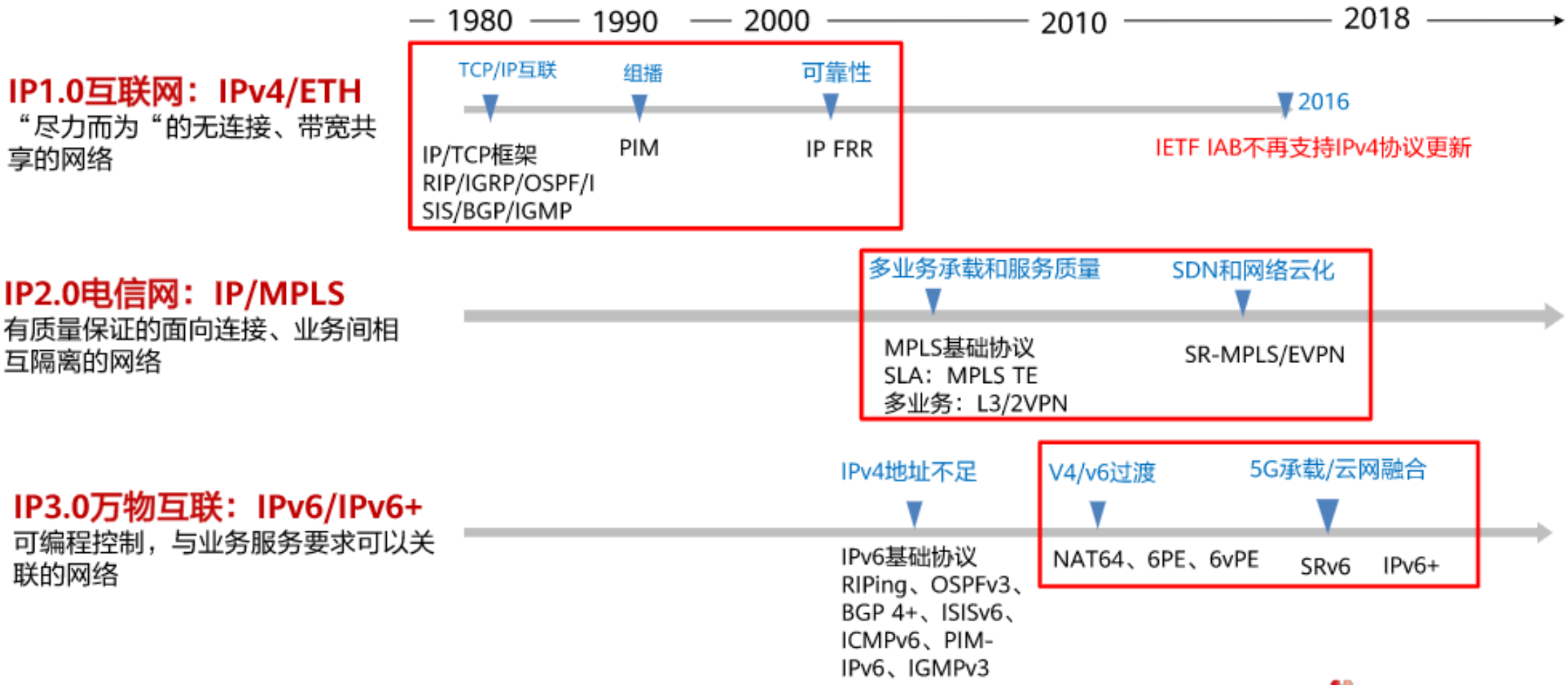
IETF、ETSI 启动了 IPv6+ 的标准化和产业创新工作

数通网络（IP网）在产业应用驱动下圈地式发展



- **IP网已经是无所不能**
  - 计算机间交换信息
  - PSTN电话语音
  - DDN数据专线
  - CATV 视频
  - 安防监控
  - 线上交互服务
  - 车联网
  - 车间生产管理与控制
  - 。。。
- **IP网络能胜任?**
  - 和业务相对分离，松耦合
  - 开放的网络架构（可弹性伸缩）
  - 技术上可不断革新进步

IP网络发展经历三个大的发展阶段



Chapter 1: summary

We've covered a "ton" of material!

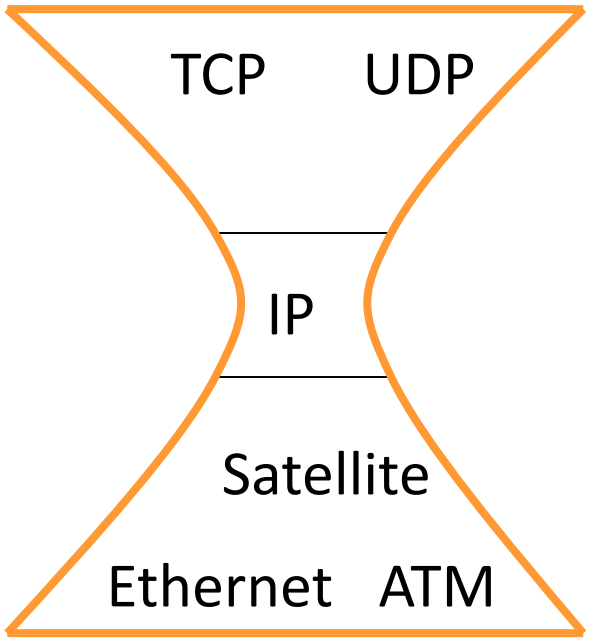
- Internet overview
- what's a protocol?
- network edge, access network, core
  - packet-switching versus circuit-switching
  - Internet structure
- performance: loss, delay, throughput
- layering, service models
- security
- history

You now have:

- context, overview, vocabulary, "feel" of networking
- more depth, detail, *and fun* to follow*!*

Summary: Internet Architecture

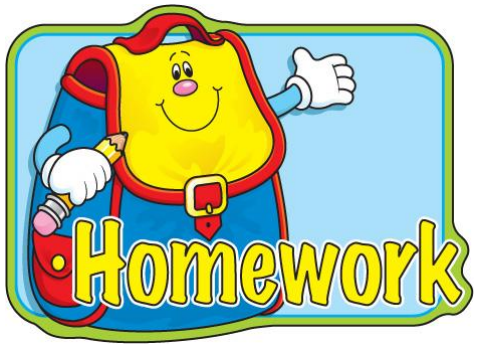
- packet-switched datagram network
- IP is the glue (network layer overlay)
- IP hourglass architecture
  - all hosts and routers run IP
- stateless architecture
  - no per flow state inside network



IP hourglass

Summary: Minimalist Approach

- Dumb network
  - IP provide minimal functionalities to support connectivity
  - addressing, forwarding, routing
- Smart end system
  - transport layer or application performs more sophisticated functionalities
  - flow control, error control, congestion control
- Advantages
  - accommodate heterogeneous technologies (Ethernet, modem, satellite, wireless)
  - support diverse applications (telnet, ftp, Web, X windows)
  - decentralized network administration



HOMEWORK

1. Give the practical examples about PAN, LAN, MAN, WAN.
2. Give us more information about IANA.
3. The present situation of DSL
4. The present situation of cable-based access network
5. Present Satellite Communication Services provided in China
6. Introduction about datacenters.
7. Some knowledge about peering, IXPs, content providers. Give some actual examples.
8. Introduce the brief history of network evolution.

Wireshark Lab 1

get acquainted with Wireshark, and make some simple packet captures and observations.

